Estimating a Rational Inattention Discrete Choice Model: A Semiparametric Approach

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Abstract. People's choices depend on both the payoffs of the available options and what they learn about those payoffs before deciding. This paper shows how choice data identify both unobserved payoff heterogeneity and the cost of information acquisition in a rational inattention discrete-choice model. I establish conditions for semiparametric point identification using exogenous payoff and information-cost shifters, without imposing parametric forms on the taste-shock distribution or on how the unit cost of information varies with observed characteristics. In the model with binary choice, I construct an unobserved choice threshold whose cumulative distribution function is the choice probability as a function of the payoff shifter. Moments of this threshold identify the payoff coefficients and the unit cost of information at each value of the information-cost shifter. A deconvolution argument then identifies the taste-shock distribution. I then extend identification to the model with multinomial choice. To estimate the model, I propose a sieve minimum-distance estimator based on observed market shares and establish its consistency. For inference, I construct confidence regions by test inversion because binding parameter constraints can make the estimator's limiting distribution nonnormal. In an application to Amazon Mechanical Turk, model counterfactuals show that lower unit costs of information raise the average task-taking probability and reduce wage elasticity.

Keywords. Rational inattention; discrete choice; semiparametric identification; payoff heterogeneity; sieve estimation; test inversion.

JEL classification. C12, C14, C25, D83, J22.

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1 Introduction

Following McFadden (1973), econometricians use random utility models to explain heterogeneity in observed choices. These models typically assume that individuals know their taste shocks, the idiosyncratic payoff components unobserved by the econometrician. In real-world situations, however, individuals may be uncertain about these components, and learning about them may require time and effort. To account for the cost of learning, Matějka and McKay (2015) apply the rational inattention (RI) framework pioneered by Sims (2003) to discrete choice. In this model, rationally inattentive individuals optimally choose what to learn by trading off the expected payoff gains from learning against the cost of acquiring information. This paper studies how the observed choices of these individuals can identify the taste-shock distribution and the unit cost of information as a function of observed characteristics.

Existing identification analyses of RI models have relied on parametric specifications (Liao, 2024; Joo, 2023; Brown and Jeon, 2024). For example, Brown and Jeon (2024) establish identification of an insurance-demand model using a Cardell distribution (Cardell, 1997) for taste shocks and a parametric specification of how the unit cost of information varies with observed characteristics. Their distributional assumptions yield closed-form choice probabilities, which they use together with the exogeneity of observed characteristics to derive identification conditions. Less is known about identification without these parametric restrictions.

Without imposing parametric forms on the taste-shock distribution or on the unit cost of information as a function of observed characteristics, this paper establishes conditions under which exogenous variation in choice data identifies both. For choices between two alternatives, the required variation comes from two observed characteristics: one changes payoffs and the other changes the unit cost of information. An observed utility shifter enters the payoff component known to the individual linearly. An observed information-cost shifter changes the unit cost of information without changing that known payoff component. The taste shocks are independent of both shifters. With this variation, I extend Lewbel's (1997) method of recovering payoff parameters from moments of an unobserved choice threshold to the RI model. I construct a similar threshold, which I call a switch point; when the utility shifter is a wage, it can be interpreted as an implied reservation wage. The switch point's cumulative distribution function gives the probability of choosing one alternative over a reference alternative, which I call alternative 0. This probability is a function of the utility shifter, holding the information-cost shifter fixed. The moments of this switch point identify the payoff coefficients and the unit cost of information at each value of the information-cost shifter. A deconvolution argument then recovers the taste-shock distribution.

Extending the argument for two alternatives establishes identification of the RI model with multiple alternatives. At positive information costs, payoff variation, such as lower wages, can leave only alternative 0 and one other alternative with positive choice probabilities. The individual's problem is then a binary RI problem. Applying the same switch-point moment identities to each other alternative paired with alternative 0 identifies the payoff coefficients and the unit cost of information common to the alternatives. I then use the probability of choosing alternative 0 and multivariate deconvolution to recover the joint distribution of taste shocks.

This paper proposes a sieve minimum-distance estimator using market shares and develops a test-inversion approach for inference that remains asymptotically valid when parameter constraints make the estimator's limiting distribution nonnormal. The estimator uses finite-dimensional approximations to the taste-shock distribution and the unit cost of information as a function of observed characteristics. The RI model gives an equation that the true parameters and population choice probabilities must satisfy. I estimate the parameters by minimizing the average squared difference between the equation's two sides after replacing population choice probabilities with observed shares. I establish consistency even though the criterion can be discontinuous at zero information cost. Inference must accommodate estimates at zero: information costs cannot be negative, so an estimated cost can reach this boundary. Similarly, when the taste-shock distribution is approximated by nonnegative probability weights on a grid extending beyond its true support, some estimated weights can be zero. I obtain intervals for functions of the model parameters by taking their ranges over parameter values that the test does not reject.

The empirical application uses data collected by Dube et al. (2020) from Amazon Mechanical Turk to study how workers' labour supply decisions vary with the difficulty of acquiring information about jobs. On this online platform, workers are paid to complete short tasks, such as transcribing conversations, answering survey questions, and entering data. Estimated information costs are lowest and close to zero when descriptions specify both what workers must do and how much work is required. The wage elasticity of the average task-taking probability, calculated over a doubling of posted rewards, falls from 0.287 at estimated information costs to 0.068 when costs are halved and 0.037 under full information. At the original rewards, the average task-taking probability rises from 2.62% at estimated costs to 8.90% when costs are halved and 12.87% under full information. These counterfactual results suggest that lower unit costs of information can increase workers' willingness to take jobs while reducing their responsiveness to higher wages in this online labour market.

Related Literature This paper contributes to three strands of literature. First, this paper contributes to RI theory by applying the framework to observational choice data when individuals are initially uncertain about their own idiosyncratic payoff components, which the researcher does not observe. Sims (2003) shows how slow responses to economic shocks can arise from limited information processing even when actions can be adjusted immediately. Individuals optimally choose the information on which to base their decisions, subject to a constraint measured by Shannon mutual information. Matějka and McKay (2015) apply this framework to discrete choice under Shannon information costs and derive a modified multinomial logit formula for choice probabilities conditional on realized payoffs. The alternative-specific weights in this formula depend on prior beliefs[1] and information costs. Fosgerau et al. (2020) show that RI models with more general information costs can generate the choice probabilities of additive random utility models such as nested logit. Caplin et al. (2022) derive restrictions on observed choices that distinguish Shannon costs from more general information costs. These restrictions concern how learning changes with prior beliefs and with different descriptions of the same payoff uncertainty.

Second, this paper contributes to the econometric literature on RI discrete-choice models. Empirical RI research has studied models with known or parametrically specified distributions of idiosyncratic taste shocks. In contrast, I establish identification without imposing a parametric form on this distribution or on the unit cost of information as a function of observed characteristics. The most closely related work is Brown and Jeon (2024), who identify and estimate an insurance-demand model using Cardell distributions and a parametric specification of information costs. In their application, consumers acquire costly information about individual-specific expected out-of-pocket costs and other plan characteristics observed by the econometrician. Therefore, in their framework, the data identify choice probabilities conditional on these characteristics and other observed covariates. By contrast, I study identification when agents acquire costly information about idiosyncratic payoff components that are unobserved by the econometrician. Joo (2023) develops a parametric RI demand model without specifying consumers' subjective priors, showing that a prior exists under which the model's choice probabilities are optimal. He uses retail promotions that affect information about products but do not enter consumption utility to test the model against full-information demand. Liao (2024) studies identification of prior beliefs and payoff parameters in an RI discrete-choice model where individuals are uncertain about market-level payoff shocks unobserved by the researcher. Unlike in my model, individuals initially know their private taste shocks, and the researcher knows their distribution. Liao also uses exogenous variation in the available alternatives to test full-information choice and shows how costly information acquisition can explain

[1]The probability distribution that describes an individual's beliefs about possible payoffs before acquiring information is called a prior belief. A prior is correct when it matches the true payoff distribution. Subjective priors reflect individuals' own assessments and may differ from that distribution.

violations of the full-information model's predictions. More generally, Gualdani and Sinha (2024) leave information structures unrestricted within a model with parameterized payoffs and prior beliefs. Using Bayes correlated equilibrium, they derive sharp identified sets for these parameters and bounds on counterfactual predictions.

Third, this paper contributes to the semiparametric discrete-choice literature by establishing identification in an RI model. Lewbel (1997) recovers moments of latent choice thresholds from discrete choices, providing the starting point for my identification argument. Fox et al. (2012) identify the random-coefficient distribution in mixed logit using local variation in product characteristics. Wang (2023) develops identification and sieve estimation of mixed-logit demand with a nonparametric random-coefficient distribution. Iaria and Wang (2025) establish real analyticity of mixed-logit and mixed-probit demand in mean utility indices for arbitrary mixing distributions, and show how this property extends identification beyond an observed open set. At positive information costs, my identification argument likewise uses analyticity and continuity to recover the choice-probability function beyond an observed interval on which both alternatives have positive choice probability, without requiring full support of the utility shifter.

The remaining discussion provides a broader overview of RI research and related literatures. Revealed-preference tests and experiments examine costly information acquisition in settings where the researcher knows individuals' prior beliefs and the realized payoffs of the alternatives, while individuals are initially uncertain about those payoffs. Using choice probabilities conditional on realized payoffs, Caplin and Dean (2015) give necessary and sufficient conditions for observed choices to be consistent with optimal learning under some information-cost function, and derive bounds on relative information costs. Dean and Neligh (2023) use laboratory experiments to show that participants' choices are broadly consistent with optimal information acquisition under an RI model with no specified functional form for information costs. They also estimate several information-cost specifications, including one based on generalized entropy.

RI provides a theory of how consideration sets, the subsets of available alternatives from which individuals actually choose, form endogenously. Caplin et al. (2019) give necessary and sufficient conditions for optimal choice under Shannon information costs, including conditions determining which alternatives receive zero choice probability. Other consideration-set models study what choices reveal about preferences when individuals overlook some available alternatives. Masatlioglu et al. (2012) assume that individuals choose their preferred considered alternative and that removing an unconsidered alternative leaves the consideration set unchanged. If removing an unchosen alternative changes the choice, the removed alternative must have been considered and ranked below the originally chosen alternative, revealing a preference

comparison. Allowing consideration to be random, Manzini and Mariotti (2014) assume independent consideration with probabilities fixed across choice sets and uniquely identify both the preference ranking and consideration probabilities from choice probabilities observed across choice sets. Cattaneo et al. (2020) allow dependent random consideration, requiring that removing an option outside a consideration set cannot reduce the probability of considering exactly that set. They characterize the preference rankings consistent with observed choices and construct confidence sets for those rankings.

RI research also examines how information acquisition affects market outcomes and welfare. Afrouzi (2024) shows that firms facing fewer competitors pay less attention to aggregate conditions, consistent with survey evidence. In his calibrated model, this reduced attention makes monetary shocks have larger effects on output. With complete markets and Shannon information costs, Angeletos and Sastry (2025) establish that competitive equilibrium is efficient. They also give conditions under which other information costs create welfare gains from simplifying prices or markets.

Finally, empirical and theoretical studies examine how attention and existing information affect responses to macroeconomic news and information treatments. Song and Stern (2025) use firms' annual reports to measure attention to macroeconomic conditions and find that attentive firms' market values rise more after monetary easing and fall less after tightening. Weber et al. (2025) find smaller changes in inflation expectations after information treatments when inflation is high, with evidence that more households and firms already know the information provided. In a dynamic RI model, Maćkowiak and Wiederholt (2026) show that stronger incentives to be informed can reduce information-treatment effects through learning before an experiment, but can increase them through learning during the experiment when individuals acquire no information beforehand.

Notation

I use $a \coloneqq b$ to define a by b, and F_V for the distribution of a random variable or vector V. For a distribution H, the subscripts in $\mathbb{E}_H$, $\mathbb{P}_H$, Var_H, and Cov_H specify the distribution used. In particular, if $V \sim H$, then

$$\mathbb{E}_H[f(V)] \coloneqq \int f(v)\, dH(v), \qquad \text{and} \qquad \mathbb{P}_H\{V \in B\} \coloneqq \int \mathbf{1}\{v \in B\}\, dH(v),$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. In Section 2, F is the joint distribution of the signal T and payoff vector U, and F_T is its marginal distribution for T. Thus $\mathbb{E}_F[U_j \mid T]$ conditions under the joint distribution, whereas $\mathbb{E}_{F_T}$ averages over signals. The operators $\mathbb{E}_G$ and $\mathbb{P}_G$ use the taste-shock distribution G, and $H_1 \otimes H_2$ denotes the product of two distributions. Unless

otherwise indicated, expectations, probabilities, variances, and covariances are evaluated at the true parameter values.

I use the same symbol for a distribution and its cumulative distribution function. In particular, $G(B)$ is the probability of a set B under the taste-shock distribution, while $G(t) \coloneqq \mathbb{P}_G\{\Omega \leq t\}$. For vectors, this inequality is componentwise.

2 The Rational Inattention Model

2.1 The Rational Inattention Problem

For ease of exposition, I begin with an RI discrete-choice model with two alternatives, 0 and 1, and then extend the analysis to multiple alternatives. The individual is initially uncertain about a component of alternative 1's payoff and can acquire information before choosing.

Each market m, such as a job posting, has observed characteristics $Z_m \in \mathbb{R}$ and $X_m \in \mathbb{R}^{d_x}$, where d_x is a fixed positive integer. Write $\mathcal{X} \coloneqq \operatorname{supp}(X_m)$. Let $\lambda : \mathcal{X} \to [0, \infty)$ give the unit cost of information as a function of the information-cost shifter X_m. For individual i in this market, I specify the payoffs and unit cost of information as

$$U_{i0m} = 0, \qquad U_{i1m} = \alpha + \delta Z_m + \Omega_{im}, \qquad \text{and} \qquad \lambda_m \coloneqq \lambda(X_m). \tag{2.1}$$

The utility shifter Z_m enters the known payoff component $\alpha + \delta Z_m$ linearly.

The individual knows the taste-shock distribution G but does not know her own taste shock Ω_{im} before acquiring information. She also knows (Z_m, X_m), the payoff component $\alpha + \delta Z_m$, and the unit cost $\lambda(X_m)$. The taste shocks are independent and identically distributed across individuals i and markets m. I also assume that the taste shock Ω_{im} is independent of the market characteristics (Z_m, X_m):

$$\Omega_{im} \perp\!\!\!\perp (Z_m, X_m).$$

The analyst observes market characteristics and market shares but observes neither the taste shocks nor the information individuals acquire. I write $\theta \coloneqq (\alpha, \delta, \lambda, G)$ for a parameter value.[2]

The individual can learn about her uncertain payoff by acquiring a *signal* T, the information produced by a learning procedure. For example, a worker considering a posting offering four dollars to "Transcribe this MP3 of a 5 minute interview" may be uncertain how demanding accurate transcription would be for her. Personal difficulty, which contributes to Ω_{im}, depends

[2]Additional covariates can be considered by conditioning the binary identification argument in Subsection 2.4 on utility characteristics H_U and cost characteristics H_λ. For the specification $\alpha(H_U) + \delta Z + \Omega$ with cost $\lambda(X, H_\lambda)$, the recovery formulas apply at conditioning values where the stated exclusion, support, and low-cost conditions hold. This gives a partially linear payoff specification with a flexible intercept. A common slope or taste-shock distribution across conditioning values is an additional restriction when imposed.

on how well the recording's pronunciation and vocabulary match her experience. The signal can be an internal assessment formed by reading the requirements, recalling comparable work, and considering how that experience applies to the assignment.

An *information structure* specifies how likely the individual is to receive each signal at each possible payoff, and the individual chooses this structure before receiving the signal (Matějka and McKay, 2015). Fix a market with characteristics (z, x). I suppress the individual and market indices and write $U := (U_0, U_1)$. Let P_U denote the prior distribution of U implied by the known payoff component $\alpha + \delta z$ and the taste-shock distribution G. Following Matějka and McKay (2015), I represent the information structure by a joint distribution $F := F_{T,U}$ of the signal T and payoffs U, with marginal distributions F_T and F_U. The individual chooses F subject to $F_U = P_U$, thereby choosing how the signal depends on the payoffs while leaving their distribution unchanged.[3] In the worker example, F specifies how likely each assessment is at each possible level of personal difficulty. If the distribution of assessments is the same at every payoff, the signal conveys no information.

After receiving the signal $T = t$, the individual chooses the alternative with the highest expected payoff given that signal. She evaluates payoffs under the distribution implied by F after observing t, called her posterior distribution. She therefore maximizes

$$\mathbb{E}_F[U_j \mid T = t]. \tag{2.2}$$

Before receiving the signal, the individual chooses the information structure to trade off expected payoff gains against information costs. Following Matějka and McKay (2015), I take these costs to be proportional to Shannon mutual information. Her problem is

$$\max_{F:\, F_U = P_U} \left\{ \mathbb{E}_{F_T}\left[\max_{j \in \{0,1\}} \mathbb{E}_F[U_j \mid T] \right] - \lambda(x) I_F(U;T) \right\}, \tag{2.3}$$

where

$$I_F(U;T) := \int \log\left(\frac{dF}{d(F_T \otimes P_U)} \right) dF. \tag{2.4}$$

[3] For a fixed distribution of signals recommending which alternative to choose, the individual solves an entropy-regularized optimal transport problem with the payoff distribution fixed. Engh (2026) shows that the RI problem is obtained by also choosing the signal distribution to maximize the value of this transport problem.

The first term in (2.3) is the expected payoff from choosing after receiving the signal; the second is the cost of the information obtained. The mutual information $I_F(U;T)$[4] defined in (2.4) measures the amount of information the signal T conveys about the payoffs U. It equals zero when the signal and payoffs are independent. Multiplying this amount by the unit cost of information $\lambda(x)$ gives the information cost $\lambda(x)I_F(U;T)$.

The unit cost of information $\lambda(x) \geq 0$ measures the payoff the individual gives up for each unit of information acquired. A lower $\lambda(x)$ makes any given amount of information cheaper. When information is clearly organized and easy to interpret, learning which option is better may require very little cognitive effort. At positive cost, the individual may leave some uncertainty unresolved because learning more would not justify its cost. At zero cost, she can make a full-information choice by learning which alternative yields the higher payoff.

2.2 Choice Probabilities of Rationally Inattentive Agents

At positive information cost, choice probabilities can be computed without solving directly for the information structure. I write $b(z,x;\theta)$ for the *ex ante choice probability*: the probability of choosing alternative 1 conditional on market characteristics (z,x), before receiving the signal. For $\lambda(x) > 0$, Matějka and McKay (2015) characterize it as the maximizer of the objective below. A nondegenerate G makes this objective strictly concave in q, so the maximizer is unique:

$$b(z,x;\theta) := \arg\max_{q\in[0,1]} \int \log\left(1-q+qe^{(\alpha+\delta z+\omega)/\lambda(x)}\right) dG(\omega). \tag{2.5}$$

The ex ante probability $b(z,x;\theta)$ averages conditional choice probabilities over the taste shocks. I call $b(z,x;\theta)$ *interior* when $0 < b(z,x;\theta) < 1$, so both alternatives have positive choice probability. At positive cost and an interior $b(z,x;\theta)$, Matějka and McKay (2015) give the probability of choosing alternative 1 conditional on $\Omega = \omega$ as

$$p_\theta(\omega;z,x) := \frac{b(z,x;\theta)e^{(\alpha+\delta z+\omega)/\lambda(x)}}{1-b(z,x;\theta)+b(z,x;\theta)e^{(\alpha+\delta z+\omega)/\lambda(x)}}. \tag{2.6}$$

[4]When payoffs take finitely many possible values, mutual information equals prior entropy minus expected posterior entropy.

$$I_F(U;T) = H(P_U) - \int H(F_{U|T=t})\, dF_T(t).$$

Here, entropy is defined by

$$H(Q) := -\sum_u Q(\{u\}) \log Q(\{u\}),$$

with $0\log 0 := 0$. The first term measures uncertainty about payoffs before receiving the signal. The second measures the uncertainty remaining after receiving the signal, averaged over the possible signals. Their difference measures how much the individual learns. An uninformative signal leaves the payoff distribution unchanged, so mutual information is zero.

Averaging $p_\theta(\omega; z, x)$ over the taste-shock distribution G gives the ex ante choice probability:

$$b(z, x; \theta) = \int p_\theta(\omega; z, x)\, dG(\omega). \tag{2.7}$$

Since the ex ante choice probability also appears in (2.6), equation (2.7) is a fixed-point restriction: the average of the conditional probabilities must equal the ex ante probability used in their formula. For $0 < b(z, x; \theta) < 1$, (2.7) is equivalent to the first-order condition for (2.5).

When information is free, the individual chooses alternative 1 when its payoff exceeds zero:

$$b(z, x; \theta) = \mathbb{P}_G\{\alpha + \delta z + \Omega > 0\}. \tag{2.8}$$

For identification, I use a right-continuous version of the zero-cost choice-probability function.[5]

2.3 Observations

The analyst observes market characteristics and the share of individuals who choose alternative 1 in each of M markets. Market m has a deterministic number $N_{m,M}$ of sampled individuals. I write its characteristics as $W_m := (Z_m, X_m)$ and its observed share as

$$\widehat{s}_m := \frac{1}{N_{m,M}} \sum_{i=1}^{N_{m,M}} Y_{im}, \tag{2.9}$$

where Y_{im} is the indicator for choosing alternative 1. Market characteristics W_m are independent and identically distributed across markets. Conditional on these characteristics, choices are independent across individuals and markets, with

$$Y_{im} \mid W_m \sim \text{Bernoulli}(s_m), \qquad \text{where} \qquad s_m := \mathbb{P}(Y_{im} = 1 \mid W_m).$$

Suppose the observed choices are generated by the model at an unknown parameter value $\theta_0 := (\alpha_0, \delta_0, \lambda_0, G_0)$. The population choice probability is

$$\pi(z, x) := \mathbb{P}(Y = 1 \mid Z = z, X = x) = b(z, x; \theta_0). \tag{2.10}$$

Thus $s_m = \pi(Z_m, X_m)$.

[5] With a nonzero slope and a continuous distribution of Z conditional on X, exact payoff ties have probability zero across markets. Choosing either alternative at a tie therefore gives the same choice probabilities for almost every market.

2.4 Identification

In this subsection, I establish conditions under which observed population choice probabilities identify θ_0 in the RI model with two alternatives. Subsection 2.4.1 extends the identification result to multiple alternatives. I write $\mathcal{Z}_x \coloneqq \text{supp}(Z \mid X = x)$ for the conditional support of Z. Identification requires a mapping from these probabilities to the parameters:

Definition 1 (Identification)**.** The parameter θ_0 is *identified* if there exists a mapping $\mathcal{T}$, known to the analyst, that recovers it from the observed population choice probabilities:

$$\theta_0 = \mathcal{T}\left(\left\{\{\pi(z,x)\}_{z\in\mathcal{Z}_x}\right\}_{x\in\mathcal{X}}\right). \tag{2.11}$$

I begin with normalization and regularity assumptions for the RI model.

Assumption ID1.

(a) The taste shocks are independent of the shifters:

$$\Omega_{im} \perp\!\!\!\perp (Z_m, X_m). \tag{2.12}$$

(b) $\mathbb{E}\Omega = 0$ and $\text{Var}(\Omega) = 1$.

(c) $\delta_0 \neq 0$.

(d) $\lambda_0 : \mathcal{X} \to [0, \infty)$ is continuous.

(e) For every $x \in \mathcal{X}$ with $\lambda_0(x) = 0$, the function $z \mapsto b(z, x; \theta_0)$ is continuous on $\mathcal{Z}_x$.

Assumption ID1(a) means that changing either shifter leaves the taste-shock distribution G_0 unchanged. Assumption ID1(b) fixes payoff location and scale, and ID1(c) requires the utility shifter to affect payoffs. Assumptions ID1(d)–(e), together with ID3(b), make $(z, x) \mapsto b(z, x; \theta_0)$ jointly continuous on the observed support. The distribution of observed choices and shifters determines conditional choice probabilities only up to changes on sets of covariate values with probability zero. Joint continuity makes their values uniquely determined at every support point, as required by Definition 1.

I next turn to the variation in information costs used for identification.

Assumption ID2 (Arbitrarily low information costs).

$$\inf_{x \in \mathcal{X}} \lambda_0(x) = 0. \tag{2.13}$$

Assumption ID2 means that some values of the cost shifter make learning nearly effortless.[6] With finitely many cost categories, the assumption requires one category to have zero cost. With a continuous shifter, costs can approach zero without reaching it.

For the utility shifter, I state conditions on its distribution and observed support conditional on the information-cost shifter.

Assumption ID3 (Support of the utility shifter).

(a) For each x, $Z \mid X = x$ has a continuous distribution with interval support $\mathcal{Z}_x$.

(b) Define

$$\mathcal{X}_0 := \{x \in \mathcal{X} : \lambda_0(x) = 0\}.$$

For every $x \in \mathcal{X}_0$,

$$\mathbb{P}\{-\Omega \in \alpha_0 + \delta_0 \operatorname{int}(\mathcal{Z}_x)\} = 1.$$

For every $x \notin \mathcal{X}_0$, there exists

$$z_x^* \in \operatorname{int}(\mathcal{Z}_x), \qquad \text{and} \qquad 0 < \pi(z_x^*, x) < 1.$$

Assumption ID3(b) requires an observed utility-shifter value at which both alternatives have positive choice probability when information is costly. For example, there must be an observed wage at which some workers take the task and others do not. When information is free, observed variation in known payoffs must cover the thresholds $-\Omega$ for choosing alternative 1. In the wage example, the observed wage range must cover workers' reservation wages. If this range has a lowest or highest value, no positive fraction of workers can have a reservation wage equal to that value.

[6] Both Brown and Jeon (2024, equation (15)) and Joo (2023, equation (16)) use exponential information-cost specifications. For a specification $\lambda_0(x) = \exp(x'\beta_0)$ with coefficient vector β_0, ID2 holds if $\operatorname{supp}(X'\beta_0) = \mathbb{R}$, since the index can then become arbitrarily negative.

The identification argument requires the choice-probability function $z \mapsto b(z, x; \theta_0)$ to be identified on $\mathbb{R}$ for each x. At positive information cost, analyticity allows probabilities on an observed interval to determine values outside that interval. Together with ID1–ID2, the support conditions in ID3 give the following result.

Theorem 1 (Identification)**.** *Under Assumptions ID1–ID3, the true parameter*

$$\theta_0 = (\alpha_0, \delta_0, \lambda_0, G_0)$$

is identified.

I briefly outline the proof; Section B.2 gives the complete argument. Throughout this explanation, assume without loss of generality that $\delta_0 > 0$. I identify the parameters by interpreting $z \mapsto b(z, x; \theta_0)$ as the CDF of a latent threshold for each fixed x. The starting point is the connection with Lewbel (1997): consider the full-information binary choice rule

$$Y = \mathbf{1}\{\alpha_0 + \delta_0 Z \geq -\Omega\}, \qquad \text{where } \Omega \perp\!\!\!\perp Z \quad \text{and} \quad \delta_0 > 0.$$

Define the threshold

$$S := -\frac{\alpha_0 + \Omega}{\delta_0}, \qquad \text{so} \qquad \alpha_0 + \delta_0 S = -\Omega.$$

Rearranging the payoff comparison gives $Y = \mathbf{1}\{S \leq Z\}$. Since S depends only on Ω, it is independent of Z, and hence

$$\mathbb{P}(Y = 1 \mid Z = z) = \mathbb{P}(S \leq z).$$

Thus the choice probability as a function of z is the CDF of S. When Z is a wage, S is a reservation wage: alternative 1 is chosen when the wage reaches this threshold. The mean-zero and unit-variance normalizations for Ω in ID1(b) give moments that recover the payoff coefficients:

$$\mathbb{E}S = -\frac{\alpha_0}{\delta_0}, \qquad \text{and} \qquad \operatorname{Var}(S) = \frac{1}{\delta_0^2}.$$

I construct a similar threshold in the RI model from the conditional choice probability $p_{\theta_0}(\omega; z, x)$ in (2.6). Write $\pi(z, x) := b(z, x; \theta_0)$ for the choice probability at every $z \in \mathbb{R}$,

including values outside the observed support. Let $\ell(p) := \log\{p/(1-p)\}$ and let ε be standard logistic, independent of Ω, with CDF $\sigma(t) := (1+e^{-t})^{-1}$. At positive cost and an interior probability, rewriting (2.6) gives

$$\begin{aligned} p_{\theta_0}(\omega; z, x) &= \sigma\left(\ell(\pi(z,x)) + \frac{\alpha_0 + \delta_0 z + \omega}{\lambda_0(x)}\right) \\ &= \mathbb{P}\left\{\varepsilon \le \ell(\pi(z,x)) + \frac{\alpha_0 + \delta_0 z + \omega}{\lambda_0(x)}\right\}. \end{aligned}$$

Averaging over $\Omega \sim G_0$, independently of ε, therefore gives

$$\pi(z,x) = \mathbb{P}\{h_x(z) \ge -\Omega + \lambda_0(x)\varepsilon\}, \tag{2.14}$$

where

$$h_x(z) := \alpha_0 + \delta_0 z + \lambda_0(x)\ell(\pi(z,x)). \tag{2.15}$$

For x with $\lambda_0(x) > 0$, define

$$I_x := \{z \in \mathbb{R} : 0 < \pi(z,x) < 1\}.$$

The RI model implies that I_x is a nonempty open interval and that $h_x : I_x \to \mathbb{R}$ is strictly increasing and onto. Its inverse therefore converts the threshold in (2.14) into units of the utility shifter.

Definition 2 (Switch point)**.** For each $x \in \mathcal{X}$, define the *switch point* S_x by

$$S_x := \begin{cases} h_x^{-1}(-\Omega + \lambda_0(x)\varepsilon), & \lambda_0(x) > 0, \\ -\dfrac{\alpha_0 + \Omega}{\delta_0}, & \lambda_0(x) = 0, \end{cases}$$

where $\Omega \sim G_0$ and ε is standard logistic and independent of Ω.

At positive cost, the switch point S_x has CDF $\pi(\cdot, x)$ because h_x is strictly increasing: for $z \in I_x$,

$$\mathbb{P}(S_x \le z) = \mathbb{P}\{-\Omega + \lambda_0(x)\varepsilon \le h_x(z)\} = \pi(z,x).$$

The same CDF equality holds for every real value of the utility shifter, at both positive and zero cost:[7]

$$\mathbb{P}(S_x \le z) = \pi(z,x), \qquad z \in \mathbb{R}. \tag{2.16}$$

[7] At positive cost, $S_x \in I_x$ almost surely. To the left of I_x, both $\mathbb{P}(S_x \le z)$ and $\pi(z,x)$ equal zero; to its right, both equal one. Continuity gives the same equalities at any finite endpoints. At zero cost, the equality follows from $S_x = -(\alpha_0 + \Omega)/\delta_0$, using the right-continuous choice-probability function described in Subsection 2.2.

The switch-point distribution provides moments that identify the payoff coefficients and information costs. These moments depend on the CDF $z \mapsto \pi(z, x)$ on $\mathbb{R}$, which ID3 determines from the observed probabilities by the support and analyticity argument above. At positive information cost, applying h_x to the switch point gives

$$-\Omega + \lambda_0(x)\varepsilon = \alpha_0 + \delta_0 S_x + \lambda_0(x)\ell(\pi(S_x, x)). \tag{2.17}$$

I write $\mu(x) \coloneqq \mathbb{E}S_x$ and $V(x) \coloneqq \mathrm{Var}(S_x)$. I also define the following covariance $C(x)$, which is positive for a nondegenerate switch-point distribution:

$$C(x) \coloneqq \mathrm{Cov}\big(S_x, \ell(\pi(S_x, x))\big). \tag{2.18}$$

At positive cost, $\pi(S_x, x)$ is uniform on $(0, 1)$, so $\ell(\pi(S_x, x))$ has the same mean and variance as the standard logistic variable ε. Taking means and variances therefore cancels the two logistic variance terms and gives

$$\mu(x) = -\frac{\alpha_0}{\delta_0}, \qquad \text{and} \qquad \delta_0^2 V(x) + 2\delta_0 \lambda_0(x) C(x) = 1. \tag{2.19}$$

At zero cost, $S_x = -(\alpha_0 + \Omega)/\delta_0$ gives the same restrictions. Equation (2.19) explains what information costs do to switch-point dispersion: at positive cost, the positive covariance term makes the variance smaller than the full-information value $1/\delta_0^2$. That value is attained exactly when information is free, since δ_0 and $C(x)$ are positive.

The supremum of the switch-point variances $V(x)$ over $x \in \mathcal{X}$ identifies δ_0. As information cost approaches zero, switch-point variance approaches its full-information value; Cauchy–Schwarz bounds the difference from that value by a constant times $\lambda_0(x)$. Under ID2, these variances can therefore be arbitrarily close to the full-information value. The common mean then identifies the intercept:

$$\delta_0 = \left(\sup_{x \in \mathcal{X}} V(x)\right)^{-1/2}, \tag{2.20}$$

$$\text{and} \qquad \alpha_0 = -\delta_0 \mu(x). \tag{2.21}$$

Once (α_0, δ_0) are known, solving the same moment identity gives $\lambda_0(x)$ at each x:

$$\lambda_0(x) = \frac{1 - \delta_0^2 V(x)}{2\delta_0 C(x)}. \tag{2.22}$$

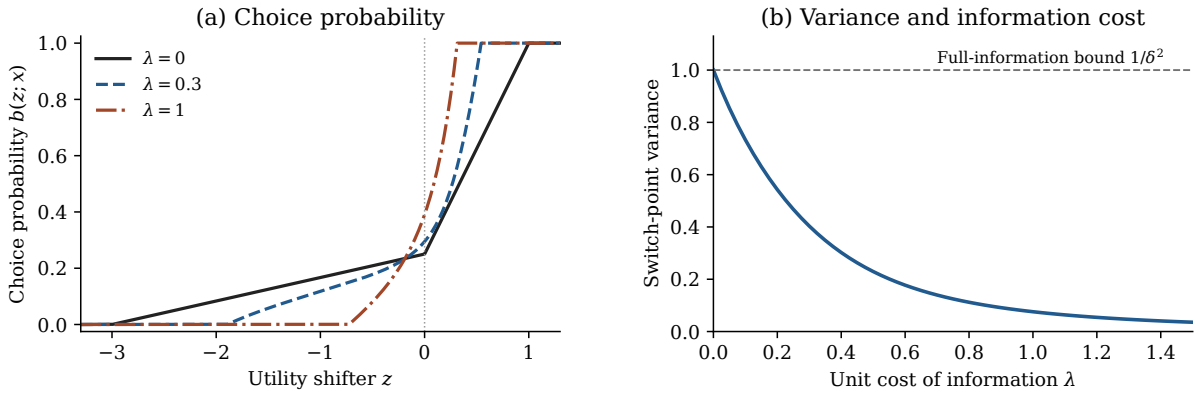


Figure 1: Choice probabilities and switch-point variances. The illustration uses $\alpha = 0$, $\delta = 1$, and an asymmetric taste density equal to $3/4$ on $[-1, 0]$ and $1/12$ on $[0, 3]$. This distribution has mean zero and variance one. The switch points have the same mean, but their variances fall as information costs rise. I compute the choice probabilities by deterministic integration of the RI model.

Deconvolution identifies G_0 if information costs are positive at some x. The identified coefficients (α_0, δ_0) and cost function λ_0 make h_x known, so the identified distribution of S_x determines that of $h_x(S_x)$. At such an x, (2.17) expresses $h_x(S_x)$ as the sum of $-\Omega$ and an independent logistic variable with known scale $\lambda_0(x)$. Writing $\varphi_V(t) \coloneqq \mathbb{E}e^{itV}$, independence gives

$$\varphi_{-\Omega}(t) = \frac{\varphi_{h_x(S_x)}(t)}{\varphi_\varepsilon(\lambda_0(x)t)}. \tag{2.23}$$

The denominator is known and has no real zeros, so the ratio identifies the characteristic function, and hence the distribution, of $-\Omega$. Changing its sign gives the distribution of Ω, namely G_0. If costs are zero everywhere, G_0 is directly the distribution of $-\alpha_0 - \delta_0 S_x$. These calculations recover θ_0 from the choice probabilities. The estimator in Section 3 uses the RI optimality equation directly, without first estimating switch-point moments or extending the observed choice-probability functions beyond the observed support.

2.4.1 Extension: Multiple Alternatives

With multiple alternatives, identification must recover both the marginal taste-shock distributions and their dependence. I consider alternatives $0, 1, \ldots, J$, with payoffs

$$U_0 = 0, \qquad \text{and} \qquad U_j = \alpha_j + \delta_j z_j + \Omega_j, \qquad j = 1, \ldots, J. \tag{2.24}$$

The payoff of alternative 0 is normalized to zero, and $z \coloneqq (z_1, \ldots, z_J)$ is the vector of utility shifters. The unit cost of information $\lambda(x)$ is common across the alternatives. The individual

learns about $\Omega := (\Omega_1, \ldots, \Omega_J)$, whose joint distribution G is independent of the shifters. In this extension, I omit subscript 0 on the true parameter and write $\theta := (\alpha, \delta, \lambda, G)$, where $\alpha := (\alpha_1, \ldots, \alpha_J)'$ and $\delta := (\delta_1, \ldots, \delta_J)'$. Dependence among taste shocks allows a worker who finds one kind of task attractive to find another kind attractive as well. I first recover the probability of choosing each alternative j when only j and alternative 0 are available. I then use the probability of choosing alternative 0 when all alternatives are available. Appendix A states the formal conditions and theorem; Section B.5 gives the proof.

To fix the common payoff scale, I normalize $\mathbb{E}_G \Omega_j = 0$ for every j and $\mathrm{Var}_G(\Omega_1) = 1$. Only the variance of the first taste shock is normalized; the other marginal variances will be recovered. Changing the sign of each utility shifter when necessary allows me to assume $\delta_j > 0$ without loss of generality. I let $b(z, x; \theta)$ collect the $J+1$ probabilities of choosing each alternative. At positive information cost, this vector maximizes the following objective:

$$b(z, x; \theta) := \arg\max_{q \in \Delta_J} \lambda(x) \mathbb{E}_G \log\left(q_0 + \sum_{j=1}^{J} q_j e^{U_j/\lambda(x)} \right), \tag{2.25}$$

where $\Delta_J := \{q : q_j \geq 0, \ \sum_{j=0}^{J} q_j = 1\}$. Assumption A2 rules out an almost-sure linear dependence among $1, e^{\Omega_1/\lambda(x)}, \ldots, e^{\Omega_J/\lambda(x)}$ and ensures uniqueness of the choice-probability vector.

At positive cost, when all alternatives except 0 and j have zero choice probability, the objective in (2.25) is the binary RI objective:

$$\max_{p \in [0,1]} \lambda(x) \mathbb{E}_G \log\left(1 - p + p e^{U_j/\lambda(x)}\right).$$

The taste-shock distribution in this binary problem is the marginal distribution of Ω_j. Write $p_j(z_j; x)$ for the probability of choosing alternative j in this binary problem. Let $S_x^{(j)}$ be the switch point with CDF $p_j(\cdot; x)$. Its moments are

$$\begin{aligned} \mu_j(x) &:= \mathbb{E} S_x^{(j)}, \qquad V_j(x) := \mathrm{Var}(S_x^{(j)}), \\ C_j(x) &:= \mathrm{Cov}\left(S_x^{(j)}, \ell(p_j(S_x^{(j)}; x))\right). \end{aligned}$$

The functions $p_j(\cdot; x)$ can be recovered from choice probabilities when all alternatives are available. At positive information cost and fixed x, the full-consideration region is the set of utility-shifter vectors z at which every alternative has positive choice probability. Analyticity determines the probabilities throughout this connected region from their values on an observed open subset. For each $u \in \mathbb{R}$ with $0 < p_j(u; x) < 1$, I construct covariate vectors z_n with $z_{n,j} = u$ for every n, such that every alternative has positive choice probability along the sequence, while the probabilities of alternatives other than 0 and j approach zero. The limiting probability of choosing alternative j solves the binary RI maximization problem and therefore

equals $p_j(u;x)$. Continuity and monotonicity then determine $p_j(\cdot;x)$ also where it equals zero or one. Section B.5 proves this result using the moment conditions in Appendix A. At zero cost, full conditional support identifies the full-information choice probabilities for every $z \in \mathbb{R}^J$.

The functions $p_1(\cdot;x)$, $x \in \mathcal{X}$, identify α_1, δ_1, and the common cost function λ by the formulas for two alternatives, since $\mathrm{Var}_G(\Omega_1) = 1$. For alternatives $j = 2,\dots,J$, the marginal taste variances have not been normalized. I write $\sigma_j^2 := \mathrm{Var}_G(\Omega_j)$ and use the corresponding moment identity

$$\delta_j^2 V_j(x) + 2\delta_j \lambda(x) C_j(x) = \sigma_j^2, \qquad \text{and} \qquad \mu_j(x) = -\frac{\alpha_j}{\delta_j}. \tag{2.26}$$

The right-hand side is an unknown marginal variance, but the cost function λ on the left-hand side has already been recovered from the functions $p_1(\cdot;x)$, $x \in \mathcal{X}$.

Subtracting (2.26) at two values of x cancels the unknown variance σ_j^2, because it is the same in both equations. Suppose $\lambda(x) > 0$ at some $x \in \mathcal{X}$. At two values x, x' with $V_j(x) \neq V_j(x')$, the difference identifies δ_j because $\delta_j > 0$ and λ is already known:

$$\delta_j = \frac{2\{\lambda(x')C_j(x') - \lambda(x)C_j(x)\}}{V_j(x) - V_j(x')}. \tag{2.27}$$

The mean then gives the intercept,

$$\alpha_j = -\delta_j \mu_j(x),$$

and substituting the coefficients into either variance equation gives σ_j^2. For a nondegenerate marginal taste-shock distribution, $V_j(x)$ approaches its full-information value as information costs approach zero. At positive cost it is smaller, so there are two values x, x' with different switch-point variances.

To identify dependence among the taste shocks, I use the probability of choosing alternative 0 when all alternatives are available. At positive information cost, I use an observed region where every alternative has positive choice probability. Write $b^j(z,x)$ for the population choice probability of alternative j, the jth component of $b(z,x;\theta)$. Let $\eta_0,\dots,\eta_J$ be independent standard Gumbel variables, independent of Ω. Conditional on Ω, the logit probability of choosing alternative j equals the probability that j maximizes

$$U_j + \lambda(x)\log b^j(z,x) + \lambda(x)\eta_j, \qquad j = 0,\dots,J.$$

Alternative 0 is selected only when its value exceeds that of every other alternative. Its choice probability therefore involves all taste-shock components jointly. I define $\zeta_j := \eta_j - \eta_0$ and the known thresholds

$$c_j(z,x) := -\alpha_j - \delta_j z_j + \lambda(x) \log \frac{b^0(z,x)}{b^j(z,x)}. \tag{2.28}$$

The probability of these simultaneous inequalities is

$$b^0(z,x) = \mathbb{P}\{\Omega_j + \lambda(x)\zeta_j \leq c_j(z,x) \text{ for every } j\}. \tag{2.29}$$

Equation (2.29) expresses the probability of choosing alternative 0 as the joint CDF of $H_x := \Omega + \lambda(x)\zeta$, evaluated at the thresholds determined by the identified coefficients and choice probabilities.

The distribution of H_x is identified from its CDF on an open set of thresholds. The observed full-consideration region gives an open set of thresholds $(c_1(z,x), \ldots, c_J(z,x))$.[8] At positive cost, adding the independent vector $\lambda(x)\zeta$ to Ω produces a random vector with a real-analytic density. The CDF in (2.29) determines this density on the open set, and real analyticity determines it on $\mathbb{R}^J$.

Deconvolution then recovers the joint taste-shock distribution G from the distribution of H_x. Independence of $\eta_0, \ldots, \eta_J$ and the standard Gumbel characteristic function (Janson, 2010, Section 3) give the characteristic function of ζ:

$$\varphi_\zeta(t) = \Gamma\left(1 + i\sum_{j=1}^{J} t_j\right) \prod_{j=1}^{J} \Gamma(1 - it_j), \qquad t \in \mathbb{R}^J.$$

Here Γ is the gamma function, and $\varphi_\zeta(t)$ is nonzero for every $t \in \mathbb{R}^J$.[9] Independence of Ω and ζ therefore gives

$$\varphi_G(t) = \frac{\varphi_{H_x}(t)}{\varphi_\zeta(\lambda(x)t)}. \tag{2.30}$$

This ratio identifies the characteristic function, and hence the joint distribution, of the taste-shock vector.[10]

If $\lambda(x) = 0$ for every $x \in \mathcal{X}$, full conditional support identifies the payoff coefficients and joint distribution G. The binary formulas identify α_1, δ_1, and the zero cost function. For $j = 2, \ldots, J$, consider $z_1 = t$ and $z_j = rt + u$, with $r > 0$, $u \in \mathbb{R}$, and the other shifters fixed. As $t \to \infty$, the ratio $r = \delta_1/\delta_j$ is the only value for which alternatives 1 and j can both have

[8] Section B.5 shows that the map $z \mapsto (c_1(z,x), \ldots, c_J(z,x))$ has a nonsingular Jacobian on this region, so its image is open.

[9] The gamma function has no zeros, and the arguments in this formula have real part one, away from its poles (NIST, 2026, Section 5.2(i)).

[10] The components of ζ are dependent because each subtracts the same Gumbel variable η_0. The deconvolution uses their known joint distribution.

positive limiting choice probabilities for some u. The known value of δ_1 fixes the remaining slopes, and the binary switch-point means identify the intercepts. The probability of choosing alternative 0 directly identifies the joint taste-shock distribution because

$$b^0(z,x) = G(-\alpha_1 - \delta_1 z_1, \ldots, -\alpha_J - \delta_J z_J), \qquad z \in \mathbb{R}^J.$$

3 Estimation

In this section, I construct a sieve minimum-distance estimator using market shares and establish its consistency. For ease of exposition, I first present the estimator and consistency argument for two alternatives. Subsection 3.2.1 extends the estimation method to multiple alternatives.

3.1 The Estimator

I construct a minimum-distance estimator by measuring how closely the observed market shares satisfy the RI equation (2.7). I write $w := (z,x)$ and $s(w) := \pi(z,x)$, and let $\sigma(u) := (1+e^{-u})^{-1}$ be the logistic CDF. At positive cost and an input probability $q \in (0,1)$, Ψ averages the conditional choice probabilities computed from θ and q. I also define it at probabilities zero and one and at zero cost:

$$\Psi(\theta; w, q) := \begin{cases} \displaystyle\int \sigma\left(\ell(q) + \frac{\alpha + \delta z + \omega}{\lambda(x)}\right) dG(\omega), & \lambda(x) > 0, \quad 0 < q < 1, \\ q, & \lambda(x) > 0, \quad q \in \{0,1\}, \\ \mathbb{P}_G\{\alpha + \delta z + \Omega > 0\}, & \lambda(x) = 0. \end{cases} \tag{3.1}$$

The RI residual is the difference between $\Psi(\theta; w, q)$ and the input probability q:

$$r(\theta; w, q) := \Psi(\theta; w, q) - q. \tag{3.2}$$

At the true parameter and population choice probability,

$$r(\theta_0; W_m, s_m) = 0. \tag{3.3}$$

I use finite-dimensional approximations to estimate λ and G. Let Θ be the set of parameters satisfying the model restrictions and $\Theta_M \subseteq \Theta$ a finite-dimensional subset, called a sieve. In the data, I replace the population choice probability s_m by the observed market share $\widehat{s}_m$ and use the average squared residual:

$$\widehat{Q}_M(\theta) := \frac{1}{M}\sum_{m=1}^{M} r(\theta; W_m, \widehat{s}_m)^2. \tag{3.4}$$

A nonnegative basis expansion can approximate the continuous function λ, which gives the unit cost of information, while probability masses on an increasingly rich grid or a histogram with increasingly many bins can approximate the taste-shock distribution. The probability weights sum to one and satisfy the payoff normalizations. I define the estimator $\widehat{\theta}_M \in \Theta_M$ as an approximate minimizer of the sample criterion, allowing an optimization error ϵ_M:

$$\widehat{Q}_M(\widehat{\theta}_M) \le \inf_{\theta\in\Theta_M} \widehat{Q}_M(\theta) + \epsilon_M, \qquad \text{where} \qquad \epsilon_M = o_p(1). \tag{3.5}$$

3.2 Consistency

The consistency analysis begins with the following conditions on sample sizes.

Assumption E1. The market sizes satisfy

$$M \to \infty \qquad \text{and} \qquad N_M := \min_{m\le M} N_{m,M} \to \infty.$$

Assumption E1 requires both the number of markets and the number of observations within each market to increase.

The following assumption collects the restrictions on the parameter space Θ.

Assumption E2 (Parameter space and regularity).

(a) Θ contains θ_0 and is compact under

$$\begin{aligned} d(\theta, \widetilde{\theta}) := {} & |\alpha - \widetilde{\alpha}| + |\delta - \widetilde{\delta}| \\ & + \|\lambda - \widetilde{\lambda}\|_\infty + d_L(G, \widetilde{G}), \end{aligned} \tag{3.6}$$

where d_L is the Lévy metric for weak convergence.

(b) For every $\theta = (\alpha, \delta, \lambda, G) \in \Theta$,

$$\int \omega \, dG(\omega) = 0, \qquad \int \omega^2 \, dG(\omega) = 1, \qquad \text{and} \qquad \delta \neq 0.$$

The function $\lambda : \mathcal{X} \to [0, \infty)$, which gives the unit cost of information, is continuous and satisfies

$$\inf_{x \in \mathcal{X}} \lambda(x) = 0.$$

(c) For every $\theta \in \Theta$ and every $x \in \mathcal{X}$ with $\lambda(x) = 0$, either $\mathcal{Z}_x = \mathbb{R}$, or the function $z \mapsto b(z, x; \theta)$ is real analytic on $\mathbb{R}$.

Under Assumption E2(a), convergence in the metric d means convergence of (α, δ), uniform convergence of λ, and weak convergence of G. Assumption E2(b) imposes the payoff normalization, nonzero-slope, and information-cost restrictions in ID1(b)–(d) and ID2 on every parameter value in Θ. At zero information cost, E2(c) requires either full conditional support of Z, or a choice-probability function that is real analytic on $\mathbb{R}$. Together with ID1–ID3, E2(b)–(c) ensure that the population criterion is uniquely minimized at θ_0.

The remaining condition concerns approximation of θ_0 by parameter values in Θ_M.

Assumption E3 (Sieve approximation). A deterministic sequence $\{\theta_M^*\}_{M=1}^{\infty}$ satisfies

$$\theta_M^* \in \Theta_M \subseteq \Theta \qquad \text{and} \qquad d(\theta_M^*, \theta_0) \to 0 \quad (M \to \infty).$$

Assumption E3 means that the finite-dimensional approximations in Θ_M can approach θ_0 as the sample grows.

Theorem 2 (Consistency)**.** *Let $\widehat{\theta}_M \in \Theta_M$ satisfy (3.5). Under ID1–ID3 and E1–E3,*

$$d(\widehat{\theta}_M, \theta_0) \to_p 0 \qquad \text{as } M \to \infty.$$

Briefly, the proof first shows that replacing population choice probabilities with market shares has a vanishing effect on the criterion. It then uses identification and compactness of Θ to establish convergence to θ_0; Section B.3 gives the full argument.

3.2.1 Extension: Multiple Alternatives

With multiple alternatives, I minimize the sum of squared residuals for alternatives 1 through J. For the $J+1$ alternatives in Section 2.4.1, let $\mathbf{q} := (q_1, \ldots, q_J)'$ collect these input probabilities and set $q_0 := 1 - \sum_{j=1}^{J} q_j$. The probability of choosing alternative 0 is determined by the others, so only J residual components are needed. Write $v_j(\omega; z) := \alpha_j + \delta_j z_j + \omega_j$ and $v_0(\omega; z) := 0$. At positive cost, the averaged conditional choice probability is

$$\Psi_j(\theta; w, \mathbf{q}) := \int \frac{q_j e^{v_j(\omega;z)/\lambda(x)}}{q_0 + \sum_{k=1}^{J} q_k e^{v_k(\omega;z)/\lambda(x)}} \, dG(\omega), \qquad j = 1, \ldots, J. \tag{3.7}$$

At zero cost, I use the full-information probabilities that each alternative has the highest payoff. When information is free at the true parameter, Assumption A6(b) rules out payoff ties. I define the column vector

$$\mathbf{r}(\theta; w, \mathbf{q}) := \left(\Psi_j(\theta; w, \mathbf{q}) - q_j\right)_{j=1}^{J}.$$

Let $\widehat{s}_{m,j}$ be the observed share choosing alternative j in market m, and set $\widehat{\mathbf{s}}_m := (\widehat{s}_{m,1}, \ldots, \widehat{s}_{m,J})'$. Summing the squared residuals gives

$$\widehat{Q}_M(\theta) := \frac{1}{M} \sum_{m=1}^{M} \|\mathbf{r}(\theta; W_m, \widehat{\mathbf{s}}_m)\|_2^2. \tag{3.8}$$

The sieve approximates the common unit cost of information as a function of the cost shifter and the joint taste-shock distribution. Positive bounded market weights can be included as in (B.3.4).

For fixed J, the consistency proof controls each residual component separately and then sums the bounds. The proof also requires continuity and identification conditions for the model with multiple alternatives; Section B.3 outlines these requirements.

4 Inference

In this section, I construct confidence regions for the model parameters and confidence intervals for functions of these parameters using test inversion. For ease of exposition, I first present the procedure for two alternatives. Subsection 4.2 establishes the asymptotic validity of the intervals, and Subsection 4.2.1 extends the inference procedure to multiple alternatives.

4.1 Confidence Intervals

Information costs and sieve probability weights can equal zero, placing estimates on the boundary of the parameter space. A normal approximation derived for interior parameters need not apply in these cases. I instead test $H_0 : \theta_0 = \theta$ for $\theta \in \Theta_M$ using the RI residuals and collect the parameter values that are not rejected. This gives a confidence region without requiring asymptotic normality of the estimator.

At the true parameter θ_0, $r(\theta_0; W_m, s_m) = 0$. The observed share $\widehat{s}_m$ need not equal the population probability s_m, so $r(\theta_0; W_m, \widehat{s}_m)$ need not be zero. For exposition, I take $N_{m,M} = N_M$ in every market and write

$$r_m(\theta, q) := r(\theta; W_m, q), \qquad \text{and} \qquad A_m(\theta, q) := \partial_q r_m(\theta, q).$$

At θ_0 and $0 < s_m < 1$, a Taylor expansion of the residual gives

$$r_m(\theta_0, \widehat{s}_m) = A_m(\theta_0, s_m)(\widehat{s}_m - s_m) + O_p((\widehat{s}_m - s_m)^2). \tag{4.1}$$

Conditional on W_m, the first-order term in (4.1) has variance

$$\frac{A_m(\theta_0, s_m)^2 s_m(1-s_m)}{N_M}.$$

For a fixed market with $0 < s_m < 1$ and $A_m(\theta_0, s_m) \neq 0$, the binomial central limit theorem therefore gives

$$\frac{N_M r_m(\theta_0, \widehat{s}_m)^2}{A_m(\theta_0, s_m)^2 s_m(1-s_m)} \Rightarrow \chi^2_1 \qquad (N_M \to \infty).$$

I sum the squared residuals divided by their estimated variances:

$$T_M(\theta) := \sum_{m \in \widehat{\mathcal{A}}_M(\theta)} \frac{N_M r_m(\theta, \widehat{s}_m)^2}{A_m(\theta, \widehat{s}_m)^2 \widehat{s}_m(1-\widehat{s}_m)}, \tag{4.2}$$

where, for fixed $\eta \in (0, 1/2)$ and $a_* > 0$,

$$\widehat{\mathcal{A}}_M(\theta) := \{m : \eta \leq \widehat{s}_m \leq 1-\eta,\ |A_m(\theta, \widehat{s}_m)| \geq a_*\}. \tag{4.3}$$

For rejection probability $\gamma \in (0,1)$, I use the critical value $\chi^2_{1-\gamma, |\widehat{\mathcal{A}}_M(\theta)|}$, the $(1-\gamma)$ quantile of the chi-square distribution with $|\widehat{\mathcal{A}}_M(\theta)|$ degrees of freedom. Restricting the sum to $\widehat{\mathcal{A}}_M(\theta)$ excludes markets with observed shares near zero or one or residual derivatives close to zero. Without trimming, small denominators can magnify individual terms in $T_M(\theta)$ and invalidate the chi-square approximation as M and N_M grow together.

I invert this test by collecting the sieve parameters that are not rejected:

$$\mathcal{C}^s_{M,\gamma} := \left\{\theta \in \Theta_M : T_M(\theta) \leq \chi^2_{1-\gamma, |\widehat{\mathcal{A}}_M(\theta)|}\right\}. \tag{4.4}$$

If $\widehat{\mathcal{A}}_M(\theta) = \varnothing$, then $T_M(\theta) = 0$, so $\theta \in \mathcal{C}^s_{M,\gamma}$. I obtain intervals for parameters and counterfactuals by minimizing and maximizing a scalar function $\phi : \Theta \to \mathbb{R}$ over the confidence region:

$$L_M := \inf_{\theta \in \mathcal{C}^s_{M,\gamma}} \phi(\theta), \qquad \text{and} \qquad U_M := \sup_{\theta \in \mathcal{C}^s_{M,\gamma}} \phi(\theta). \tag{4.5}$$

For example, $\phi(\theta)$ can equal a payoff coefficient such as δ, the information cost $\lambda(x)$ at a fixed x, or a counterfactual change in task-taking probability.

4.2 Asymptotic Validity of the Confidence Intervals

The following assumptions give asymptotic coverage for the intervals in (4.5). I use the sampling model in Section 2.3, with a common within-market sample size N_M and the fixed thresholds (η, a_*) in (4.3).

For a market with $0 < s_m < 1$ and $\lambda_0(X_m) > 0$, the derivative of the RI residual at the true parameter satisfies

$$A_m(\theta_0, s_m) = -\frac{\mathrm{Var}\{p_{\theta_0}(\Omega; W_m)\}}{s_m(1-s_m)}. \tag{4.6}$$

At zero cost, the residual subtracts its input probability from a full-information probability, so the derivative is -1.

For the inference result, I first specify how M and N_M grow together.

Assumption F1 (Joint growth). Each market has N_M choices, with

$$M, N_M \to \infty \qquad \text{and} \qquad M/\sqrt{N_M} \to 0.$$

F1 controls errors in the Taylor approximation and estimated variances as the number of markets increases. Together with F3, it makes trimming based on estimated shares agree with trimming based on population probabilities with probability approaching one.

Write $s(w) := \pi(z, x)$. For w with $0 < s(w) < 1$, let

$$A_0(w) := \partial_q r(\theta_0; w, s(w)).$$

Here $A_0(w)$ is the derivative of the residual with respect to the input probability q, evaluated at θ_0 and $q = s(w)$. I next state a condition on the proportion of markets retained by the test.

Assumption F2 (Positive probability of satisfying the trimming conditions).

$$\mathbb{P}\{\eta < s(W) < 1-\eta,\ |A_0(W)| > a_*\} > 0.$$

Assumption F2 ensures that a positive proportion of the population of markets satisfies both trimming conditions strictly at θ_0.

Let $\theta_M^* = (\alpha_M^*, \delta_M^*, \lambda_M^*, G_M^*) \in \Theta_M$ be a deterministic approximation to θ_0, with $d(\theta_M^*, \theta_0) \to 0$, as in E3. For $0 < s(w) < 1$, define

$$A_M(w) := \partial_q r(\theta_M^*; w, s(w))$$

and the set of covariate values retained when the test uses population probabilities:

$$\mathcal{R}_M := \{w : \eta \le s(w) \le 1-\eta,\ |A_M(w)| \ge a_*\}. \tag{4.7}$$

I next consider how share-estimation errors can change which residuals are used to calculate the test statistic. For $0 < s(w) < 1$, I define the distance to the nearest trimming cutoff:

$$\Delta_M(w) := \min\left\{|s(w)-\eta|,\ |s(w)-(1-\eta)|, \left||A_M(w)|-a_*\right|\right\}. \tag{4.8}$$

Set $\Delta_M(w) := \eta$ when $s(w) \in \{0,1\}$. When $s(w) = 0$ or $s(w) = 1$, the observed share equals $s(w)$ almost surely, so sampling error cannot change whether the market is retained.

The following assumption gives a probability bound for $\Delta_M(W)$.

Assumption F3 (Little probability near the trimming cutoffs). There exist $C < \infty$, $t_0 > 0$ and an integer M_0 such that

$$\mathbb{P}\{\Delta_M(W) \le t\} \le Ct \quad \text{for every } M \ge M_0 \text{ and } 0 < t \le t_0.$$

F3 limits the probability that a small error in an estimated market share changes whether that market's residual is used to calculate the test statistic.

Replacing θ_0 by its sieve approximation θ_M^* can produce a nonzero residual even when population choice probabilities are used. I therefore impose the following assumption.

Assumption F4 (Negligible residual bias). Define

$$\rho_M := \sup_{w \in \mathcal{R}_M} |r(\theta_M^*; w, s(w))|. \tag{4.9}$$

Require $M^{1/4}\sqrt{N_M}\,\rho_M \to 0$.

F4 is an undersmoothing condition. It makes the sieve approximation bias negligible for the normal approximation to the centered and scaled test statistic.

Recall that $\phi : \Theta \to \mathbb{R}$ is the scalar function of interest and $[L_M, U_M]$ is the interval defined in (4.5). The test is inverted over the sieve Θ_M, while the reported interval is intended to cover the true value $\phi(\theta_0)$. I therefore impose the following assumption.

Assumption F5 (Negligible approximation error in the reported function). Let $\phi_M^* := \phi(\theta_M^*)$ and $E_M := \{\theta_M^* \in \mathcal{C}_{M,\gamma}^s\}$. For some deterministic $\varepsilon_M \downarrow 0$,

$$|\phi_M^* - \phi(\theta_0)| = o(\varepsilon_M),$$

$$\text{and} \qquad \mathbb{P}\big\{[\phi_M^* - \varepsilon_M, \phi_M^* + \varepsilon_M] \subseteq [L_M, U_M] \mid E_M\big\} \to 1.$$

Here E_M is the event that the test does not reject θ_M^*. F1–F4 establish coverage for the sieve approximation $\phi(\theta_M^*)$. F5 ensures that, when θ_M^* is accepted, the difference between $\phi(\theta_M^*)$ and $\phi(\theta_0)$ is small enough for the same interval to contain $\phi(\theta_0)$ as well.

Theorem 3 (Coverage for functions of the parameters)**.** *Under the stated sampling assumptions, E3, ID1(c), ID3(a), and F1–F5, with compact Θ and continuous $\phi : \Theta \to \mathbb{R}$,*

$$\liminf_{M\to\infty} \mathbb{P}\{\phi(\theta_0) \in [L_M, U_M]\} \geq 1 - \gamma.$$

Section B.4 gives the proof. Theorem 3 requires no asymptotic normality of the sieve estimator, so parameter constraints may bind.

4.2.1 Extension: Multiple Alternatives

With multiple alternatives, share-estimation errors are correlated, so I use a covariance matrix to standardize the residuals. Choosing one alternative excludes choosing another. I use the J shares and residuals for alternatives 1 through J from Section 3.2.1, and let $\mathbf{s}_m$ denote the vector of their population choice probabilities. For conditionally independent multinomial choices, the covariance of $\widehat{\mathbf{s}}_m$ conditional on W_m is

$$\frac{\Sigma_m}{N_M}, \qquad \text{where} \qquad \Sigma_m := \operatorname{diag}(\mathbf{s}_m) - \mathbf{s}_m \mathbf{s}_m'.$$

Let $B_m(\theta, \mathbf{q}) := \partial_{\mathbf{q}'} \mathbf{r}(\theta; W_m, \mathbf{q})$ be the Jacobian of the residual vector with respect to $\mathbf{q}$. At the true parameter, the leading residual covariance is

$$\frac{1}{N_M} B_m(\theta_0, \mathbf{s}_m) \Sigma_m B_m(\theta_0, \mathbf{s}_m)'.$$

An estimate of N_M times this covariance is

$$\widehat{V}_m(\theta) := B_m(\theta, \widehat{\mathbf{s}}_m) \widehat{\Sigma}_m B_m(\theta, \widehat{\mathbf{s}}_m)',$$
$$\text{where} \qquad \widehat{\Sigma}_m := \operatorname{diag}(\widehat{\mathbf{s}}_m) - \widehat{\mathbf{s}}_m \widehat{\mathbf{s}}_m'.$$

The trimming rule now excludes markets with small choice probabilities or a nearly singular residual covariance matrix. The shares of every alternative, including alternative 0, must be bounded away from zero, and the smallest eigenvalue of $\widehat{V}_m(\theta)$ must be bounded away from zero. For retained markets, the squared standardized residual in the binary statistic is replaced by

$$N_M \, \mathbf{r}(\theta; W_m, \widehat{\mathbf{s}}_m)' \widehat{V}_m(\theta)^{-1} \mathbf{r}(\theta; W_m, \widehat{\mathbf{s}}_m).$$

Extending the proof requires the Jacobian $B_m(\theta, \mathbf{q})$ in place of the scalar derivative $A_m(\theta, q)$ and the covariance matrix above to standardize the residual vector. The counterparts of F1–F4 must control sampling and sieve-approximation errors, retain a positive fraction of markets, and limit how often errors in estimated shares change whether a market's residual is used to calculate the statistic. These conditions permit the Taylor-expansion and trimming comparisons used in the binary proof. The test then uses a chi-square critical value with J degrees of freedom per retained market. Test inversion gives a confidence region, and minimizing and maximizing ϕ over this region gives its interval. As in F5, when the test does not reject the sieve approximation θ_M^*, the interval must contain $\phi(\theta_0)$ as well as $\phi(\theta_M^*)$, with probability approaching one.

5 Empirical Application: Assessing Tasks in an Online Labour Market

5.1 Data and Specification

I use workers' decisions on Amazon Mechanical Turk, an online market for short assignments, to examine wage responses under different information costs. A worker sees a posting's reward and description but may still be uncertain how demanding the assignment would be for her. I relate the estimated unit cost of information to how concretely the description specifies the work.

The posting data collected by Dube et al. (2020) record rewards, requester identifiers, titles, descriptions, and the number of assignments still available at different times. I retain postings with a positive reward, an identified requester, at least two recorded counts of available assignments within the first 30 minutes, and at least ten assignments available at the first observation. To compare similar work at different rewards, I then select groups from the same requester with matching task descriptions that contain at least four postings, at least three distinct rewards, and a highest reward at least twice the lowest. These restrictions yield 174 postings. Requiring a recorded count of available assignments at the common three-hour endpoint leaves 137

postings in 35 groups, observed between June 1 and August 20, 2017. The repeated tasks provide within-group reward variation, and the common endpoint gives a comparable observation window for task taking.

Workers choose whether to take a task after viewing its posting. To estimate the model parameters, I need to estimate the probability that a worker takes the task. I use each posting's task-taking share, or market share, to estimate its task-taking probability. I add up each decrease in the number of available assignments during the three-hour window, ignoring increases. I divide this total by the calibrated number of workers who viewed the posting.[11]

I classify each posting's description by whether it specifies what workers must do and how much work is required. *Task specificity* requires a concrete action and its target, such as finding an agent's email address. *Workload specificity* requires a stated duration, a numerical amount of work, or an effort limit, such as inspecting at most three search results. A broad label such as "survey" need not specify a concrete task, and "quick" alone does not specify a numerical workload. Titles and descriptions are read together, with identical text assigned the same code. Combining the two criteria determines the description category x_m in Table 1. The two middle categories need not have a predetermined cost ranking: one specifies the activity, while the other specifies the workload.

Table 1: Task description categories

x	Category	Postings	Illustrative description
0	Both specific	29	Find the agent's email address; inspect up to three search results.
1	Task only	83	Find the agent's email address.
2	Workload only	9	Complete a two-minute study.
3	Neither specific	16	Complete a survey.

Notes: Examples illustrate the coding criteria. Task specificity requires a concrete action and target; workload specificity requires a numerical duration, quantity, or effort limit. The sample contains 44 distinct posting texts. Category values are labels: they impose neither equal distances nor a ranking of the two middle categories.

[11] I assume a common rate of ten worker arrivals per initial assignment over 90 minutes. For the three-hour window, the calibrated worker count is therefore twenty times the posting's initial number of assignments. Chilton et al. (2010) document that workers sort postings by the number of available assignments and concentrate on the first few pages of results. Larger batches can therefore attract more visits by appearing earlier in these results.

I estimate how the posted reward affects the probability that a worker takes a task. Worker i decides whether to take the task in posting m, with the payoff from not taking it normalized to zero. The payoff from taking it and the unit cost of information are

$$v_{im} = a_{c(m)} + \delta z_m + \Omega_{im}, \qquad \text{and} \qquad \lambda_m := \lambda(x_m). \tag{5.1}$$

The index $c(m)$ groups postings from the same requester with matching task descriptions. The intercept $a_{c(m)}$ allows average payoffs to differ across these groups, while Ω_{im} represents uncertain worker-specific suitability. I estimate the reward coefficient from reward variation within groups, defining the reward index as

$$z_m := \log r_m - \overline{\log r}_{c(m)},$$

where r_m is the posted reward and the group average is computed from the estimation sample. Within each task group, I treat the posted reward as independent of Ω_{im}.

Descriptions that specify both the activity and the workload can make it easier for workers to assess personal difficulty. I estimate a separate information cost for each description category, without imposing a ranking or setting any category's cost to zero. I approximate the taste-shock distribution by a 41-bin histogram with mean zero and variance one. I jointly estimate the bin probabilities, the 35 task-group intercepts, the reward coefficient, and four information costs by minimum distance.

5.2 Estimated Information Costs

The estimated unit cost of information is lowest when both the task and workload are specific (Figure 2). It is approximately 0.0033 when both are specific, 0.0164 when only the task is specific, 0.0079 when only the workload is specific, and 0.0134 when neither criterion is met.

Table 2 reports nominal 95% intervals computed by simulation-based test inversion for the posting counts. The last column reports the corresponding intervals for a larger simulated sample generated at the estimated parameters.

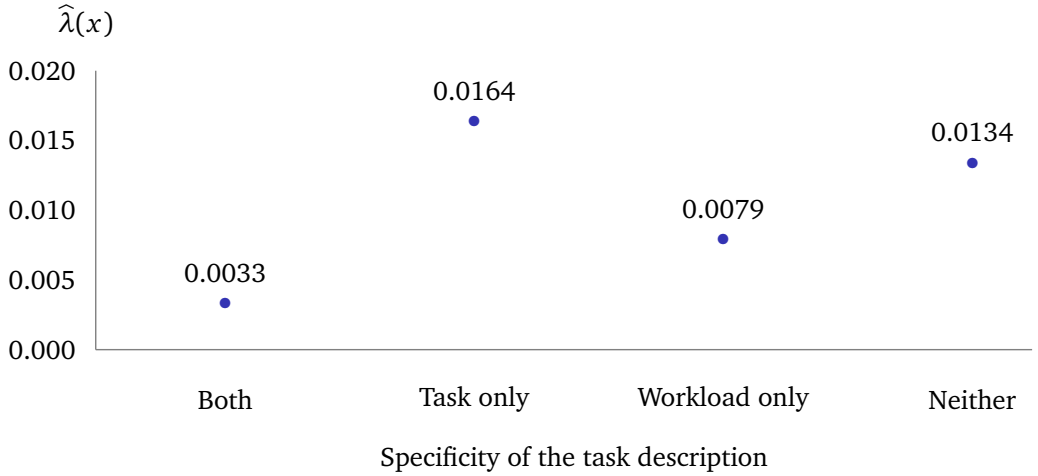


Figure 2: Estimated unit costs of information by task-description category.

Table 2: Information costs and uncertainty

Quantity	Estimate	Actual data	Simulated sample
Both specific	0.0033	$[0, 8]$	$[0, 0.0555]$
Task only	0.0164	$[0, 0.5953]$	$[0.0123, 0.3017]$
Workload only	0.0079	$[0, 2.1085]$	$[0.0031, 0.1107]$
Neither specific	0.0134	$[0, 8]$	$[0.0043, 0.2540]$
Wage coefficient δ	0.0017	$[10^{-6}, 1.2694]$	$[0.0015, 0.0421]$

Notes: The first four rows report unit costs of information; the last reports the wage coefficient. Brackets report nominal 95% confidence intervals. "Actual data" uses the 137 observed postings. The simulated-sample intervals use a larger simulated sample, treating the estimated model parameters as true.

5.3 Wage Responses under Different Information Costs

I compare wage responses under estimated, halved, and zero information costs, holding the payoff coefficients and taste-shock distribution fixed. For all 137 postings, I double the posted reward and compute choice probabilities at both the original and doubled reward under each cost specification. I let workers choose their information acquisition separately at each reward. I report the percentage-point change in the average task-taking probability and its log-arc elasticity.[12]

[12]Let $\bar{b}_1^\kappa$ and $\bar{b}_2^\kappa$ be the equally weighted average task-taking probabilities at the original and doubled reward, with information costs $\kappa\widehat{\lambda}$ for $\kappa \in \{1, 1/2, 0\}$. The percentage-point change is $100(\bar{b}_2^\kappa - \bar{b}_1^\kappa)$, and the log-arc elasticity is

$$\mathcal{E}^\kappa := \frac{\log(\bar{b}_2^\kappa / \bar{b}_1^\kappa)}{\log 2}. \tag{5.2}$$

Table 3: Wage doubling for all 137 postings

Information regime	Before (%)	After (%)	Change (pp)	Elasticity
Estimated costs	2.62	3.20	0.577	0.287
Half costs	8.90	9.32	0.426	0.068
Full information	12.87	13.21	0.335	0.037

Notes: Before and After are average task-taking probabilities at the original and doubled rewards. Change is measured in percentage points (pp). The same postings and estimated payoffs enter every regime. Rewards double for all postings; the within-group mean log rewards calculated from the original sample remain fixed. Probabilities are averaged with equal posting weights. Elasticity is defined by (5.2).

With the estimated payoff distribution, lower information costs reduce wage elasticity and raise the task-taking probability at the original reward. In Table 3, the elasticity falls from 0.287 at estimated costs to 0.068 at half costs and 0.037 under full information. Doubling rewards raises the average task-taking probability by about 22.0%, 4.8%, and 2.6% in these three cases. At the original reward, the average task-taking probability rises from 2.62% to 8.90% and 12.87% as costs fall.

Learning at a lower unit cost can help workers recognize that a task suits them and take it at the original reward, leaving fewer of these workers to change their decision when the wage increases. Under full information and the estimated payoff distribution, relatively few additional workers find the task worthwhile when the reward doubles. The absolute increase in the average task-taking probability falls from 0.577 to 0.335 percentage points, and the higher starting probability further reduces the percentage response. Learning can also reveal that an assignment is unsuitable and lead a worker to decline a task she would otherwise take. The net effects on task-taking probabilities and elasticities therefore depend on the payoff distribution and rewards.

Table 4 reports nominal 95% confidence intervals computed by the same numerical test-inversion procedure. The full-information effect compares task-taking probabilities at zero and estimated costs at the original reward; the elasticity difference is the estimated-cost elasticity minus the full-information elasticity. The last column reports intervals for the larger simulated sample.

5.4 Wage Responses under a Cardell Taste-Shock Distribution

I compare the wage responses predicted by the flexible specification with those predicted under a Cardell taste-shock distribution, a family used by Brown and Jeon (2024). I normalize the Cardell distribution to mean zero and variance one and estimate its shape jointly with the other parameters. The postings, utility controls, and cost categories are the same as in the flexible specification.

Table 4: Counterfactual estimates and uncertainty

Quantity	Estimate	Actual data	Simulated sample
Wage effect (pp)	0.577	[0.000, 18.675]	[0.527, 1.299]
Full-information effect (pp)	10.253	[−4.390, 14.945]	[3.970, 14.532]
Estimated-cost elasticity	0.287	[0.000, 2.353]	[0.072, 0.305]
Half-cost elasticity	0.068	[0.000, 2.196]	[0.025, 0.196]
Full-information elasticity	0.037	[0.000, 1.931]	[0.010, 0.131]
Elasticity difference	0.250	[−0.504, 1.627]	[0.000, 0.277]

Notes: Brackets report nominal 95% confidence intervals. "Actual data" uses the 137 observed postings. The simulated-sample intervals use a larger simulated sample, treating the estimated model parameters as true. "pp" denotes percentage points. The wage effect is the change in average task-taking probability when rewards double at estimated information costs. The full-information effect is the change in average task-taking probability from estimated costs to zero costs at the original rewards. Elasticities measure the log change in average task-taking probability divided by log 2 when rewards double. The elasticity difference is the estimated-cost elasticity minus the full-information elasticity.

The Cardell estimates assign zero information cost to every category. They predict a 14.5% increase in the average task-taking probability when rewards double, with log-arc elasticity approximately 0.195. Since the estimated costs are already zero, halving or removing them leaves that prediction unchanged. The flexible specification produces distinct responses under estimated, halved, and zero information costs in Figure 3.

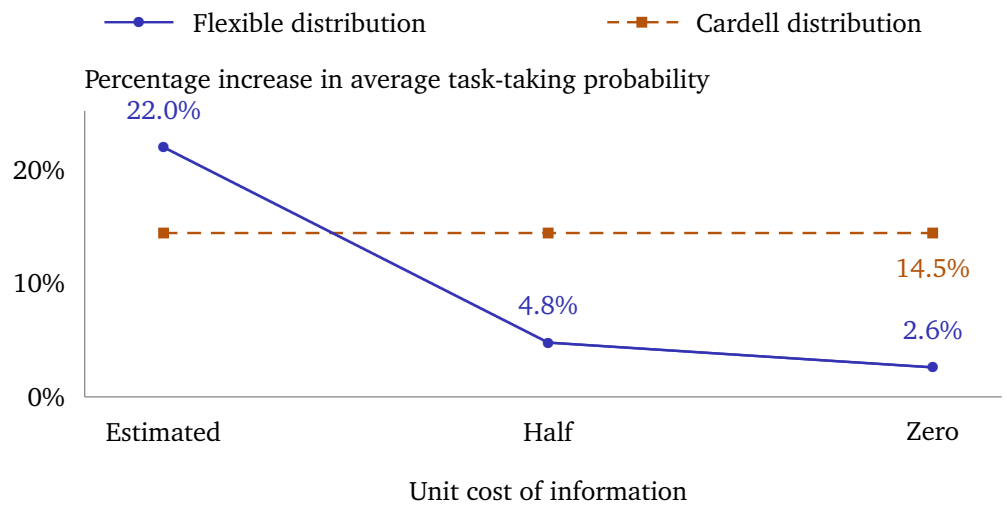


Figure 3: Percentage response to wage doubling under flexible and Cardell taste-shock distributions. Each model is estimated on the same sample. The Cardell estimates assign zero cost to all four categories, so its three information regimes coincide.

The flexible specification also attains a smaller minimum-distance criterion than the Cardell specification. The values are 7.711×10^{-5} and 2.550×10^{-4}, respectively, a reduction of approximately 70%. This difference is not a formal test of the Cardell family.

6 Conclusion

I establish semiparametric identification of the taste-shock distribution and the unit cost of information as a function of observed characteristics in an RI discrete choice model. For binary choice, I interpret the probability of choosing alternative 1, as a function of the utility shifter, as the CDF of a switch point. The switch point's mean, its variance when information is free, and the covariance in (2.18) identify the payoff coefficients and information costs. Deconvolution then recovers the taste-shock distribution. The argument uses optimal learning under Shannon information costs without imposing parametric forms on the taste-shock distribution or on how the unit cost of information varies with observed characteristics.

I estimate the model from market shares using the RI optimality equation. Test inversion gives confidence regions that allow zero information costs and zero probability weights in the taste-shock distribution sieve. The resulting intervals for functions of the parameters have coverage under the sampling, trimming, and approximation conditions stated in the paper.

In the Mechanical Turk application, the estimated unit cost of information varies across description categories. With a flexible taste-shock distribution, lower unit costs of information raise the average task-taking probability at the original reward while reducing wage responsiveness. The Cardell specification assigns zero costs and gives different counterfactual predictions. These estimates illustrate why the taste-shock distribution matters for empirical RI analysis.

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A Identification for Choices among Multiple Alternatives

This appendix states the model, assumptions, and identification theorem for the extension in Section 2.4.1. The proof is in Section B.5.

The alternatives are $\{0, 1, \ldots, J\}$, with $J \geq 1$, and

$$v_j(\Omega; z) \coloneqq a_j(z) + \Omega_j,$$
$$\text{where} \qquad a_j(z) \coloneqq \alpha_j + \delta_j z_j \quad (j = 1, \ldots, J), \qquad \text{and} \qquad a_0(z) = \Omega_0 = 0.$$

Write $\alpha \coloneqq (\alpha_1, \ldots, \alpha_J)' \in \mathbb{R}^J$, $\delta \coloneqq (\delta_1, \ldots, \delta_J)' \in \mathbb{R}^J$, and $a(z) \coloneqq (a_1(z), \ldots, a_J(z))' \in \mathbb{R}^J$. The joint taste-shock distribution G has finite second moments, $\mathbb{E}_G \Omega = 0$, and $\mathrm{Var}_G(\Omega_1) = 1$. I assume $\delta_j \neq 0$ for every j. I orient each shifter so that $\delta_j > 0$. The unit cost of information is a continuous function $\lambda : \mathcal{X} \to [0, \infty)$ of the cost shifter x. Write $\mathcal{Z}_x \coloneqq \mathrm{supp}(Z \mid X = x)$ and $\mathcal{X}_+ \coloneqq \{x \in \mathcal{X} : \lambda(x) > 0\}$.

At positive cost, $b(z; x) \coloneqq (b^0(z; x), \ldots, b^J(z; x))$ solves

$$\max_{q \in \Delta_J} \Phi(q; a(z), x),$$
$$\text{where} \qquad \Phi(q; a, x) \coloneqq \lambda(x) \mathbb{E}_G \log \left\{ q_0 + \sum_{j=1}^{J} q_j e^{(a_j + \Omega_j)/\lambda(x)} \right\}, \tag{A.1}$$

where $\Delta_J \coloneqq \{q \in [0,1]^{J+1} : \sum_{j=0}^{J} q_j = 1\}$. At zero cost, individuals select the alternative with the highest payoff. The consideration set is

$$\mathcal{A}(z; x) \coloneqq \{j \in \{0, \ldots, J\} : b^j(z; x) > 0\}.$$

The observed population choice probabilities are collected in

$$\mathcal{B}_{\text{obs}} \coloneqq \left\{ \left\{ (b^0(z; x), \ldots, b^J(z; x)) \right\}_{z \in \mathcal{Z}_x} \right\}_{x \in \mathcal{X}}.$$

For the extension to multiple alternatives, I first state the assumptions on taste shocks and individuals' prior beliefs.

Assumption A1 (Independence from shifters and a common prior). $\Omega \perp\!\!\!\perp (Z, X)$, $\Omega \sim G$, and individuals use G as their common prior.

Individuals know the joint distribution G, but initially do not know their own taste-shock vector Ω. Changing either shifter leaves this distribution unchanged.

I next state a condition on the taste-shock distribution G.

Assumption A2 (Exponential non-collinearity). For every $x \in \mathcal{X}_+$, if a vector $u = (u_0, \ldots, u_J) \in \mathbb{R}^{J+1}$ satisfies

$$u_0 + \sum_{j=1}^{J} u_j e^{\Omega_j/\lambda(x)} = 0 \quad G\text{-almost surely,}$$

then $u = 0$.

Assumption A2 makes the optimal probability vector $b(z;x)$ unique at positive information cost. For example, a joint taste density that is positive on a nonempty open set satisfies A2, including when the taste shocks are correlated.

Identification of the payoff coefficients (α, δ) also uses the following condition on observed shifter values.

Assumption A3 (Positive choice probabilities in the observed support).

$$\text{For every } x \in \mathcal{X}_+, \quad \exists z_x^\circ \in \operatorname{int}(\mathcal{Z}_x): \quad b^j(z_x^\circ; x) > 0 \quad (j = 0, \ldots, J).$$

For fixed $x \in \mathcal{X}_+$, continuity implies that every alternative also has positive choice probability at shifter vectors sufficiently close to z_x°. Some alternatives may have zero probability at other observed shifter values.

The condition on variation in information costs carries over from the binary model.

Assumption A4 (Arbitrarily low information costs).

$$\inf_{x \in \mathcal{X}} \lambda(x) = 0.$$

I also use the following moment conditions on G.

Assumption A5 (Exponential moments). For every $t > 0$,

$$\mathbb{E}_G e^{t\Omega_k} < \infty \qquad \text{and} \qquad \mathbb{E}_G e^{t(\Omega_k - \Omega_j)} < \infty \quad (j \neq k,\ j, k \in \{1, \ldots, J\}).$$

Any bounded taste-shock distribution satisfies these moment conditions.

At zero information cost, I state conditions on the observed support $\mathcal{Z}_x$ and the pairwise taste differences $\Omega_j - \Omega_k$.

Assumption A6 (Choice at zero information cost).

(a) For every $x \in \mathcal{X}$ with $\lambda(x) = 0$, $\mathcal{Z}_x = \mathbb{R}^J$.

(b) If the unit cost of information satisfies $\lambda(x) = 0$ at some cost-shifter value $x \in \mathcal{X}$, every pairwise taste difference has a continuous distribution:

$$\mathbb{P}_G(\Omega_j - \Omega_k = t) = 0 \quad \text{for every } t \in \mathbb{R} \text{ and } 0 \le j \ne k \le J.$$

At zero information cost, Assumption A6(a) gives full joint support $\mathcal{Z}_x = \mathbb{R}^J$, and A6(b) rules out payoff ties with positive probability at each fixed z. The choice probabilities therefore do not depend on how ties are broken.

The identification argument differs according to whether information has a positive cost at any x. If some x has $\lambda(x) > 0$, I recover the payoff coefficients using the choice-probability functions when only alternatives 0 and j are available, together with variation in information costs. I then recover G from the probability of choosing alternative 0 when all alternatives are available. If $\lambda(x) = 0$ for every x, full conditional support identifies the payoff coefficients and G from full-information choices. The following theorem covers both cases; Section B.5 gives its proof.

Theorem 4 (Identification with multiple alternatives)**.** *Suppose A1–A6 hold,* $\delta_j \ne 0$ *for* $j = 1, \ldots, J$*,* $\mathbb{E}_G\|\Omega\|^2 < \infty$*,* $\mathbb{E}_G\Omega = 0$*, and* $\mathrm{Var}_G(\Omega_1) = 1$*. The observed family* $\mathcal{B}_{\text{obs}}$ *identifies*

$$\theta := (\alpha, \delta, \lambda, G).$$

B Proofs

B.1 Auxiliary Results

I first state and prove lemmas for the binary RI model used in the proofs of Theorems 1–3. The first two establish properties of choice-probability functions and switch-point moments for identification. The next four concern residual continuity, replacing population choice probabilities with sample shares, binomial sampling moments, and a conditional central limit theorem. I restate the binary choice problem and residual so that these lemmas can be read independently of the main text.

I write $w := (z, x)$, $\theta := (\alpha, \delta, \lambda, G)$, and

$$v(\omega; z) := \alpha + \delta z + \omega, \qquad \sigma(u) := (1 + e^{-u})^{-1}, \qquad \ell(q) := \log\{q/(1-q)\} \quad (0 < q < 1).$$

For a fixed value x of the cost shifter with $\lambda(x) > 0$, the population choice probability $b(z, x; \theta)$ maximizes

$$q \longmapsto \lambda(x) \int \log\{1 - q + q e^{v(\omega;z)/\lambda(x)}\}\, dG(\omega), \qquad q \in [0, 1]. \tag{B.1.1}$$

At zero cost, I use the full-information comparison and its right-continuous CDF version when defining switch points. When discussing market shares, $s(w) := b(z, x; \theta_0)$ denotes the true population choice probability. The residual used below is $r(\theta; w, q) := \Psi(\theta; w, q) - q$, where

$$\Psi(\theta; w, q) := \begin{cases} \displaystyle\int \sigma\left(\ell(q) + \frac{v(\omega; z)}{\lambda(x)}\right) dG(\omega), & \lambda(x) > 0,\ 0 < q < 1, \\ q, & \lambda(x) > 0,\ q \in \{0, 1\}, \\ \mathbb{P}_G\{v(\Omega; z) > 0\}, & \lambda(x) = 0. \end{cases}$$

The uses of dominated convergence and interchange of integrals below follow Billingsley (1995, Theorems 16.4 and 18.3).

Lemma B.1.1 (Binary RI choice probabilities)**.** *Fix x, write $\lambda_x := \lambda(x) > 0$, and suppose G is nondegenerate with $\mathbb{E}_G|\Omega| < \infty$ and $\delta > 0$. The maximizer $b(z, x; \theta)$ in (B.1.1) is unique and continuous in z. Define its set of utility-shifter values giving interior probabilities by*

$$\mathcal{I}_{\theta,x} := \{z \in \mathbb{R} : 0 < b(z, x; \theta) < 1\}.$$

The set $\mathcal{I}_{\theta,x}$ is a nonempty open interval. The function $z \mapsto b(z,x;\theta)$ is real analytic and strictly increasing on $\mathcal{I}_{\theta,x}$, with range $(0,1)$. It equals zero to the left of this interval and one to the right. Moreover,

$$\begin{aligned} h_{\theta,x} : \mathcal{I}_{\theta,x} &\longrightarrow \mathbb{R}, \\ z &\longmapsto \alpha + \delta z + \lambda_x \ell(b(z,x;\theta)) \end{aligned}$$

is an increasing bijection. If two choice-probability functions at positive information costs $z \mapsto b(z,x;\theta)$ and $z \mapsto b(z,x;\widetilde{\theta})$ agree on a nonempty open interval where both take values in $(0,1)$, then $b(z,x;\theta) = b(z,x;\widetilde{\theta})$ for every $z \in \mathbb{R}$.

Proof. For this proof only, I suppress the fixed x and write $\lambda := \lambda_x$, $b(z) := b(z,x;\theta)$, and $h(z) := h_{\theta,x}(z)$.

I first describe the solution with $0 < b(z) < 1$ as the inverse of an increasing function. This will also determine where $b(z)$ equals zero or one, and why observations on an interval with $0 < b(z) < 1$ determine $b(z)$ for every $z \in \mathbb{R}$.

Write $\bar{a} := \alpha + \delta z$ and let $L := -\Omega + \lambda\varepsilon$, where ε is standard logistic and independent of Ω. With $\sigma(u) := (1+e^{-u})^{-1}$,

$$F_L(h) := \mathbb{P}(L \le h) = \int \sigma\left(\frac{h+\omega}{\lambda}\right) dG(\omega).$$

The integrand is strictly increasing in h, lies in $(0,1)$, and tends to zero and one at the two ends of $\mathbb{R}$. Bounded convergence gives the same limits for F_L. Differentiation under the integral gives

$$F_L'(h) = \frac{1}{\lambda} \int \sigma\left(\frac{h+\omega}{\lambda}\right)\left\{1 - \sigma\left(\frac{h+\omega}{\lambda}\right)\right\} dG(\omega) > 0.$$

The derivative is bounded by $1/(4\lambda)$. Thus F_L is continuous and strictly increasing from zero to one, and its inverse $F_L^{-1}(q)$ is defined for every $q \in (0,1)$.

For real analyticity, apply Lemma B.1.7 with $J = 1$ and mixing variable $-\Omega$. The difference of two independent standard Gumbel variables is standard logistic (Train, 2002, Section 3.1), so that lemma applies directly to $L := -\Omega + \lambda\varepsilon$. In particular, F_L is real analytic. This smoothing property does not require G to have a density.

I next relate each interior probability to the known payoff index that makes it optimal.

For $q \in (0,1)$ and $\omega \in \mathbb{R}$, define

$$p_q(\omega) := \sigma\left(\frac{F_L^{-1}(q) + \omega}{\lambda}\right),$$

and define the payoff-index map

$$a : (0,1) \longrightarrow \mathbb{R}, \qquad q \longmapsto a(q) := F_L^{-1}(q) - \lambda \ell(q).$$

By construction $\mathbb{E}_G p_q(\Omega) = q$. Hence

$$\begin{aligned}\mathbb{E}_G[p_q(\Omega)(1-p_q(\Omega))] &= \mathbb{E}_G p_q(\Omega) - \mathbb{E}_G[p_q(\Omega)^2] \\ &= q - q^2 - \mathrm{Var}_G(p_q(\Omega)) = q(1-q) - \mathrm{Var}_G(p_q(\Omega)).\end{aligned}$$

Since $\omega \mapsto p_q(\omega)$ is strictly increasing and G is nondegenerate, $\mathrm{Var}_G(p_q(\Omega)) > 0$. Consequently

$$0 < F_L'(F_L^{-1}(q)) = \frac{q(1-q) - \mathrm{Var}_G(p_q(\Omega))}{\lambda} < \frac{q(1-q)}{\lambda}.$$

Differentiating $a(q)$ now yields

$$a'(q) = \frac{1}{F_L'(F_L^{-1}(q))} - \frac{\lambda}{q(1-q)} > 0. \tag{B.1.2}$$

The real-analytic inverse-function theorem (Krantz and Parks, 2002, Theorem 1.5.3) applies because $F_L'(h) > 0$ for every $h \in \mathbb{R}$. Together with closure under composition, it makes $a(q)$ analytic. Equation (B.1.2) makes it strictly increasing. Its image $I_a := a((0,1))$ is a nonempty open interval, which may be bounded or unbounded on either side.

The function $a(q)$ also determines when the maximizing probability equals zero or one.

Let the choice-probability objective, in payoff units, be

$$J_{\bar{a}}(q) := \lambda \int \log\{1 - q + q e^{(\bar{a}+\omega)/\lambda}\}\, dG(\omega), \qquad 0 \le q \le 1.$$

The bound

$$\left|\lambda \log\{1 - q + q e^{(\bar{a}+\omega)/\lambda}\}\right| \le |\bar{a}| + |\omega|$$

is integrable because $\mathbb{E}_G|\Omega| < \infty$. Lemma B.1.8, with $J = 1$, therefore gives finiteness and continuity of $q \mapsto J_{\bar{a}}(q)$ on $[0,1]$. Nondegeneracy of G also gives A2 in this binary case: a relation $u_0 + u_1 e^{\Omega/\lambda} = 0$ with $u_1 \neq 0$ would force Ω to be constant, while $u_1 = 0$ requires $u_0 = 0$. Lemma B.1.9 then gives strict concavity and a unique optimal choice probability.

For $q \in (0,1)$, choose $\eta > 0$ so that $q \in [\eta, 1-\eta]$. With $R(\omega) := e^{(\bar{a}+\omega)/\lambda} > 0$,

$$\left|\lambda \frac{R(\omega) - 1}{1 - q + qR(\omega)}\right| \le \frac{\lambda}{\eta}.$$

This bound justifies differentiation under the integral on any such interval of input probabilities. The logit identity then gives

$$\begin{aligned} J'_{\bar{a}}(q) &= \lambda \int \frac{e^{(\bar{a}+\omega)/\lambda} - 1}{1-q+qe^{(\bar{a}+\omega)/\lambda}}\, dG(\omega) \\ &= \frac{\lambda}{q(1-q)}\{F_L(\bar{a}+\lambda\ell(q)) - q\}. \end{aligned}$$

Because $F_L(a(q)+\lambda\ell(q)) = q$ and F_L is strictly increasing,

$$\operatorname{sign} J'_{\bar{a}}(q) = \operatorname{sign}\{\bar{a} - a(q)\}.$$

If $\bar{a} \in I_a$, there is exactly one q with $\bar{a} = a(q)$. The derivative is positive below that probability and negative above it, so the unique optimum is $q = a^{-1}(\bar{a})$. If $\bar{a}$ is at or below a finite left endpoint of I_a, $J'_{\bar{a}}(q) < 0$ for every $q \in (0,1)$ and the optimum is zero. If $\bar{a}$ is at or above a finite right endpoint, the optimum is one.

The probabilities $a^{-1}(\bar{a})$ approach zero and one at finite endpoints of I_a. To prove convergence to zero at a finite left endpoint $L := \inf I_a$, take any sequence $\bar{a}_n \in I_a$ with $\bar{a}_n \to L$ and put $q_n := a^{-1}(\bar{a}_n)$. If q_n did not tend to zero, some subsequence would satisfy $q_n \geq \varepsilon$ for a fixed $\varepsilon \in (0,1)$. Monotonicity of a would give $\bar{a}_n = a(q_n) \geq a(\varepsilon) > L$ along that subsequence, contradicting $\bar{a}_n \to L$. Thus $q_n \to 0$. At a finite right endpoint $R := \sup I_a$, the same reasoning uses $q_n \leq 1-\varepsilon$ and $\bar{a}_n \leq a(1-\varepsilon) < R$ to show $q_n \to 1$.

Thus the interior utility-shifter interval is

$$I := \{z : \alpha + \delta z \in I_a\},$$

and on this interval

$$b(z) = a^{-1}(\alpha + \delta z) \quad (z \in I).$$

The inverse-function theorem and (B.1.2) make b analytic and strictly increasing on I. As q ranges over $(0,1)$, the corresponding z ranges over I, so $b(I) = (0,1)$. The limits just proved show that b remains continuous where it equals zero or one. Finally, on I,

$$h(z) = F_L^{-1}(b(z)).$$

Since F_L^{-1} maps $(0,1)$ onto $\mathbb{R}$, h maps I bijectively and increasingly onto $\mathbb{R}$.

It remains to show why equality on an observed open interval where both probabilities lie in $(0,1)$ implies equality for every real z.

Consider two such functions $z \mapsto b_1(z)$ and $z \mapsto b_2(z)$ with interior intervals I_1 and I_2, agreeing on a nonempty observed open interval. The identity theorem says that two real-analytic functions agreeing on an open interval agree throughout their connected common domain. By Krantz and Parks (2002, Corollary 1.2.6), they therefore agree on $I_1 \cap I_2$.

Write $I_i = (L_i, R_i)$, allowing an endpoint to be infinite. If $L_1 < L_2$, their nonempty intersection implies $L_2 < R_1$. Thus L_2 is finite and lies in I_1. Choose $z_n \in I_1 \cap I_2$ with $z_n \downarrow L_2$. The second function satisfies $b_2(z_n) \to 0$, whereas continuity of the first gives $b_1(z_n) \to b_1(L_2) > 0$. This contradicts their equality on the intersection. Interchanging the functions rules out $L_2 < L_1$, so $L_1 = L_2$. The corresponding argument at the right endpoints uses the limit one and gives $R_1 = R_2$. Hence $I_1 = I_2$, and their zero and one extensions also agree. □

Lemma B.1.2 (Switch-point moments)**.** *Suppose $\mathbb{E}_G\Omega = 0$, $\mathrm{Var}_G(\Omega) = \tau^2 \in (0, \infty)$, and $\delta > 0$. Fix x with $\lambda_x := \lambda(x) \geq 0$, and write $\pi(z, x) := b(z, x; \theta)$ for every $z \in \mathbb{R}$. If $\lambda_x = 0$, suppose this function is continuous. If $\lambda_x > 0$, set*

$$I_x := \{z : 0 < \pi(z, x) < 1\}, \qquad h_x : I_x \to \mathbb{R}, \quad h_x(z) := \alpha + \delta z + \lambda_x \ell(\pi(z, x)).$$

For $\Omega \sim G$ and an independent standard logistic ε, define the switch point

$$S_x := \begin{cases} h_x^{-1}(-\Omega + \lambda_x \varepsilon), & \lambda_x > 0, \\ -(\alpha + \Omega)/\delta, & \lambda_x = 0. \end{cases}$$

Its CDF is $\pi(\cdot, x)$. The moments

$$\mu(x) := \mathbb{E}S_x, \qquad V(x) := \mathrm{Var}(S_x), \qquad C(x) := \mathrm{Cov}(S_x, \ell(\pi(S_x, x)))$$

are finite, $V(x) > 0$, $C(x) > 0$, and

$$\mu(x) = -\frac{\alpha}{\delta}, \qquad \delta^2 V(x) + 2\delta\lambda_x C(x) = \tau^2. \tag{B.1.3}$$

Moreover,

$$0 \leq \tau^2 - \delta^2 V(x) \leq \frac{2\pi\tau}{\sqrt{3}}\lambda_x, \tag{B.1.4}$$

and $V(x) = \tau^2/\delta^2$ if and only if $\lambda_x = 0$.

Proof. I first verify that the switch point has the stated CDF. At positive cost and $z \in I_x$, the Matějka–McKay conditional choice formula and its average give

$$\begin{aligned}\pi(z,x) &= \int \frac{\pi(z,x)e^{(\alpha+\delta z+\omega)/\lambda_x}}{1-\pi(z,x)+\pi(z,x)e^{(\alpha+\delta z+\omega)/\lambda_x}}\,dG(\omega)\\ &= \int \sigma\left(\frac{h_x(z)+\omega}{\lambda_x}\right)dG(\omega)\\ &= \mathbb{P}\{h_x(z) \geq -\Omega + \lambda_x\varepsilon\}.\end{aligned}$$

Lemma B.1.1 makes h_x an increasing bijection from I_x onto $\mathbb{R}$. Thus

$$\mathbb{P}(S_x \leq z) = \pi(z,x) \qquad (z \in I_x).$$

Since S_x lies in I_x almost surely, the equality also holds to its left and right, where both probabilities are zero and one, respectively. Continuity gives equality at finite endpoints. At zero cost, $S_x = -(\alpha+\Omega)/\delta$ gives

$$\mathbb{P}(S_x \leq z) = \mathbb{P}_G(\alpha + \delta z + \Omega \geq 0) = \pi(z,x),$$

where continuity rules out a positive probability of a payoff tie at any z. This also shows that S_x is nondegenerate: at positive cost its CDF increases from zero to one on I_x, and at zero cost its variance is $\tau^2/\delta^2 > 0$.

The continuous CDF implies that $\pi(S_x,x)$ is uniform on $(0,1)$. To verify this directly, fix $u \in (0,1)$ and choose z_u with $\pi(z_u,x) = u$. Any interval on which the CDF equals u has probability zero. Consequently

$$\mathbb{P}\{\pi(S_x,x) \leq u\} = \mathbb{P}(S_x \leq z_u) = u.$$

In particular, the variable

$$\eta_x := \ell(\pi(S_x,x))$$

is well defined almost surely and is standard logistic. Both ε and η_x have mean zero and variance $\pi^2/3$. These moments follow, for example, from the representation of a standard logistic variable as the difference of two independent standard Gumbel variables, whose variance is $\pi^2/6$ (Train, 2002, Section 3.1). The variables ε and η_x need not be independent.

Let $L_x := -\Omega + \lambda_x\varepsilon$. Applying h_x to the definition of S_x at positive cost gives

$$L_x = \alpha + \delta S_x + \lambda_x\eta_x. \tag{B.1.5}$$

At zero cost, the same identity follows directly from $S_x = -(\alpha+\Omega)/\delta$. Independence of Ω and ε gives

$$\mathbb{E}L_x = 0, \qquad \mathbb{E}L_x^2 = \mathrm{Var}(L_x) = \tau^2 + \lambda_x^2 \frac{\pi^2}{3} < \infty.$$

Rearranging (B.1.5) bounds the switch-point second moment:

$$\mathbb{E}S_x^2 \le \frac{3}{\delta^2}\left\{\mathbb{E}L_x^2 + \alpha^2 + \lambda_x^2 \mathbb{E}\eta_x^2\right\} < \infty.$$

Thus $\mu(x)$, $V(x)$, and $C(x)$ are finite. Taking means in (B.1.5) gives $0 = \alpha + \delta\mu(x)$. Taking variances gives

$$\tau^2 + \lambda_x^2 \frac{\pi^2}{3} = \delta^2 V(x) + 2\delta\lambda_x C(x) + \lambda_x^2 \frac{\pi^2}{3}.$$

The common logistic variance terms cancel, proving (B.1.3). The covariance term remains because S_x and η_x generally depend on each other.

I next establish $C(x) > 0$. Let S_x' be an independent copy of S_x. Expansion and independence give

$$2C(x) = \mathbb{E}\big[(S_x - S_x')\{\ell(\pi(S_x, x)) - \ell(\pi(S_x', x))\}\big].$$

The product is nonnegative because the CDF and log odds are increasing. The two CDF values are independent continuous uniform variables, so they differ almost surely. Monotonicity then makes the product strictly positive almost surely. Its expectation is finite by the second-moment bound, and hence $C(x) > 0$.

It follows from (B.1.3) that

$$\tau^2 - \delta^2 V(x) = 2\delta\lambda_x C(x) \ge 0,$$

with equality exactly when $\lambda_x = 0$. Cauchy–Schwarz and this inequality give

$$C(x) \le \frac{\pi}{\sqrt{3}}\sqrt{V(x)}, \qquad \delta\sqrt{V(x)} \le \tau.$$

Therefore

$$0 \le \tau^2 - \delta^2 V(x) = 2\delta\lambda_x C(x) \le \frac{2\pi}{\sqrt{3}}\lambda_x \delta\sqrt{V(x)} \le \frac{2\pi\tau}{\sqrt{3}}\lambda_x,$$

which proves (B.1.4). □

Lemma B.1.3 (Residual derivatives and continuity)**.** *For each $\eta \in (0, 1/2)$ there is a finite constant K_η, independent of θ and w, such that*

$$|r(\theta; w, q)| \le 1, \qquad |\partial_q r(\theta; w, q)| + |\partial_q^2 r(\theta; w, q)| \le K_\eta \tag{B.1.6}$$

for $q \in [\eta, 1-\eta]$. In particular, $q \mapsto \partial_q r(\theta; w, q)$ is uniformly Lipschitz on this interval. Fix $w := (z, x)$ and $q \in (0,1)$, and suppose $d(\theta_n, \theta) \to 0$, where $\theta := (\alpha, \delta, \lambda, G)$. If $\lambda(x) > 0$, or if $\lambda(x) = 0$ and $G\{-\alpha - \delta z\} = 0$, then

$$r(\theta_n; w, q) \longrightarrow r(\theta; w, q),$$
$$\partial_q r(\theta_n; w, q) \longrightarrow \partial_q r(\theta; w, q).$$

For each fixed $\theta := (\alpha, \delta, \lambda, G)$ with $\delta \neq 0$, ID3(a) implies

$$\mathbb{P}\{\lambda(X) = 0,\ G\{-\alpha - \delta Z\} > 0\} = 0.$$

Proof. Recall $r(\theta; w, q) := \Psi(\theta; w, q) - q$. Because $\Psi(\theta; w, q)$ and q lie in $[0,1]$, $|r(\theta; w, q)| \le 1$. I next bound derivatives with respect to the input probability and then examine continuity in the parameter, including a zero-cost limit.

When $\lambda(x) > 0$, define

$$p_q(\omega) := \sigma\left(\ell(q) + \frac{\alpha + \delta z + \omega}{\lambda(x)}\right).$$

Using $\sigma'(u) = \sigma(u)(1 - \sigma(u))$ and $\ell'(q) = 1/\{q(1-q)\}$ gives

$$\partial_q p_q(\omega) = \frac{p_q(\omega)(1 - p_q(\omega))}{q(1-q)}.$$

Differentiating once more and collecting terms gives

$$\begin{aligned}\partial_q^2 p_q(\omega) &= \frac{p_q(\omega)(1 - p_q(\omega))\{(1 - 2p_q(\omega)) - (1 - 2q)\}}{q^2(1-q)^2} \\ &= \frac{2p_q(\omega)(1 - p_q(\omega))(q - p_q(\omega))}{q^2(1-q)^2}.\end{aligned}$$

Let $d_\eta := \eta(1-\eta) > 0$. For $q \in [\eta, 1-\eta]$ and $\omega \in \mathbb{R}$,

$$|\partial_q p_q(\omega)| \le \frac{1}{4d_\eta}, \qquad |\partial_q^2 p_q(\omega)| \le \frac{1}{2d_\eta^2}.$$

These bounds do not involve the payoff, the cost, or G. They justify differentiation under the integral. Since

$$\partial_q r(\theta; w, q) = \int \partial_q p_q(\omega)\, dG(\omega) - 1,$$
$$\partial_q^2 r(\theta; w, q) = \int \partial_q^2 p_q(\omega)\, dG(\omega),$$

the constant $K_\eta := 1 + 1/(4d_\eta) + 1/(2d_\eta^2)$ suffices in (B.1.6). When $\lambda(x) = 0$, $\Psi(\theta; w, q)$ is independent of q, so $\partial_q r(\theta; w, q) = -1$ and $\partial_q^2 r(\theta; w, q) = 0$. The mean-value theorem also gives

$$|\partial_q r(\theta; w, t) - \partial_q r(\theta; w, q)| \le K_\eta |t - q|$$

for two shares in the trimmed interval.

When $\lambda_0(x) > 0$, set $\theta = \theta_0$ and $q = s(w) \in (0, 1)$. The population choice probability q satisfies $\mathbb{E} p_q(\Omega) = q$. Thus

$$\begin{aligned}\partial_q r(\theta_0; w, q) &= \frac{\mathbb{E}[p_q(\Omega)(1 - p_q(\Omega))]}{q(1-q)} - 1 \\ &= \frac{q(1-q) - \operatorname{Var}(p_q(\Omega))}{q(1-q)} - 1 = -\frac{\operatorname{Var}(p_q(\Omega))}{q(1-q)}.\end{aligned}$$

This verifies (4.6).

I next show that the residual $r(\theta_n; w, q)$ and its derivative $\partial_q r(\theta_n; w, q)$ converge to their values at θ, under the conditions in the lemma.

For the sequence in the statement, the representation theorem in Billingsley (1995, Theorem 25.6) allows weak convergence $G_n \Rightarrow G$ to be represented by random variables $\Omega_n \sim G_n$ and $\Omega \sim G$ on a common probability space with $\Omega_n \to \Omega$ almost surely. This representation lets me compare the integrands directly. Metric convergence also gives

$$\alpha_n \to \alpha, \qquad \delta_n \to \delta, \qquad \lambda_n(x) \to \lambda(x).$$

Write $v_n := \alpha_n + \delta_n z + \Omega_n$ and $v := \alpha + \delta z + \Omega$. The integrand defining $\Psi(\theta_n; w, q)$ is

$$p_n(q) := \begin{cases} \sigma(\ell(q) + v_n/\lambda_n(x)), & \lambda_n(x) > 0, \\ \mathbf{1}\{v_n > 0\}, & \lambda_n(x) = 0. \end{cases}$$

If $\lambda(x) > 0$, then $\lambda_n(x) > 0$ eventually. Ordinary continuity gives, almost surely,

$$p_n(q) \longrightarrow p(q) := \sigma(\ell(q) + \nu/\lambda(x)),$$
$$\partial_q p_n(q) = \frac{p_n(q)(1-p_n(q))}{q(1-q)} \longrightarrow \frac{p(q)(1-p(q))}{q(1-q)}.$$

The integrands satisfy $0 \le p_n(q) \le 1$ and $|\partial_q p_n(q)| \le 1/\{4q(1-q)\}$. Because q is fixed in $(0,1)$, bounded convergence applies to both expectations. Subtracting q from the first expectation and 1 from the second proves the two limits in the statement.

If $\lambda(x) = 0$, then, outside the payoff-tie event $\nu = 0$, $\nu_n \to \nu$ with the same sign eventually. Along terms with $\lambda_n(x) > 0$,

$$\nu > 0 \implies \ell(q) + \nu_n/\lambda_n(x) \to +\infty, \quad p_n(q) \to 1,$$
$$\nu < 0 \implies \ell(q) + \nu_n/\lambda_n(x) \to -\infty, \quad p_n(q) \to 0.$$

In either case

$$\partial_q p_n(q) = \frac{p_n(q)(1-p_n(q))}{q(1-q)} \to 0.$$

Along terms with $\lambda_n(x) = 0$, $p_n(q) = \mathbf{1}\{\nu_n > 0\}$ and $\partial_q p_n(q) = 0$. Therefore $p_n(q) \to \mathbf{1}\{\nu > 0\}$ and $\partial_q p_n(q) \to 0$ along the full sequence. The condition $G\{-\alpha - \delta z\} = 0$ makes these limits almost sure. Applying bounded convergence with the same bounds gives

$$r(\theta_n; w, q) \longrightarrow \mathbb{P}_G(\nu > 0) - q = r(\theta; w, q),$$
$$\partial_q r(\theta_n; w, q) \longrightarrow -1 = \partial_q r(\theta; w, q).$$

To apply this continuity argument across markets for a fixed $\theta := (\alpha, \delta, \lambda, G)$, I must show that $G\{-\alpha - \delta Z\} > 0$ has probability zero.

Let $\mathcal{D}_G := \{\omega : G\{\omega\} > 0\}$ be the set of point masses of a fixed G. This set is countable. With $\delta \neq 0$, a payoff tie requires

$$Z \in \mathcal{T}_\theta := \{(-\alpha - \omega)/\delta : \omega \in \mathcal{D}_G\}.$$

The set $\mathcal{T}_\theta$ is also countable. ID3(a) gives $\mathbb{P}(Z \in \mathcal{T}_\theta \mid X = x) = 0$ at each x, and integration over X gives probability zero across markets. This assertion is for each fixed candidate; the argument does not require a single tie-free event for every candidate simultaneously. □

For the share-sampling results, markets have characteristics W_m, population choice probabilities $s_m := s(W_m)$, and deterministic sizes $N_{m,M}$. Conditional on the characteristics, the count $N_{m,M}\widehat{s}_m$ is binomial with success probability s_m. The two criteria are

$$\widehat{Q}_M(\theta) := M^{-1}\sum_m r(\theta; W_m, \widehat{s}_m)^2, \qquad Q_M(\theta) := M^{-1}\sum_m r(\theta; W_m, s_m)^2.$$

To bound the change in a squared residual when its input probability changes from s to t, define

$$H(t,s) := \begin{cases} \min\{1, \frac{1}{2}|\ell(t)-\ell(s)| + 2|t-s|\}, & t,s \in (0,1), \\ 0, & t = s \in \{0,1\}, \\ 1, & \text{otherwise.} \end{cases} \tag{B.1.7}$$

Lemma B.1.4 (Share substitution)**.** *Under the sampling conditions and E1, the function $H : [0,1]^2 \to [0,1]$ defined above satisfies*

$$\sup_{\theta\in\Theta} |\widehat{Q}_M(\theta) - Q_M(\theta)| \le \frac{1}{M}\sum_{m=1}^{M} H(\widehat{s}_m, s_m) \to_p 0. \tag{B.1.8}$$

Proof. I first bound $|r(\theta;w,t)^2 - r(\theta;w,s)^2|$ by a function of t and s that does not depend on θ. I then show that the average bound $M^{-1}\sum_m H(\widehat{s}_m, s_m)$ tends to zero.

For two input probabilities t, s, write $r_t := r(\theta;w,t)$ and $r_s := r(\theta;w,s)$. Since $|r_t|, |r_s| \le 1$,

$$\begin{aligned} |r_t^2 - r_s^2| &= |r_t - r_s||r_t + r_s| \\ &\le 2|r_t - r_s| \le 2|\Psi(\theta;w,t) - \Psi(\theta;w,s)| + 2|t-s|. \end{aligned}$$

When $\lambda(x) > 0$ and $t, s \in (0,1)$, $\sup_u |\sigma'(u)| = 1/4$ implies

$$|\Psi(\theta;w,t) - \Psi(\theta;w,s)| \le \tfrac{1}{4}|\ell(t) - \ell(s)|.$$

Thus $|r_t^2 - r_s^2| \le \frac{1}{2}|\ell(t)-\ell(s)| + 2|t-s|$. When $\lambda(x) = 0$, $\Psi(\theta;w,q)$ does not depend on q, so $|r_t^2 - r_s^2| \le 2|t-s|$. Since $r_t^2, r_s^2 \in [0,1]$, $|r_t^2 - r_s^2| \le 1$ in all cases. If $t = s \in \{0,1\}$, then $r_t^2 = r_s^2$. If either input equals zero or one and $t \ne s$, I use the bound one. These inequalities give exactly $|r_t^2 - r_s^2| \le H(t,s)$.

Applying this market by market,

$$|\widehat{Q}_M(\theta) - Q_M(\theta)| \le \frac{1}{M}\sum_m H(\widehat{s}_m, s_m).$$

The right-hand side contains no candidate parameter, so taking the supremum over θ leaves the same bound.

It remains to show $\mathbb{E}[M^{-1}\sum_m H(\widehat{s}_m, s_m)] \to 0$.

For fixed market characteristics w, let n denote its within-market sample size and define

$$h_n(w) := \mathbb{E}[H(\widehat{s}, s(w)) \mid W = w], \qquad \text{where} \qquad n\widehat{s} \mid W = w \sim \text{Binomial}(n, s(w)).$$

When $0 < s(w) < 1$, binomial share consistency gives $\widehat{s} \to_p s(w)$ as $n \to \infty$. Since $H(t, s(w)) \to 0$ as $t \to s(w)$ and $0 \le H \le 1$, for every $\varepsilon > 0$,

$$h_n(w) \le \varepsilon + \mathbb{P}\{H(\widehat{s}, s(w)) > \varepsilon \mid W = w\}, \qquad \text{so} \qquad \limsup_{n\to\infty} h_n(w) \le \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives $h_n(w) \to 0$. When $s(w) \in \{0, 1\}$, the sampled share equals $s(w)$ exactly, so $h_n(w) = 0$.

The minimum sample size $N_M := \min_m N_{m,M}$ tends to infinity by E1. Even though the sizes can differ across markets,

$$\begin{aligned}\mathbb{E}\left[\frac{1}{M}\sum_m H(\widehat{s}_m, s_m)\right] &= \int \frac{1}{M}\sum_m h_{N_{m,M}}(w)\, dF_W(w) \\ &\le \int \sup_{n \ge N_M} h_n(w)\, dF_W(w) \longrightarrow 0.\end{aligned}$$

The inequality uses $N_{m,M} \ge N_M$ for every market. For each fixed w, $h_n(w) \to 0$ means that all sufficiently late terms are small, so $\sup_{n \ge N_M} h_n(w) \to 0$. This supremum is bounded by one; bounded convergence justifies the last integral limit. Finally, Markov's inequality gives

$$\mathbb{P}\left\{\frac{1}{M}\sum_m H(\widehat{s}_m, s_m) > \varepsilon\right\} \le \frac{1}{\varepsilon}\mathbb{E}\left[\frac{1}{M}\sum_m H(\widehat{s}_m, s_m)\right] \longrightarrow 0. \qquad \square$$

Lemma B.1.5 (Binomial sampling moments)**.** *Fix $\eta \in (0, 1/2)$ and $N \in \mathbb{N}$. Let $N\widehat{s} \sim$ Binomial(N, s) with $s \in [\eta, 1-\eta]$, and set $\xi := \sqrt{N}(\widehat{s} - s)/\sqrt{s(1-s)}$. Then*

$$\mathbb{E}\xi = 0, \qquad \mathbb{E}\xi^2 = 1,$$
$$\mathbb{E}\xi^3 = \frac{1-2s}{\sqrt{Ns(1-s)}}, \qquad \mathbb{E}\xi^4 = 3 + \frac{1-6s(1-s)}{Ns(1-s)}.$$

For each fixed $p \ge 2$, $\mathbb{E}|\xi|^p$ is uniformly bounded and $\mathbb{E}|\widehat{s} - s|^p \le C_p N^{-p/2}$, uniformly over this share interval. The same statements hold conditionally on market characteristics.

Proof. Write $N\widehat{s} = \sum_{i=1}^{N} Y_i$, with independent Bernoulli variables of success probability s, and let $B_i := Y_i - s$. Put $D := s(1-s)$. The two possible values of B_i give

$$\mathbb{E}B_i = 0, \quad \mathbb{E}B_i^2 = D, \quad \mathbb{E}B_i^3 = D(1-2s), \quad \mathbb{E}B_i^4 = D(1-3D).$$

Also $\xi = (ND)^{-1/2}\sum_i B_i$. Independence and centering immediately give $\mathbb{E}\xi = 0$ and $\mathbb{E}\xi^2 = 1$.

In the expansion of the third power, any term with an index appearing only once has expectation zero. Only terms with all three indices equal remain:

$$\mathbb{E}\left(\sum_i B_i\right)^3 = N\mathbb{E}B_1^3.$$

In the fourth power, the nonzero terms have either all four indices equal or two pairs of equal indices. There are $3N(N-1)$ terms of the latter type, so

$$\mathbb{E}\left(\sum_i B_i\right)^4 = N\mathbb{E}B_1^4 + 3N(N-1)(\mathbb{E}B_1^2)^2.$$

Dividing by $(ND)^{3/2}$ and $(ND)^2$, respectively, gives

$$\mathbb{E}\xi^3 = \frac{1-2s}{\sqrt{ND}}, \qquad \mathbb{E}\xi^4 = \frac{1-3D}{ND} + 3\frac{N-1}{N} = 3 + \frac{1-6D}{ND}.$$

For higher absolute moments, Hoeffding (1963, Theorem 2), applied to both tails of the Bernoulli average, and $D \geq \eta(1-\eta)$ give

$$\mathbb{P}(|\xi| > t) = \mathbb{P}\Big(|\widehat{s}-s| > t\sqrt{D/N}\Big) \leq 2e^{-2Dt^2} \leq 2e^{-2\eta(1-\eta)t^2}.$$

Using the tail-integral formula,

$$\mathbb{E}|\xi|^p = \int_0^\infty p t^{p-1}\mathbb{P}(|\xi| > t)\,dt \leq 2p\int_0^\infty t^{p-1}e^{-2\eta(1-\eta)t^2}\,dt < \infty.$$

The bound depends only on p and η. Since $\widehat{s} - s = \sqrt{D/N}\,\xi$, it also gives the stated share-error bound. Conditional on the market characteristics, s is fixed and the observations remain independent Bernoulli variables, so every calculation applies conditionally as well. □

Lemma B.1.6 (Conditional central limit theorem). *Let $\mathcal{F}_M := \sigma(W_1, \ldots, W_M)$. For each M, suppose there are k_M conditionally independent variables $X_{M,1}, \ldots, X_{M,k_M}$, centered given $\mathcal{F}_M$. The number of summands k_M is finite and $\mathcal{F}_M$-measurable. Set*

$$v_M^2 := \sum_{j=1}^{k_M} \mathbb{E}[X_{M,j}^2 \mid \mathcal{F}_M].$$

All conditional moments used below are finite. Suppose $\mathbb{P}(v_M > 0) \to 1$ and, for some $p > 2$,

$$\Lambda_M := \frac{\sum_{j=1}^{k_M} \mathbb{E}[|X_{M,j}|^p \mid \mathcal{F}_M]}{v_M^p} \to_p 0,$$

with the ratio defined as zero when $v_M = 0$. Define

$$S_M := \begin{cases} v_M^{-1} \sum_{j=1}^{k_M} X_{M,j}, & v_M > 0, \\ 0, & v_M = 0 \end{cases}.$$

Then

$$S_M \Rightarrow N(0,1).$$

Proof. Once the market characteristics are fixed, the number of summands and their conditional moments are fixed; the sampled choices, and hence the summands themselves, remain random. The ordinary Lyapunov theorem (Billingsley, 1995, Theorem 27.3) can then be applied to these conditional distributions. I spell out why convergence of the random Lyapunov ratio in probability is sufficient.

For a real number t, define the conditional CDF

$$F_M(t) := \mathbb{P}(S_M \le t \mid \mathcal{F}_M).$$

I will first show $F_M(t) \to_p \Phi(t)$, where Φ is the standard normal CDF. The ordinary Lyapunov theorem concerns deterministic arrays of distributions. Here Λ_M, k_M, and v_M depend on market characteristics, so I first obtain almost-sure convergence along a subsequence.

Take any deterministic subsequence of sample sizes (M_ℓ). Since

$$\Lambda_M + \mathbf{1}\{v_M = 0\} \to_p 0,$$

there is a further deterministic subsequence (M_{ℓ_r}) along which

$$\Lambda_{M_{\ell_r}} + \mathbf{1}\{v_{M_{\ell_r}} = 0\} \longrightarrow 0 \quad \text{almost surely.}$$

This follows from the subsequence characterization of convergence in probability (Billingsley, 1995, Theorem 20.5). The indicator implies $v_{M_{\ell_r}} > 0$ eventually, almost surely.

Choose the joint conditional distributions of the summands $X_{M,1}, \ldots, X_{M,k_M}$ given $\mathcal{F}_M$. In this application they are supplied directly by the finite product of binomial sampling distributions after $W_1, \ldots, W_M$ have been fixed. On a probability-one set, conditional independence, centering, and the moment identities hold for every sample size; a single such set can be used because there are countably many sample sizes.

Now fix a realization ω in this set and in the almost-sure convergence event above. For sample size M_{ℓ_r}, the conditional distribution has exactly $k_{M_{\ell_r}}(\omega)$ independent centered summands and total variance $v_{M_{\ell_r}}(\omega)^2$. Its ordinary Lyapunov ratio is the number $\Lambda_{M_{\ell_r}}(\omega)$, which tends to zero. Theorem 27.3 of Billingsley (1995) therefore gives

$$F_{M_{\ell_r}}(t)(\omega) \longrightarrow \Phi(t) \qquad \text{for every real } t.$$

Thus every subsequence of $F_M(t)$ has a further subsequence converging almost surely to $\Phi(t)$. This proves

$$F_M(t) \to_p \Phi(t).$$

No independence between samples of different sizes is needed: the Lyapunov theorem uses independence among the summands within each sample.

The conditional convergence just obtained also implies unconditional convergence.

Because $0 \le F_M(t) \le 1$, for every $\varepsilon > 0$,

$$\mathbb{E}|F_M(t) - \Phi(t)| \le \varepsilon + \mathbb{P}\{|F_M(t) - \Phi(t)| > \varepsilon\}.$$

The probability tends to zero. Letting $\varepsilon \downarrow 0$ gives $\mathbb{E}|F_M(t) - \Phi(t)| \to 0$. Finally, the tower property of conditional expectation (Billingsley, 1995, Theorem 34.4) yields

$$\mathbb{P}(S_M \le t) = \mathbb{E}F_M(t), \qquad \text{so} \qquad |\mathbb{P}(S_M \le t) - \Phi(t)| \le \mathbb{E}|F_M(t) - \Phi(t)| \longrightarrow 0.$$

The standard normal CDF is continuous at every real t, so this is precisely $S_M \Rightarrow N(0,1)$. □

I now state and prove lemmas for the RI model with multiple alternatives, used in the identification proof of Theorem 4. Throughout these lemmas, I use the setup and notation from Appendix A. The first gives the joint distribution of the Gumbel differences used in that proof. Its special case $J = 1$ also gives the logistic distribution used in the binary model. The subsequent lemmas establish continuity and uniqueness of the optimal choice probability vector and show how an interior probability vector determines the known payoff indices.

Lemma B.1.7 (The Gumbel-difference distribution and its convolution). *Let $\eta_0, \ldots, \eta_J$ be independent standard Gumbel variables, with CDF $\mathbb{P}(\eta_j \le y) = \exp(-e^{-y})$, and put $\zeta := (\zeta_1, \ldots, \zeta_J)'$, where $\zeta_j := \eta_j - \eta_0$. Its CDF F_ζ, density f_ζ, and characteristic function φ_ζ satisfy, for $y, t \in \mathbb{R}^J$,*

$$F_\zeta(y) = \frac{1}{1 + \sum_{j=1}^J e^{-y_j}}, \tag{B.1.9}$$

$$f_\zeta(y) = \frac{J!\, e^{-\sum_j y_j}}{(1 + \sum_j e^{-y_j})^{J+1}}, \tag{B.1.10}$$

$$\varphi_\zeta(t) = \Gamma\left(1 + i\sum_{j=1}^J t_j\right)\prod_{j=1}^J \Gamma(1 - it_j) \neq 0 \quad (t \in \mathbb{R}^J). \tag{B.1.11}$$

If $\Omega \in \mathbb{R}^J$ is any random vector independent of $\eta_0, \ldots, \eta_J$ and $\lambda > 0$, then the CDF F_H and density f_H of $H := \Omega + \lambda\zeta$ are real analytic on $\mathbb{R}^J$.

Proof. The inequality $\zeta \le y$ means $\eta_j - \eta_0 \le y_j$ for every $j = 1, \ldots, J$. Thus $\{\zeta \le y\}$ is the event that alternative zero is chosen when the known payoffs are $0, -y_1, \ldots, -y_J$ and the payoff shocks are $\eta_0, \ldots, \eta_J$. The multinomial logit formula (Train, 2002, Eq. (3.6)) therefore gives (B.1.9). Differentiating once in each coordinate gives (B.1.10).

A standard Gumbel variable has characteristic function $\Gamma(1 - it)$ (Janson, 2010, Section 3). Independence gives

$$\varphi_\zeta(t) = \mathbb{E}e^{-i(\sum_j t_j)\eta_0}\prod_j \mathbb{E}e^{it_j\eta_j} = \Gamma\left(1 + i\sum_j t_j\right)\prod_j \Gamma(1 - it_j).$$

The gamma function Γ has no zeros, and all its arguments here have real part one, away from its poles (NIST, 2026, Section 5.2(i)). This proves (B.1.11), including nonvanishing.

For the analyticity claim, let G denote the distribution of Ω. Conditioning on Ω gives

$$F_H(h) = \int \frac{1}{1 + \sum_{j=1}^J \exp\{(\omega_j - h_j)/\lambda\}}\, dG(\omega).$$

This is a mixed logit probability for the reference alternative, with known payoff indices $-h_j/\lambda$ and random intercepts Ω_j/λ. By Iaria and Wang (2025, Theorem 1(a)), the non-reference

choice probabilities are real analytic for any mixing distribution. Since F_H is one minus their sum, it is real analytic on $\mathbb{R}^J$. Its density $f_H = \partial_{h_1} \cdots \partial_{h_J} F_H$ is real analytic as well, because derivatives preserve real analyticity. No density or moment restriction on G is needed. □

Lemma B.1.8 (Continuity of the choice-probability objective and its unique maximizer)**.** *Fix $J \geq 1$ and x with $\lambda(x) > 0$. Let $\Omega \in \mathbb{R}^J$ have distribution G satisfying*

$$\mathbb{E}_G \|\Omega\|_1 = \sum_{j=1}^{J} \mathbb{E}_G |\Omega_j| < \infty.$$

For $a \in \mathbb{R}^J$ and $\Delta_J := \{q \in [0,1]^{J+1} : \sum_{j=0}^{J} q_j = 1\}$, define

$$\Phi(q; a, x) := \lambda(x) \mathbb{E}_G \log \left\{ q_0 + \sum_{j=1}^{J} q_j e^{(a_j + \Omega_j)/\lambda(x)} \right\}.$$

Then $(q, a) \mapsto \Phi(q; a, x)$ is finite and continuous on $\Delta_J \times \mathbb{R}^J$, including the boundary of Δ_J. For every a, a maximizer over Δ_J exists. If that maximizer is unique for every a, the map

$$q^*(\cdot; x) : \mathbb{R}^J \longrightarrow \Delta_J, \qquad a \longmapsto q^*(a; x) := \arg\max_{q \in \Delta_J} \Phi(q; a, x)$$

is continuous.

Proof. Write $\lambda := \lambda(x)$ and define the integrand

$$g(q, a, \omega) := \lambda \log \left\{ q_0 + \sum_{j=1}^{J} q_j e^{(a_j + \omega_j)/\lambda} \right\}.$$

Because the nonnegative weights $q_0, \ldots, q_J$ sum to one,

$$\min\{0, a_1 + \omega_1, \ldots, a_J + \omega_J\} \leq g(q, a, \omega) \leq \max\{0, a_1 + \omega_1, \ldots, a_J + \omega_J\}.$$

In particular,

$$|g(q, a, \omega)| \leq \|a\|_\infty + \|\omega\|_1.$$

Since $\mathbb{E}_G \|\Omega\|_1 < \infty$, the bound is integrable for fixed a, so $\Phi(q; a, x)$ is finite for every (q, a).

To prove joint continuity, take arbitrary sequences $q_n \to q$ in Δ_J and $a_n \to a$ in $\mathbb{R}^J$. The sequence a_n is bounded, so there is $R < \infty$ with $\|a_n\|_\infty \leq R$ for all n. For each fixed ω, the sum inside the logarithm is strictly positive, even if some coordinates of q are zero. Thus

$$g(q_n, a_n, \omega) \longrightarrow g(q, a, \omega), \qquad |g(q_n, a_n, \omega)| \leq R + \|\omega\|_1.$$

The bound is independent of n and integrable under G. Dominated convergence gives $\Phi(q_n; a_n, x) \to \Phi(q; a, x)$, proving joint continuity. For fixed a, continuity on the compact simplex gives existence of a maximizer by the extreme value theorem (Munkres, 2000, Theorem 27.4).

Now suppose the maximizer is unique for every a. Fix any $a_n \to a$ and write $q_n^* := q^*(a_n; x)$. I show $q_n^* \to q^*(a; x)$. Take any subsequence $\{q_{n_k}^*\}$. Compactness of Δ_J gives a further subsequence $q_{n_{k_\ell}}^* \to \bar{q} \in \Delta_J$. For every fixed $q \in \Delta_J$, optimality at $a_{n_{k_\ell}}$ gives

$$\Phi(q_{n_{k_\ell}}^*; a_{n_{k_\ell}}, x) \geq \Phi(q; a_{n_{k_\ell}}, x).$$

Joint continuity permits passage to the limit on both sides:

$$\Phi(\bar{q}; a, x) \geq \Phi(q; a, x) \qquad \text{for every } q \in \Delta_J.$$

Hence $\bar{q}$ maximizes the objective at a. Uniqueness gives $\bar{q} = q^*(a; x)$. If the original sequence q_n^* did not converge to $q^*(a; x)$, some subsequence would stay at least a fixed positive distance from $q^*(a; x)$. The compactness argument just given would produce a further subsequence converging to $q^*(a; x)$, a contradiction. Therefore $q^*(a_n; x) \to q^*(a; x)$. □

Lemma B.1.9 (Uniqueness of optimal choice probabilities)**.** *For the distribution G, cost $\lambda(x) > 0$, simplex Δ_J, and objective $\Phi(q; a, x)$ in Lemma B.1.8, suppose A2 holds at $\lambda(x)$: for $u \in \mathbb{R}^{J+1}$,*

$$u_0 + \sum_{j=1}^{J} u_j e^{\Omega_j/\lambda(x)} = 0 \quad G\text{-almost surely} \quad \Longrightarrow \quad u = 0.$$

For every $a \in \mathbb{R}^J$, the function $q \mapsto \Phi(q; a, x)$ is strictly concave on Δ_J. Its maximizer $q^*(a; x)$ is unique, and the map $a \mapsto q^*(a; x)$ is continuous.

Proof. Fix $a \in \mathbb{R}^J$ and write $\lambda := \lambda(x)$. For $q \in \Delta_J$, put

$$D_q(\Omega) := q_0 + \sum_{j=1}^{J} q_j e^{a_j/\lambda} e^{\Omega_j/\lambda} > 0.$$

Take $q, \widetilde{q} \in \Delta_J$ with $q \neq \widetilde{q}$. If $D_q(\Omega) = D_{\widetilde{q}}(\Omega)$ almost surely, then

$$(q_0 - \widetilde{q}_0) + \sum_{j=1}^{J} (q_j - \widetilde{q}_j) e^{a_j/\lambda} e^{\Omega_j/\lambda} = 0 \quad G\text{-almost surely.}$$

Apply A2 with $u_0 := q_0 - \widetilde{q}_0$ and $u_j := (q_j - \widetilde{q}_j)e^{a_j/\lambda}$ for $j = 1, \dots, J$. It gives $u_j = 0$ for every j. Since $e^{a_j/\lambda} > 0$, this would imply $q = \widetilde{q}$, a contradiction. Therefore

$$\mathbb{P}_G\{D_q(\Omega) \neq D_{\widetilde{q}}(\Omega)\} > 0.$$

For any $t \in (0,1)$, strict concavity of the logarithm gives

$$\log\{tD_q(\Omega) + (1-t)D_{\widetilde{q}}(\Omega)\} \geq t \log D_q(\Omega) + (1-t)\log D_{\widetilde{q}}(\Omega),$$

with a strict inequality on the event just displayed. All three logarithms are integrable by Lemma B.1.8. Taking expectations and multiplying by $\lambda > 0$ therefore gives

$$\Phi(tq + (1-t)\widetilde{q}; a, x) > t\Phi(q; a, x) + (1-t)\Phi(\widetilde{q}; a, x).$$

This is strict concavity. Since a maximizer exists, it is unique. The continuity conclusion now follows from Lemma B.1.8. □

Lemma B.1.10 (Interior shares determine a unique payoff vector)**.** *Use the cost $\lambda(x) > 0$, distribution G, and objective $\Phi(q; a, x)$ in Lemma B.1.8, and suppose A2 holds at $\lambda(x)$. Write*

$$\begin{aligned}
\mathcal{D}_J &:= \{p \in [0,1]^J : \textstyle\sum_{j=1}^J p_j \leq 1\}, \\
\mathcal{D}_J^\circ &:= \{p \in \mathcal{D}_J : p_j > 0 \ (j = 1, \dots, J),\ \textstyle\sum_{j=1}^J p_j < 1\}, \\
p_0 &:= 1 - \textstyle\sum_{j=1}^J p_j, \qquad q(p) := (p_0, p_1, \dots, p_J)',
\end{aligned}$$

so that $\mathcal{D}_J$ uses the J free coordinates of Δ_J. Let $\bar{b}(a; x)$ be the unique maximizer of $\Phi(\cdot; a, x)$, and set

$$\mathcal{W}_x := \{a \in \mathbb{R}^J : \bar{b}^j(a; x) > 0 \ (j = 0, \dots, J)\}.$$

For every $p \in \mathcal{D}_J^\circ$, the function

$$H(a; p, x) := \Phi(q(p); a, x) - p'a \qquad (a \in \mathbb{R}^J)$$

is strictly convex and tends to $+\infty$ as $\|a\|_\infty \to \infty$. It has a unique minimizer $a^*(p; x)$. The set $\mathcal{W}_x$ is nonempty and open, and the maps

$$\begin{aligned}
B_x : \mathcal{W}_x &\longrightarrow \mathcal{D}_J^\circ, \quad a \longmapsto B_x(a) := (\bar{b}^1(a; x), \dots, \bar{b}^J(a; x))', \\
a^*(\cdot; x) : \mathcal{D}_J^\circ &\longrightarrow \mathcal{W}_x, \quad p \longmapsto a^*(p; x)
\end{aligned}$$

are mutually inverse bijections.

Proof. Lemmas B.1.8 and B.1.9 give a unique choice-probability vector $\bar{b}(a;x)$ and continuity of $a \mapsto \bar{b}(a;x)$ on $\mathbb{R}^J$. Consequently $\mathcal{W}_x$ is open: if every $\bar{b}^j(a;x) > 0$, continuity keeps every $\bar{b}^j(\widetilde{a};x)$ positive for $\widetilde{a}$ sufficiently close to a. I show that each $p \in \mathcal{D}_J^\circ$ is attained at exactly one payoff vector a by minimizing $a \mapsto H(a;p,x)$.

Fix $p \in \mathcal{D}_J^\circ$, write $\lambda := \lambda(x)$, and define

$$\begin{aligned} D(a,\omega;p) &:= p_0 + \sum_{j=1}^{J} p_j e^{(a_j+\omega_j)/\lambda}, \\ \pi_0(a,\omega;p) &:= p_0 / D(a,\omega;p), \\ \pi_j(a,\omega;p) &:= p_j e^{(a_j+\omega_j)/\lambda} / D(a,\omega;p) \quad (j=1,\dots,J). \end{aligned}$$

All $\pi_j(a,\omega;p)$ are positive and sum to one. Differentiating with respect to a gives

$$\begin{aligned} \nabla_a H(a;p,x) &= \mathbb{E}_G \pi_{1:J}(a,\Omega;p) - p, \\ \nabla_a^2 H(a;p,x) &= \frac{1}{\lambda} \mathbb{E}_G \{ \operatorname{diag}(\pi_{1:J}) - \pi_{1:J} \pi_{1:J}' \}, \end{aligned}$$

where $\pi_{1:J} := (\pi_1,\dots,\pi_J)'$, evaluated at $(a,\Omega;p)$ in the second line. These differentiations under the expectation are valid because $0 < \pi_j < 1$ and

$$\partial_{a_k} \pi_j = \lambda^{-1} \pi_j (\mathbf{1}\{j=k\} - \pi_k) \quad (j,k=1,\dots,J)$$

is bounded independently of Ω.

For $u \in \mathbb{R}^J$, put $u_0 := 0$. The Hessian's quadratic form is

$$u' \nabla_a^2 H(a;p,x) u = \frac{1}{\lambda} \mathbb{E}_G \left\{ \sum_{j=0}^{J} \pi_j u_j^2 - \left(\sum_{j=0}^{J} \pi_j u_j \right)^2 \right\}.$$

For each Ω, the expression in braces is the variance of the numbers $u_0,\dots,u_J$ under the positive probabilities $\pi_0,\dots,\pi_J$. It can be zero only if all these numbers are equal. Since $u_0 = 0$, it is strictly positive for every $u \neq 0$. Thus $\nabla_a^2 H(a;p,x)$ is positive definite and $H(\cdot;p,x)$ is strictly convex.

To establish existence of its minimizer, put $p_{\min} := \min_{0 \le j \le J} p_j > 0$ and $a_0 = 0$. Choosing the largest of $a_0,\dots,a_J$ in the denominator gives

$$\lambda \log D(a,\Omega;p) \ge \lambda \log p_{\min} + \max_{0 \le j \le J} a_j - \|\Omega\|_\infty.$$

The first-moment assumption implies $\mathbb{E}_G\|\Omega\|_\infty < \infty$. Writing $a_{\max} := \max_{0\le j\le J} a_j$ and $a_{\min} := \min_{0\le j\le J} a_j$, I obtain

$$H(a;p,x) \ge a_{\max} - \sum_{j=0}^{J} p_j a_j + \lambda \log p_{\min} - \mathbb{E}_G\|\Omega\|_\infty.$$

The first difference on the right satisfies

$$\begin{aligned} a_{\max} - \sum_{j=0}^{J} p_j a_j &= \sum_{j=0}^{J} p_j (a_{\max} - a_j) \\ &\ge p_{\min}(a_{\max} - a_{\min}) \ge p_{\min}\|a\|_\infty. \end{aligned}$$

The last inequality uses $a_0 = 0$. Therefore $H(a;p,x) \to +\infty$ as $\|a\|_\infty \to \infty$. In particular, outside a sufficiently large closed ball its value exceeds $H(0;p,x)$. Continuity and the extreme value theorem on that ball give a minimizer; strict convexity makes it unique. Denote it by $a^*(p;x)$.

I next show that the maximizing shares at $a^*(p;x)$ are exactly $q(p)$. The first-order condition for minimizing H gives $\mathbb{E}_G \pi_j(a^*(p;x),\Omega;p) = p_j$ for $j = 1,\dots,J$. Summing these equations and taking complements gives the same equality for $j = 0$. Since all p_j are positive, division gives

$$\mathbb{E}_G \frac{e^{(a_j^*(p;x)+\Omega_j)/\lambda}}{D(a^*(p;x),\Omega;p)} = 1 \quad (j = 1,\dots,J), \qquad \mathbb{E}_G \frac{1}{D(a^*(p;x),\Omega;p)} = 1.$$

The derivative of the choice-probability objective in its free coordinates is

$$\partial_{p_j} \Phi(q(p);a,x) = \lambda \mathbb{E}_G \frac{e^{(a_j+\Omega_j)/\lambda} - 1}{D(a,\Omega;p)}.$$

This derivative is valid for $p \in \mathcal{D}_J^\circ$: each ratio $e^{(a_j+\Omega_j)/\lambda}/D$ is bounded by $1/p_j$, and $1/D \le 1/p_0$, with locally uniform bounds. The preceding equalities make this derivative zero for every j at $(a^*(p;x),p)$. Strict concavity in the shares then gives

$$\bar{b}(a^*(p;x);x) = q(p), \qquad B_x(a^*(p;x)) = p.$$

Thus $a^*(p;x) \in \mathcal{W}_x$, and $\mathcal{W}_x$ is nonempty.

Conversely, take $a \in \mathcal{W}_x$ and put $p := B_x(a)$. Since every component of $q(p)$ is positive, the equations $\partial_{p_j}\Phi(q(p);a,x) = 0$ give a common number k:

$$\mathbb{E}_G \frac{e^{(a_j+\Omega_j)/\lambda}}{D(a,\Omega;p)} = \mathbb{E}_G \frac{1}{D(a,\Omega;p)} =: k \quad (j = 1,\dots,J).$$

Multiplying by p_j, including the equation for alternative 0 multiplied by p_0, and summing shows $k = \mathbb{E}_G[D/D] = 1$. Hence $\mathbb{E}_G \pi_j(a, \Omega; p) = p_j$ for every j and $\nabla_a H(a; p, x) = 0$. The strictly convex function $H(\cdot; p, x)$ has only one stationary point, so $a = a^*(p; x)$. This proves both inverse identities and the claimed bijection. □

Lemma B.1.11 (Analytic choice-probability maps and the value function)**.** *Under the conditions and notation of Lemma B.1.10, the mutually inverse maps B_x and $a^*(\cdot; x)$ are real analytic. Define the optimized value and its convex conjugate by*

$$V_x(a) := \max_{q \in \Delta_J} \Phi(q; a, x), \qquad V_x^*(p) := \sup_{a \in \mathbb{R}^J} \{p'a - V_x(a)\}.$$

The function V_x is finite and convex on $\mathbb{R}^J$. For $p \in \mathcal{D}_J^\circ$ and $a \in \mathcal{W}_x$,

$$V_x^*(p) = -H(a^*(p; x); p, x),$$
$$\nabla_a V_x(a) = B_x(a), \qquad \nabla_p V_x^*(p) = a^*(p; x).$$

The matrix $D_a B_x(a)$ is symmetric and positive definite for every $a \in \mathcal{W}_x$.

Proof. I first establish the differentiability needed for the gradient identities. Fix x and write $\lambda := \lambda(x)$. For $(a, p) \in \mathbb{R}^J \times \mathcal{D}_J^\circ$, define

$$c_j(a, p) := -a_j + \lambda \log(p_0/p_j), \qquad Q_j(c) := \mathbb{E}_G \frac{e^{(\Omega_j - c_j)/\lambda}}{1 + \sum_{k=1}^J e^{(\Omega_k - c_k)/\lambda}} \quad (j = 1, \ldots, J),$$

and $Q_0(c) := 1 - \sum_{j=1}^J Q_j(c)$. These mixed logit probabilities are real analytic in c by Iaria and Wang (2025, Theorem 1(a)). Since $(a, p) \mapsto c(a, p)$ is real analytic on the stated domain, the following two first-order-condition maps are jointly real analytic:

$$F_j(a, p) := \partial_{p_j} \Phi(q(p); a, x) = \lambda\{Q_j(c(a, p))/p_j - Q_0(c(a, p))/p_0\},$$
$$L_j(a, p) := \partial_{a_j} H(a; p, x) = Q_j(c(a, p)) - p_j, \quad j = 1, \ldots, J.$$

Write $F := (F_1, \ldots, F_J)'$ and $L := (L_1, \ldots, L_J)'$.

To apply the implicit-function theorem to $F(a, p) = 0$, I verify that $D_p F(a, p)$ is nonsingular. Put

$$t_j(a, \Omega) := e^{(a_j + \Omega_j)/\lambda} - 1, \qquad t(a, \Omega) := (t_1(a, \Omega), \ldots, t_J(a, \Omega))',$$

and recall the denominator

$$D(a, \Omega; p) = p_0 + \sum_{j=1}^J p_j e^{(a_j + \Omega_j)/\lambda}.$$

Differentiating the choice-probability objective gives

$$D_p F(a,p) = -\lambda \mathbb{E}_G \frac{t(a,\Omega)t(a,\Omega)'}{D(a,\Omega;p)^2}.$$

The ratios t_j/D are locally uniformly bounded by $1/p_j + 1/p_0$, which justifies differentiation under the expectation. For a nonzero $u \in \mathbb{R}^J$, the equality $u't(a,\Omega) = 0$ almost surely would imply

$$-\sum_{j=1}^{J} u_j + \sum_{j=1}^{J} u_j e^{a_j/\lambda} e^{\Omega_j/\lambda} = 0 \quad G\text{-almost surely.}$$

A2 would make every $u_j e^{a_j/\lambda}$ zero, contradicting $u \neq 0$. Thus $u't$ is nonzero with positive probability, and

$$u' D_p F(a,p) u = -\lambda \mathbb{E}_G\{(u't)^2/D^2\} < 0.$$

The matrix $D_p F(a,p)$ is therefore negative definite. The real-analytic implicit-function theorem (Krantz and Parks, 2002, Theorem 2.5.1), applied at every $(a, B_x(a))$ with $a \in \mathcal{W}_x$, makes $B_x(a)$ real analytic on $\mathcal{W}_x$.

For the inverse, $D_a L(a,p) = \nabla_a^2 H(a;p,x)$ is positive definite by the preceding lemma. The same theorem applied to $L(a,p) = 0$ makes $a^*(p;x)$ real analytic on $\mathcal{D}_J^\circ$. The local solutions agree on overlaps because $B_x(a)$ and $a^*(p;x)$ are unique at each a and p, respectively.

I now relate these inverse maps to V_x and V_x^*. For fixed q, the function $a \mapsto \Phi(q;a,x)$ is convex, because the logarithm of a sum of exponentials with nonnegative weights is convex. Its maximum over q is convex as well. The bound in Lemma B.1.8 gives

$$|V_x(a)| \leq \|a\|_\infty + \mathbb{E}_G \|\Omega\|_1 < \infty.$$

On $\mathcal{W}_x$, differentiate $V_x(a) = \Phi(q(B_x(a)); a, x)$. The derivative through $B_x(a)$ vanishes because the share first-order conditions are $F(a, B_x(a)) = 0$. The remaining derivative is

$$\nabla_a V_x(a) = \mathbb{E}_G \pi_{1:J}(a,\Omega;B_x(a)) = B_x(a),$$

where the final equality was proved in Lemma B.1.10.

Fix $p \in \mathcal{D}_J^\circ$. For every $a \in \mathbb{R}^J$,

$$p'a - V_x(a) \leq p'a - \Phi(q(p);a,x) = -H(a;p,x) \leq -H(a^*(p;x);p,x).$$

Both inequalities are equalities at $a = a^*(p;x)$ because $\bar{b}(a^*(p;x);x) = q(p)$. Taking the supremum proves the expression for $V_x^*(p)$. Equivalently,

$$V_x^*(p) = p'a^*(p;x) - V_x(a^*(p;x)).$$

The chain rule gives

$$\nabla_p V_x^*(p) = a^*(p;x) + D_p a^*(p;x)'\{p - \nabla_a V_x(a^*(p;x))\} = a^*(p;x),$$

since $\nabla_a V_x(a^*(p;x)) = B_x(a^*(p;x)) = p$.

Finally, $D_a B_x(a) = \nabla_a^2 V_x(a)$ is symmetric and positive semidefinite by convexity of V_x. Differentiating the inverse identity gives

$$D_p a^*(B_x(a);x)\, D_a B_x(a) = I_J.$$

Hence $D_a B_x(a)$ is nonsingular. A symmetric positive semidefinite matrix with no zero eigenvalue is positive definite, proving the last claim. □

I return to the binary RI model to justify the joint continuity of $(z,x) \mapsto b(z,x;\theta_0)$ used in the discussion preceding Theorem 1. Assumption ID1(e) imposes continuity only in z. The next result derives joint continuity from the RI problem, continuity of λ_0, and the existing zero-cost support condition in ID3(b).

Lemma B.1.12 (Joint continuity of the true choice probability)**.** *Suppose ID1 and ID3 hold. If $\lambda_0(x) = 0$ for some x, the distribution G_0 has no atoms. At every such x, $z \mapsto b(z,x;\theta_0)$ is continuous on $\mathbb{R}$ and*

$$\inf_{z \in \mathcal{Z}_x} b(z,x;\theta_0) = 0, \qquad \sup_{z \in \mathcal{Z}_x} b(z,x;\theta_0) = 1.$$

The map

$$\operatorname{supp}(Z,X) \longrightarrow [0,1], \qquad (z,x) \longmapsto b(z,x;\theta_0)$$

is jointly continuous. In particular, for every sequence (z_n, x_n) in this support converging to $(z_\star, x_\star)$ in the support,

$$b(z_n, x_n; \theta_0) \longrightarrow b(z_\star, x_\star; \theta_0). \tag{B.1.12}$$

Proof. First fix x with $\lambda_0(x) = 0$. The switch point is $S_x = -(\alpha_0 + \Omega)/\delta_0$, and ID3(b) gives

$$\mathbb{P}\{S_x \in \operatorname{int}(\mathcal{Z}_x)\} = 1.$$

An atom of S_x at any $s \in \operatorname{int}(\mathcal{Z}_x)$ would make the full-information probability $z \mapsto \mathbb{P}(\alpha_0+\delta_0 z+\Omega > 0)$ discontinuous at s, contrary to ID1(e). Thus S_x has no atoms. Since $\Omega = -\alpha_0 - \delta_0 S_x$ and $\delta_0 \neq 0$, G_0 has no atoms either.

Write $F_{S_x}(z) := \mathbb{P}(S_x \leq z)$. The full-information rule gives

$$b(z, x; \theta_0) = \begin{cases} F_{S_x}(z), & \delta_0 > 0, \\ 1 - F_{S_x}(z), & \delta_0 < 0, \end{cases} \qquad z \in \mathbb{R}.$$

Both expressions are continuous on $\mathbb{R}$. Since S_x lies in $\operatorname{int}(\mathcal{Z}_x)$ almost surely, F_{S_x} is zero to the left of $\mathcal{Z}_x$ and one to its right. Its continuity and its limits zero and one as $z \to -\infty$ and $z \to +\infty$ therefore give

$$\inf_{z \in \mathcal{Z}_x} F_{S_x}(z) = 0, \qquad \sup_{z \in \mathcal{Z}_x} F_{S_x}(z) = 1.$$

The same infimum and supremum hold for $b(\cdot, x; \theta_0)$ under either sign of δ_0.

I now prove joint continuity. Take a sequence (z_n, x_n) and a limit $(z_\star, x_\star)$ as in the statement. The taste-shock distribution is the same G_0 along the sequence by ID1(a). Write

$$\begin{aligned} a_n &:= \alpha_0 + \delta_0 z_n, \quad \lambda_n := \lambda_0(x_n), \quad b_n := b(z_n, x_n; \theta_0), \\ a_\star &:= \alpha_0 + \delta_0 z_\star, \quad \lambda_\star := \lambda_0(x_\star), \quad b_\star := b(z_\star, x_\star; \theta_0). \end{aligned}$$

Then $a_n \to a_\star$ and $\lambda_n \to \lambda_\star$ by ID1(d). The normalization in ID1(b) gives $\mathbb{E}|\Omega| \leq (\mathbb{E}\Omega^2)^{1/2} = 1$.

First suppose $\lambda_\star > 0$. Then $\lambda_n > 0$ eventually. For $q \in [0,1]$ and $\lambda_0(x) > 0$, define the choice-probability objective

$$J(q; z, x) := \lambda_0(x) \int \log\left\{1 - q + q e^{(\alpha_0 + \delta_0 z + \omega)/\lambda_0(x)}\right\} dG_0(\omega).$$

If $q_n \to q$, the integrand at (q_n, z_n, x_n) converges pointwise to the integrand at $(q, z_\star, x_\star)$. It also satisfies

$$\left|\lambda_n \log\{1 - q_n + q_n e^{(a_n+\omega)/\lambda_n}\}\right| \leq |a_n| + |\omega|.$$

The sequence (a_n) is bounded, so the right-hand side has an integrable bound independent of n. Dominated convergence gives $J(q_n; z_n, x_n) \to J(q; z_\star, x_\star)$.

The variance normalization makes G_0 nondegenerate, so the binary maximizer is unique by Lemma B.1.9. The same continuity-of-maximizers argument proved in Lemma B.1.8, using the compact set $[0,1]$ and the objective convergence just established, gives $b_n \to b_\star$.

Now suppose $\lambda_\star = 0$. The full-information rule in Matějka and McKay (2015, Theorem 1) gives

$$b(z, x_\star; \theta_0) = \mathbb{P}\{\alpha_0 + \delta_0 z + \Omega > 0\}.$$

The absence of atoms established at the start of the proof implies

$$\mathbb{P}\{a_\star + \Omega = 0\} = 0.$$

I first consider terms with $\lambda_n = 0$. At these terms, $b_n = \mathbb{P}\{a_n + \Omega > 0\}$. Since $a_n \to a_\star$ and the limiting payoff tie has probability zero, bounded convergence gives $b_n \to \mathbb{P}\{a_\star + \Omega > 0\} = b_\star$.

For terms with $\lambda_n > 0$, fix any trial probability $q \in (0,1)$. Outside the probability-zero limiting payoff tie, the logistic integrand converges to $\mathbf{1}\{a_\star + \omega > 0\}$. Since it lies in $[0,1]$, bounded convergence gives

$$r(\theta_0; (z_n, x_n), q) = \int \sigma\left(\ell(q) + \frac{a_n + \omega}{\lambda_n}\right) dG_0(\omega) - q \longrightarrow b_\star - q. \tag{B.1.13}$$

The derivative identity proved in Lemma B.1.1 gives

$$\partial_q J(q; z_n, x_n) = \frac{\lambda_n}{q(1-q)} r(\theta_0; (z_n, x_n), q).$$

For $q < b_\star$, the derivative is positive eventually and concavity gives $b_n > q$; for $q > b_\star$, it gives $b_n < q$. Letting these fixed trial probabilities approach $b_\star$ proves $b_n \to b_\star$. If $b_\star = 0$, only the upper bounds $b_n < q$ are needed; if $b_\star = 1$, only the lower bounds $b_n > q$ are needed. Combining the terms with zero and positive costs proves (B.1.12) and completes the proof. □

Joint continuity also determines the conditional choice probabilities at every point of the observed support. The distribution of the data initially determines these probabilities only up to sets of covariate values with probability zero. If two continuous versions differed at a support point (z, x), continuity would make them differ throughout a neighborhood of that point. Such a neighborhood has positive probability by the definition of support, contradicting almost-everywhere agreement. Thus the choice-probability function in Definition 1 is uniquely determined on the observed support.

B.2 Proof of Theorem 1

Proof. Without loss of generality, assume $\delta_0 > 0$. Lemma B.1.12 shows that, at every x with $\lambda_0(x) = 0$, the function $z \mapsto b(z, x; \theta_0)$ is continuous on $\mathbb{R}$ and its values on $\mathcal{Z}_x$ can be arbitrarily close to zero and arbitrarily close to one.

Fix $x \in \mathcal{X}$ and define from the observed probabilities

$$p_-(x) := \inf_{z \in \mathcal{Z}_x} \pi(z,x), \qquad p_+(x) := \sup_{z \in \mathcal{Z}_x} \pi(z,x).$$

If $p_-(x) = 0$ and $p_+(x) = 1$, monotonicity gives $\pi(z,x) = 0$ to the left of $\mathcal{Z}_x$ and $\pi(z,x) = 1$ to its right. This argument applies whether information costs are zero or positive.

If $p_-(x) > 0$ or $p_+(x) < 1$, then $\lambda_0(x) > 0$, because zero cost would imply $p_-(x) = 0$ and $p_+(x) = 1$. ID3(b) and continuity of $z \mapsto b(z,x;\theta_0)$ then supply a nonempty open interval $J_x \subseteq \text{int}(\mathcal{Z}_x)$ such that

$$0 < \pi(z,x) < 1 \qquad (z \in J_x).$$

Lemma B.1.1 then identifies $z \mapsto b(z,x;\theta_0)$ on $\mathbb{R}$ from its values on J_x. Thus, for every $x \in \mathcal{X}$, the function

$$\pi(\cdot,x) : \mathbb{R} \longrightarrow [0,1], \qquad z \longmapsto b(z,x;\theta_0)$$

is identified. I use this notation also for $z \notin \mathcal{Z}_x$. The function is continuous on $\mathbb{R}$: at positive cost this follows from Lemma B.1.1, and at zero cost from Lemma B.1.12.

For x with $\lambda_0(x) > 0$, define

$$I_x := \{z \in \mathbb{R} : 0 < \pi(z,x) < 1\}, \qquad \begin{aligned} h_x : I_x &\longrightarrow \mathbb{R}, \\ z &\longmapsto \alpha_0 + \delta_0 z + \lambda_0(x)\ell(\pi(z,x)). \end{aligned}$$

For $z \in I_x$, the conditional choice probability in (2.6) can be written as

$$\begin{aligned} p_{\theta_0}(\omega;z,x) &= \frac{\pi(z,x)e^{(\alpha_0+\delta_0 z+\omega)/\lambda_0(x)}}{1-\pi(z,x)+\pi(z,x)e^{(\alpha_0+\delta_0 z+\omega)/\lambda_0(x)}} \\ &= \sigma\left(\frac{h_x(z)+\omega}{\lambda_0(x)}\right). \end{aligned}$$

Let ε be standard logistic and independent of $\Omega \sim G_0$. Since σ is its CDF, averaging this equation over G_0 gives

$$\begin{aligned} \pi(z,x) &= \int \mathbb{P}\left\{\varepsilon \le \frac{h_x(z)+\omega}{\lambda_0(x)}\right\} dG_0(\omega) \\ &= \mathbb{P}\{h_x(z) \ge -\Omega + \lambda_0(x)\varepsilon\}. \end{aligned} \tag{B.2.1}$$

Lemma B.1.1 makes $h_x : I_x \to \mathbb{R}$ an increasing bijection, so the switch point in Definition 2 is well defined:

$$S_x := \begin{cases} h_x^{-1}(-\Omega + \lambda_0(x)\varepsilon), & \lambda_0(x) > 0, \\ -(\alpha_0+\Omega)/\delta_0, & \lambda_0(x) = 0. \end{cases}$$

For positive cost and $z \in I_x$, (B.2.1) gives

$$\mathbb{P}(S_x \le z) = \mathbb{P}\{-\Omega + \lambda_0(x)\varepsilon \le h_x(z)\} = \pi(z, x).$$

Since $S_x \in I_x$ almost surely, this equality extends to $z \notin I_x$, with equality at any finite endpoints by continuity. At zero cost, continuity of $\pi(\cdot, x)$ and $\delta_0 > 0$ imply

$$\mathbb{P}(\Omega = -\alpha_0 - \delta_0 z) = 0 \qquad (z \in \mathbb{R}).$$

Consequently, the definition of S_x gives

$$\mathbb{P}(S_x \le z) = \mathbb{P}(\alpha_0 + \delta_0 z + \Omega \ge 0) = \pi(z, x) \qquad (z \in \mathbb{R})$$

at zero cost as well. Thus the identified function $\pi(\cdot, x)$ is the CDF of S_x for every x.

Lemma B.1.2 gives the finite, identified moments

$$\mu(x) := \mathbb{E}S_x, \qquad V(x) := \mathrm{Var}(S_x), \qquad C(x) := \mathrm{Cov}(S_x, \ell(\pi(S_x, x))) > 0.$$

They are integrals with respect to the identified CDF $\pi(\cdot, x)$. The lemma also gives

$$\mu(x) = -\frac{\alpha_0}{\delta_0}, \qquad \text{and} \qquad \delta_0^2 V(x) + 2\delta_0 \lambda_0(x) C(x) = 1. \tag{B.2.2}$$

In particular, $V(x) \le 1/\delta_0^2$ at every x. For each positive integer n, ID2 supplies $x_n \in \mathcal{X}$ with $\lambda_0(x_n) < 1/n$. The variance bound in Lemma B.1.2 gives

$$0 \le 1 - \delta_0^2 V(x_n) \le \frac{2\pi}{\sqrt{3}} \lambda_0(x_n) < \frac{2\pi}{\sqrt{3}\, n} \longrightarrow 0.$$

Hence

$$\sup_{x \in \mathcal{X}} V(x) = \frac{1}{\delta_0^2}.$$

The supremum need not be attained. This equality and the common mean identify

$$\delta_0 = \left(\sup_{x \in \mathcal{X}} V(x) \right)^{-1/2}, \qquad \text{and} \qquad \alpha_0 = -\delta_0 \mu(x). \tag{B.2.3}$$

The sequence x_n proves the equality but is not an input to these formulas. With δ_0 identified, solving (B.2.2) gives

$$\lambda_0(x) = \frac{1 - \delta_0^2 V(x)}{2\delta_0 C(x)}, \qquad x \in \mathcal{X}. \tag{B.2.4}$$

The denominator is positive, so this formula also identifies which values of x have zero cost.

I next define a map from the identified choice-probability functions $\pi(\cdot, x)$, $x \in \mathcal{X}$, to G_0. For every x with $\lambda_0(x) > 0$, both h_x and the distribution of S_x are now identified. The equality $h_x(S_x) = -\Omega + \lambda_0(x)\varepsilon$ and independence of Ω and ε give

$$\begin{aligned}\int_{I_x} e^{-ith_x(s)}\, d\pi(s,x) &= \mathbb{E}e^{-ith_x(S_x)} \\ &= \mathbb{E}\left[e^{it\Omega}e^{-it\lambda_0(x)\varepsilon}\right] \\ &= \varphi_{G_0}(t)\varphi_\varepsilon(-\lambda_0(x)t),\end{aligned}$$

where $\varphi_{G_0}(t) := \int e^{it\omega}\, dG_0(\omega)$. The integral in the first line is with respect to the switch-point distribution with CDF $s \mapsto \pi(s,x)$. The standard logistic characteristic function φ_ε is nonzero at every real argument. Specifically, Janson (2013, Example B.13, Eq. (B.47)) gives

$$\varphi_\varepsilon(t) = \begin{cases} \pi t/\sinh(\pi t), & t \neq 0, \\ 1, & t = 0. \end{cases}$$

For every x with $\lambda_0(x) = 0$, the definition of S_x instead gives

$$\Omega = -\alpha_0 - \delta_0 S_x, \qquad \text{so} \qquad \varphi_{G_0}(t) = \int_{\mathbb{R}} e^{-it(\alpha_0+\delta_0 s)}\, d\pi(s,x).$$

In this case continuity also gives the CDF formula

$$G_0(w) = 1 - \pi\left(\frac{-\alpha_0 - w}{\delta_0}, x\right) \qquad (w \in \mathbb{R}).$$

These equalities apply at every x and do not require selecting one value of the cost shifter. To state the resulting map precisely, define from the identified functions

$$\psi_x(t) := \begin{cases} \dfrac{\int_{I_x} e^{-ith_x(s)}\, d\pi(s,x)}{\varphi_\varepsilon(-\lambda_0(x)t)}, & \lambda_0(x) > 0, \\ \displaystyle\int_{\mathbb{R}} e^{-it(\alpha_0+\delta_0 s)}\, d\pi(s,x), & \lambda_0(x) = 0. \end{cases} \tag{B.2.5}$$

All terms in (B.2.5) are determined by $\pi(\cdot, x)$ and the parameter formulas (B.2.3)–(B.2.4). The preceding equalities give $\psi_x = \varphi_{G_0}$ for every $x \in \mathcal{X}$.

Define[13]

$$\mathcal{K}\big(\{\{\pi(z,x)\}_{z\in\mathcal{Z}_x}\}_{x\in\mathcal{X}}\big) := \psi. \tag{B.2.6}$$

Since $\psi = \varphi_{G_0}$, characteristic-function uniqueness identifies G_0 (Billingsley, 1995, Theorem 26.2).

Finally, let m denote the common value of $\mu(x)$ and set $v_* := \sup_{x\in\mathcal{X}} V(x)$. Both are identified, and $0 < v_* < \infty$. The calculations above give the following explicit map in Definition 1:

$$\begin{aligned}\mathcal{T}\big(\{\{\pi(z,x)\}_{z\in\mathcal{Z}_x}\}_{x\in\mathcal{X}}\big) &:= \left(-\frac{m}{\sqrt{v_*}},\quad \frac{1}{\sqrt{v_*}},\quad \left[x \mapsto \frac{v_* - V(x)}{2\sqrt{v_*}\,C(x)}\right],\quad G_0\right)\\ &= (\alpha_0, \delta_0, \lambda_0, G_0).\end{aligned} \tag{B.2.7}$$

Every function on the right is obtained from the input probabilities by the preceding constructions. This proves the theorem. □

B.3 Proof of Theorem 2

The estimator $\widehat{\theta}_M$ approximately minimizes $\theta \mapsto \widehat{Q}_M(\theta)$, which uses estimated shares and can be discontinuous when $\lambda(x) = 0$. Fix $w := (z,x)$ and the payoff parameters (α, δ, G), and write $r_t(w,q)$ for the residual when the unit cost at x is $t \geq 0$. At an input probability of zero,

$$r_t(w,0) = \begin{cases} 0, & t > 0,\\ \mathbb{P}_G\{\alpha + \delta z + \Omega > 0\}, & t = 0.\end{cases}$$

If this full-information probability is positive, then

$$\lim_{t\downarrow 0} r_t(w,0)^2 = 0 < r_0(w,0)^2. \tag{B.3.1}$$

Thus the market's squared residual is zero at positive cost but positive at zero cost. Continuity of the true choice-probability function does not prevent this jump in a candidate's squared residual.

I compare $\widehat{Q}_M$ with the two criteria formed using population choice probabilities $s_m := s(W_m)$:

$$\begin{aligned} Q_M(\theta) &:= \frac{1}{M}\sum_m r(\theta; W_m, s_m)^2, \qquad \text{and} \qquad Q_M^\circ(\theta) := \frac{1}{M}\sum_m f_\theta(W_m),\\ &\text{where} \qquad f_\theta(w) := \mathbf{1}\{0 < s(w) < 1\} r(\theta; w, s(w))^2,\end{aligned} \tag{B.3.2}$$

[13] Let Π denote the input probability family in Definition 1. Applying (B.2.5) at each $x \in \mathcal{X}$ defines the map $\Pi \mapsto \bigcup_x \{\psi_x\}$, where ψ_x denotes the function $t \mapsto \psi_x(t)$. The preceding calculation shows that this union is $\{\varphi_{G_0}\}$. Composing this map with $\{f\} \mapsto f$ defines $\mathcal{K}$, with output denoted by ψ.

and the population counterpart of Q_M° is

$$Q^\circ(\theta) := \mathbb{E} f_\theta(W).$$

The criterion $Q_M^\circ(\theta)$ excludes markets with $s_m \in \{0,1\}$ and satisfies $0 \le Q_M^\circ(\theta) \le Q_M(\theta)$ for every $\theta \in \Theta$. Following Chen (2007, Section 3.1, proof of Theorem 3.1 and Remark 3.1(4)), I show that $Q_M^\circ(\widehat{\theta}_M) \to_p 0$ and that $Q^\circ(\theta)$ is bounded away from zero when $d(\theta, \theta_0)$ is bounded away from zero. The following lemmas establish continuity and uniform convergence for Q_M°, the unique zero of Q°, and $Q_M(\theta_M^*) \to_p 0$ for the sieve approximant.

E2(c) applies to the entire compact parameter space Θ, including limits of parameter sequences, rather than only to the finite-dimensional sieves. [14] It allows $\lambda(x) > 0$ for every $x \in \mathcal{X}$ at every $\theta \in \Theta$: these costs may approach zero as x varies, as ID2 requires, without having a positive lower bound uniform in x.[15]

The sieve approximations must also satisfy E2(c). Discrete taste-shock distributions are allowed when that condition is satisfied through positive information costs or full conditional support. If a distribution used at zero cost must generate a real-analytic choice-probability function, finite Gaussian mixtures provide one possible approximation.

Lemma B.3.1 (Continuity and uniform convergence of the auxiliary criterion)**.** *Under the sampling conditions, ID3(a), and E2, Q° is continuous on Θ and*

$$\sup_{\theta \in \Theta} |Q_M^\circ(\theta) - Q^\circ(\theta)| \to_p 0.$$

Proof. Fix $\theta = (\alpha, \delta, \lambda, G) \in \Theta$ and a sequence $\vartheta_n \to \theta$ in the metric d. On $\{0 < s(W) < 1\}$, Lemma B.1.3 gives continuity of $\vartheta \mapsto r(\vartheta; W, s(W))$ at θ, except possibly on

$$\{\lambda(X) = 0,\ G\{-\alpha - \delta Z\} > 0\}.$$

This event means that, at zero information cost, the candidate distribution puts positive probability on equal payoffs for the two alternatives. Since $\delta \neq 0$, ID3(a) and Lemma B.1.3 give it probability zero. On $\{s(W) \in \{0,1\}\}$, $f_\vartheta(W) = 0$ for every ϑ, so continuity holds automatically. Therefore

$$f_{\vartheta_n}(W) \to f_\theta(W) \quad \text{almost surely,} \qquad 0 \le f_\vartheta(W) \le 1.$$

[14] One way to obtain compactness is to use a uniformly bounded, equicontinuous set of cost functions and a weakly closed set of taste-shock distributions with a common moment bound of order greater than two, provided the restrictions in E2(b)–(c) are preserved under limits.

[15] If $\mathcal{X}$ is compact, continuity and strict positivity would imply a positive minimum, so this everywhere-positive special case would be incompatible with ID2.

Bounded convergence yields $Q^\circ(\vartheta_n) \to Q^\circ(\theta)$. This proves continuity of Q° on the compact parameter space.

For each fixed θ, the preceding argument gives almost-sure continuity of $\vartheta \mapsto f_\vartheta(W)$ at θ. The bound $0 \le f_\vartheta(W) \le 1$ gives an integrable envelope, the market characteristics are i.i.d., and E2 makes Θ compact under d. Lemma 2.4 of Newey and McFadden (1994) therefore gives

$$\sup_{\theta\in\Theta} |Q_M^\circ(\theta) - Q^\circ(\theta)| \to_p 0.$$

The result uses continuity at each fixed parameter with probability one; it does not require $\vartheta \mapsto r(\vartheta; w, s(w))^2$ to be continuous at every $\theta \in \Theta$ for every w. As usual, the supremum may be interpreted in outer probability if needed, as in the qualification following that lemma.

□

The next lemma shows that $Q^\circ(\theta) = 0$ identifies θ_0. Its proof must account for the exclusion of markets with $s(W) \in \{0, 1\}$ and for candidates satisfying $\lambda(x) = 0$.

Lemma B.3.2 (The auxiliary criterion has a unique zero)**.** *Under ID1–ID3 and E2, $Q^\circ(\theta) = 0$ for $\theta \in \Theta$ implies $\theta = \theta_0$.*

Proof. Suppose $Q^\circ(\theta) = 0$. Since its integrand is nonnegative,

$$r(\theta; W, s(W)) = 0 \quad \text{almost surely on } \{0 < s(W) < 1\}.$$

To make the conditional statement precise, define

$$g_\theta(x) := \mathbb{E}\big[\mathbf{1}\{0 < s(W) < 1\} r(\theta; W, s(W))^2 \mid X = x\big].$$

The tower property gives $\mathbb{E} g_\theta(X) = Q^\circ(\theta) = 0$. Because $g_\theta(X) \ge 0$, it follows that $g_\theta(X) = 0$ almost surely. Hence the set $\mathcal{B} := \{x \in \mathcal{X} : g_\theta(x) = 0\}$ has probability one. For each $x \in \mathcal{B}$, the conditional nonnegative integrand is zero almost surely, giving the stated residual equality for almost every z with $0 < s(z, x) < 1$ under $Z \mid X = x$.

At a point with $0 < s(z, x) < 1$ and $\lambda(x) > 0$, $r(\theta; (z, x), s(z, x)) = 0$ is the first-order condition for the strictly concave choice-probability objective. It implies $b(z, x; \theta) = s(z, x)$. When $\lambda(x) = 0$, the same conclusion follows from

$$r(\theta; (z, x), s(z, x)) = \mathbb{P}_G\{\alpha + \delta z + \Omega > 0\} - s(z, x).$$

For $x \in \mathcal{B}$, the equality $b(z, x; \theta) = s(z, x)$ therefore holds at a dense set of points in every observed open interval with $0 < s(z, x) < 1$: every nonempty open subinterval has positive conditional probability under $Z \mid X = x$.

At zero information cost, global real analyticity of $z \mapsto b(z,x;\theta)$ implies $0 < b(z,x;\theta) < 1$ for every $z \in \mathbb{R}$. To see this, first suppose $\delta > 0$. Then $b(z,x;\theta) = \mathbb{P}_G\{\alpha + \delta z + \Omega > 0\}$ is nondecreasing, with limits zero as $z \to -\infty$ and one as $z \to +\infty$. If it equals zero at some $z_0 \in \mathbb{R}$, monotonicity makes it zero on $(-\infty, z_0)$. The identity theorem (Krantz and Parks, 2002, Corollary 1.2.6) then makes it zero on $\mathbb{R}$, contradicting its limit at $+\infty$. If it equals one at z_0, the same argument on (z_0, ∞) contradicts its limit at $-\infty$. For $\delta < 0$, the directions of monotonicity and the two limits reverse, and the same argument applies. E2(b) excludes $\delta = 0$.

Replace Z by $-Z$ if necessary so that $\delta_0 > 0$. Under E2, $\mathcal{Z}_x = \mathbb{R}$ whenever $\lambda_0(x) = 0$. To see this, E2(a) puts θ_0 in Θ. If $\lambda_0(x) = 0$ and $\mathcal{Z}_x \neq \mathbb{R}$, E2(c) would make $z \mapsto b(z,x;\theta_0)$ real analytic on $\mathbb{R}$. The preceding argument would then give $0 < b(z,x;\theta_0) < 1$ for every $z \in \mathbb{R}$. By ID3(b), $-\Omega$ lies in $\alpha_0 + \delta_0 \operatorname{int}(\mathcal{Z}_x)$ almost surely, so $b(z,x;\theta_0)$ equals zero or one at every z outside $\mathcal{Z}_x$. This is a contradiction. Thus $\lambda_0(x) = 0$ implies $\mathcal{Z}_x = \mathbb{R}$ under E2.

Fix $x \in \mathcal{B}$. There are observed values $z_1 < z_2$ at which $b(z_i,x;\theta) = s(z_i,x)$ for $i = 1,2$, with $0 < s(z_1,x) < s(z_2,x) < 1$. When $\lambda_0(x) > 0$, Lemma B.1.1 and ID3(b) supply an observed interval where $s(\cdot,x)$ is strictly increasing. When $\lambda_0(x) = 0$, $\mathcal{Z}_x = \mathbb{R}$ by E2, and the continuous, nonconstant full-information CDF also supplies values z_1, z_2 with these properties. If $\delta < 0$, $b(\cdot,x;\theta)$ would be nonincreasing and could not equal s at both z_1 and z_2. Thus $\delta > 0$ as well.

Now take any open interval contained in $\operatorname{int}(\mathcal{Z}_x)$ on which $0 < s(z,x) < 1$. For any point z in the interval, choose observed values $z_n^- \uparrow z$ and $z_n^+ \downarrow z$ at which $b(\cdot,x;\theta)$ and $s(\cdot,x)$ agree. Since $\delta > 0$ and $\delta_0 > 0$, monotonicity gives

$$s(z_n^-,x) = b(z_n^-,x;\theta) \leq b(z,x;\theta) \leq b(z_n^+,x;\theta) = s(z_n^+,x).$$

The true choice-probability function $s(\cdot,x)$ is continuous, so both outer terms tend to $s(z,x)$. Therefore $b(z,x;\theta) = s(z,x)$ at every point of the interval. This argument also rules out a jump in $z \mapsto b(z,x;\theta)$ at such a point when $\lambda(x) = 0$.

I next show $b(z,x;\theta) = s(z,x)$ for every $z \in \mathbb{R}$, at each $x \in \mathcal{B}$. Define the interval

$$I_x := \{z \in \mathbb{R} : 0 < s(z,x) < 1\}.$$

The argument depends on the support $\mathcal{Z}_x$ and the candidate cost $\lambda(x)$.

If $\mathcal{Z}_x = \mathbb{R}$, the preceding argument gives equality on all of I_x. At a finite left boundary of I_x, $s(z,x)$ tends to zero; at a finite right boundary it tends to one. Monotonicity and $0 \leq b(z,x;\theta) \leq 1$ then force equality also where $s(z,x) \in \{0,1\}$. This case includes every x with $\lambda_0(x) = 0$, by E2.

If $\mathcal{Z}_x \neq \mathbb{R}$, E2 implies $\lambda_0(x) > 0$. ID3(b) then provides a nonempty open interval in $\mathcal{Z}_x$ on which $0 < s(z,x) < 1$. When $\lambda(x) > 0$ also, Lemma B.1.1 gives $b(z,x;\theta) = b(z,x;\theta_0)$ for every $z \in \mathbb{R}$.

It remains to consider $\mathcal{Z}_x \neq \mathbb{R}$ and $\lambda(x) = 0$. Because $\lambda(x) = 0$ and $\mathcal{Z}_x \neq \mathbb{R}$, E2(c) makes $z \mapsto b(z,x;\theta)$ real analytic on $\mathbb{R}$. The preceding monotonicity argument then gives $0 < b(z,x;\theta) < 1$ for every $z \in \mathbb{R}$. The functions $z \mapsto b(z,x;\theta)$ and $z \mapsto s(z,x)$ are analytic on I_x, and they agree on an observed open subinterval. The identity theorem (Krantz and Parks, 2002, Corollary 1.2.6) gives agreement throughout I_x. If its left endpoint a were finite, continuity would imply

$$b(a,x;\theta) = \lim_{z \downarrow a} b(z,x;\theta_0) = 0,$$

contradicting $b(a,x;\theta) > 0$. A finite right endpoint similarly contradicts $b(z,x;\theta) < 1$ for every $z \in \mathbb{R}$. Thus $I_x = \mathbb{R}$, and $b(z,x;\theta) = b(z,x;\theta_0)$ for every $z \in \mathbb{R}$ in this case too.

The choice-probability functions agree, so I write their common value as

$$F_x(z) := b(z,x;\theta) = b(z,x;\theta_0) \quad (z \in \mathbb{R},\ x \in \mathcal{B}).$$

The function F_x is continuous. When $\lambda_0(x) > 0$, continuity follows from Lemma B.1.1; when $\lambda_0(x) = 0$, it follows from ID1(e) and ID3(b). Hence the switch points constructed in Lemma B.1.2 for θ and θ_0 have the same continuous CDF F_x. In particular, their means, variances, and covariances

$$\mu(x) := \mathbb{E}S_x, \qquad V(x) := \mathrm{Var}(S_x), \qquad C(x) := \mathrm{Cov}(S_x, \ell(F_x(S_x)))$$

are common to both parameter values on $\mathcal{B}$. The lemma's zero-cost continuity condition therefore holds for both θ and θ_0 wherever it is needed.

To apply the moment identities on $\mathcal{B}$, I need costs to approach zero on this set. Let $\lambda^\dagger \in \{\lambda, \lambda_0\}$. For every $\varepsilon > 0$, ID2 gives $x_0 \in \mathcal{X}$ with $\lambda^\dagger(x_0) < \varepsilon/2$. Continuity of $\lambda^\dagger$ gives a relative open neighborhood U of x_0 on which $\lambda^\dagger(x) < \varepsilon$. Because $x_0 \in \mathrm{supp}(X)$,

$$\mathbb{P}(X \in U) > 0, \qquad \mathbb{P}(X \in U \cap \mathcal{B}) = \mathbb{P}(X \in U) > 0.$$

Thus $\mathcal{B}$ contains a point x with $\lambda^\dagger(x) < \varepsilon$. Applying this argument to λ and λ_0 proves

$$\inf_{x \in \mathcal{B}} \lambda(x) = \inf_{x \in \mathcal{B}} \lambda_0(x) = 0.$$

Lemma B.1.2, applied separately at θ and θ_0, now gives

$$\sup_{x\in\mathcal{B}} V(x) = \frac{1}{\delta^2} = \frac{1}{\delta_0^2}.$$

Since $\delta > 0$ and $\delta_0 > 0$, this equality yields $\delta = \delta_0$. The common mean gives $\alpha = \alpha_0$, and the covariance identity gives

$$\lambda(x) = \frac{1-\delta_0^2 V(x)}{2\delta_0 C(x)} = \lambda_0(x) \qquad (x \in \mathcal{B}).$$

If there is $x \in \mathcal{B}$ with $\lambda(x) = \lambda_0(x) > 0$, the deconvolution argument in Section B.2 gives $G = G_0$. If instead $\lambda(x) = \lambda_0(x) = 0$ for every $x \in \mathcal{B}$, the common switch-point distribution gives $G = G_0$ directly through $\Omega \overset{d}{=} -\alpha_0 - \delta_0 S_x$.

Finally, λ and λ_0 agree on the full-measure set $\mathcal{B} \subseteq \mathrm{supp}(X)$, hence everywhere on that support. If they differed at some point, continuity would give a neighborhood with a nonzero difference and positive support probability, contradicting their almost-everywhere equality. Consequently $\theta = \theta_0$, and Q° has a unique zero. □

To compare $\widehat{\theta}_M$ with the feasible sieve approximant $\theta_M^* \in \Theta_M$, I also need $Q_M(\theta_M^*) \to_p 0$. This criterion includes markets with $s_m \in \{0,1\}$.

Lemma B.3.3 (Vanishing loss at the sieve approximant)**.** *Under the sampling conditions, ID1–ID3, and E3,*

$$\mathbb{E}[r(\theta_M^*; W, s(W))^2] \to 0, \qquad Q_M(\theta_M^*) \to_p 0.$$

Proof. Write

$$\theta_M^* = (\alpha_M^*, \delta_M^*, \lambda_M^*, G_M^*).$$

On $\{0 < s(W) < 1\}$, E3 and Lemma B.1.3 give

$$r(\theta_M^*; W, s(W)) \longrightarrow 0.$$

For $w := (z, x)$ with $s(w) \in \{0,1\}$, consider $\lambda_0(x) > 0$ and $\lambda_0(x) = 0$ separately. If $\lambda_0(x) > 0$, $\lambda_M^*(x) > 0$ eventually, and $r(\theta_M^*; w, s(w)) = 0$ by (3.1). If $\lambda_0(x) = 0$, terms with $\lambda_M^*(x) > 0$ also have $r(\theta_M^*; w, s(w)) = 0$. Along terms with $\lambda_M^*(x) = 0$, use the indicator-convergence argument in the proof of Lemma B.1.3. It uses convergence of $(\alpha_M^*, \delta_M^*, G_M^*)$ and $G_0\{-\alpha_0 - \delta_0 z\} = 0$. This part of the argument compares the indicators of $\alpha_M^* + \delta_M^* z + \Omega > 0$ and $\alpha_0 + \delta_0 z + \Omega > 0$ and therefore also applies when $s(w) \in \{0,1\}$. It gives

$$\Psi(\theta_M^*; w, s(w)) = \mathbb{P}_{G_M^*}\{\alpha_M^* + \delta_M^* z + \Omega > 0\} \longrightarrow \mathbb{P}\{\alpha_0 + \delta_0 z + \Omega > 0\} = s(w).$$

Subtracting $s(w)$ gives $r(\theta_M^*; w, s(w)) \to 0$ along these terms as well. By ID1(c), ID3(a), and Lemma B.1.3,

$$\mathbb{P}\{\lambda_0(X) = 0,\ G_0\{-\alpha_0 - \delta_0 Z\} > 0\} = 0.$$

Thus the limits just proved hold almost surely over W.

Every population choice probability satisfies either $0 < s(W) < 1$ or $s(W) \in \{0, 1\}$. The preceding arguments therefore establish

$$r(\theta_M^*; W, s(W)) \longrightarrow 0 \quad \text{almost surely.}$$

Lemma B.1.3 also gives

$$|r(\theta_M^*; W, s(W))| \leq 1.$$

Since θ_M^* is deterministic and every W_m has the same distribution as W, bounded convergence yields

$$\mathbb{E}Q_M(\theta_M^*) = \mathbb{E}[r(\theta_M^*; W, s(W))^2] \longrightarrow 0.$$

Markov's inequality therefore gives $Q_M(\theta_M^*) \to_p 0$. This argument evaluates Q_M only at θ_M^*; it uses no uniform convergence of $\theta \mapsto Q_M(\theta)$. □

Proof of Theorem 2. The preceding lemmas allow me to show that $Q_M^\circ(\widehat{\theta}_M) \to_p 0$, then use the unique minimum of Q° to obtain $\widehat{\theta}_M \to_p \theta_0$ in the metric d. The inequalities below connect the criterion $\widehat{Q}_M$ minimized by the estimator to Q_M°. Let

$$\zeta_M := \sup_{\theta \in \Theta} |\widehat{Q}_M(\theta) - Q_M(\theta)|.$$

Lemma B.1.4 gives $\zeta_M = o_p(1)$. Since $\theta_M^* \in \Theta_M$, minimization up to the allowed error $\epsilon_M = o_p(1)$ gives

$$\begin{aligned} 0 \leq Q_M^\circ(\widehat{\theta}_M) &\leq Q_M(\widehat{\theta}_M) \\ &\leq \widehat{Q}_M(\widehat{\theta}_M) + \zeta_M \\ &\leq \widehat{Q}_M(\theta_M^*) + \zeta_M + \epsilon_M \\ &\leq Q_M(\theta_M^*) + 2\zeta_M + \epsilon_M = o_p(1). \end{aligned} \tag{B.3.3}$$

The two copies of ζ_M arise from replacing $\widehat{s}_m$ by s_m at $\widehat{\theta}_M$ and at θ_M^*. Uniform convergence of Q_M° to Q° then gives

$$0 \leq Q^\circ(\widehat{\theta}_M) \leq Q_M^\circ(\widehat{\theta}_M) + \sup_{\theta \in \Theta} |Q_M^\circ(\theta) - Q^\circ(\theta)| \to_p 0.$$

For $\varepsilon > 0$, let $K_\varepsilon := \{\theta \in \Theta : d(\theta, \theta_0) \geq \varepsilon\}$. If this set is empty, the desired probability is zero. Otherwise, the extreme value theorem (Munkres, 2000, Theorem 27.4), applied to the compact set K_ε, and the unique zero imply

$$\kappa_\varepsilon := \min_{\theta \in K_\varepsilon} Q^\circ(\theta) > 0.$$

Therefore

$$\mathbb{P}\{d(\widehat{\theta}_M, \theta_0) \geq \varepsilon\} \leq \mathbb{P}\{Q^\circ(\widehat{\theta}_M) \geq \kappa_\varepsilon\} \longrightarrow 0.$$

This is $d(\widehat{\theta}_M, \theta_0) \to_p 0$. □

The same comparison proves consistency with bounded positive weights. The weighted criterion is

$$\widehat{Q}_M^w(\theta) := \frac{1}{M} \sum_{m=1}^{M} w(\theta; W_m, \widehat{s}_m) r(\theta; W_m, \widehat{s}_m)^2. \tag{B.3.4}$$

Let $\widehat{\theta}_M^w$ approximately minimize (B.3.4), with error $\epsilon_M^w = o_p(1)$, and suppose $0 < c \leq w(\theta; w, q) \leq C < \infty$ uniformly over (θ, w, q). The weight bounds give

$$c\widehat{Q}_M(\widehat{\theta}_M^w) \leq \widehat{Q}_M^w(\widehat{\theta}_M^w) \leq \widehat{Q}_M^w(\theta_M^*) + \epsilon_M^w \leq C\widehat{Q}_M(\theta_M^*) + \epsilon_M^w.$$

Share substitution and Lemma B.3.3 imply

$$\widehat{Q}_M(\widehat{\theta}_M^w) \leq \frac{C}{c}\{Q_M(\theta_M^*) + \zeta_M\} + \frac{\epsilon_M^w}{c} \to_p 0.$$

Hence

$$Q_M^\circ(\widehat{\theta}_M^w) \leq \widehat{Q}_M(\widehat{\theta}_M^w) + \zeta_M \to_p 0.$$

Lemma B.3.1 then gives $Q^\circ(\widehat{\theta}_M^w) \to_p 0$. For $\varepsilon > 0$ with $K_\varepsilon \neq \varnothing$,

$$\mathbb{P}\{d(\widehat{\theta}_M^w, \theta_0) \geq \varepsilon\} \leq \mathbb{P}\{Q^\circ(\widehat{\theta}_M^w) \geq \kappa_\varepsilon\} \longrightarrow 0.$$

If $K_\varepsilon = \varnothing$, the probability on the left is zero.

For the estimator with multiple alternatives in Section 3.2.1, bound the change in each squared residual when population probabilities are replaced by market shares. Summing these J bounds controls the change in the criterion when J is fixed. The auxiliary criterion includes only markets where every alternative has positive population probability. If its summand is almost surely continuous in the parameter, the uniform law of large numbers gives uniform convergence on the compact parameter space. Provided identification gives a unique zero, the preceding approximate-minimization argument proves consistency.

B.4 Proof of Theorem 3

I first show that the test accepts the deterministic sieve approximant θ_M^* with probability approaching $1-\gamma$. I then use F5 to show that the reported interval covers $\phi(\theta_0)$ with at least this limiting probability. The first part requires three comparisons. Trimming with estimated probabilities must retain the same markets as trimming with population probabilities, with probability approaching one. A centered and scaled statistic that uses population probabilities for trimming and standardization must have a normal limit. Finally, replacing those population quantities by their estimates must have a negligible effect.

In the sample with M markets, N_M is the common within-market sample size in F1. Write $w \coloneqq (z,x)$, $s(w) \coloneqq \pi(z,x)$, and $s_m \coloneqq s(W_m)$ for population choice probabilities, and $\widehat{s}_m$ for the observed market shares. Fix the trimming thresholds $\eta \in (0,1/2)$ and $a_* > 0$. The sigma-field

$$\mathcal{F}_M \coloneqq \sigma(W_1, \ldots, W_M)$$

contains the observed market characteristics. Conditional on $\mathcal{F}_M$, the population choice probabilities are fixed and the sampled choices are independent across markets. In particular, $N_M \widehat{s}_m \mid \mathcal{F}_M$ is binomial with success probability s_m.

For $q \in (0,1)$, write $A_m(\theta, q) \coloneqq \partial_q r(\theta; W_m, q)$. For $0 < s(w) < 1$, define

$$A_M(w) \coloneqq \partial_q r(\theta_M^*; w, s(w)), \qquad \text{and} \qquad \mathcal{R}_M \coloneqq \{w : \eta \le s(w) \le 1-\eta,\ |A_M(w)| \ge a_*\}.$$

The set $\mathcal{R}_M$ contains the covariate values retained using population choice probabilities. Their observed indices and number are

$$\mathcal{A}_M \coloneqq \{m \le M : W_m \in \mathcal{R}_M\}, \qquad k_M \coloneqq |\mathcal{A}_M|.$$

I also write

$$\widehat{\mathcal{A}}_M \coloneqq \widehat{\mathcal{A}}_M(\theta_M^*), \qquad \widehat{k}_M \coloneqq |\widehat{\mathcal{A}}_M|,$$

for the retained indices and number obtained from estimated shares when testing θ_M^*. Thus $\mathcal{A}_M$ depends on market characteristics and population choice probabilities, whereas $\widehat{\mathcal{A}}_M$ also depends on share-estimation error. For the proof only, put $d_M \coloneqq \max(k_M, 1)$ so that the denominators $\sqrt{d_M}$ and $\sqrt{2d_M}$ remain positive. The implemented test continues to use $\widehat{k}_M$, and accepts when that count is zero.

For each $m \in \mathcal{A}_M$, define

$$\xi_{m,M} \coloneqq \frac{\sqrt{N_M}(\widehat{s}_m - s_m)}{\sqrt{s_m(1-s_m)}}, \qquad \beta_{m,M} \coloneqq \frac{\sqrt{N_M}\, r(\theta_M^*; W_m, s_m)}{A_M(W_m)\sqrt{s_m(1-s_m)}}. \tag{B.4.1}$$

The variable $\xi_{m,M}$ rescales the share error $\widehat{s}_m - s_m$. Conditional on $\mathcal{F}_M$, it has mean zero and variance one. The quantity $\beta_{m,M}$ rescales $r(\theta_M^*; W_m, s_m)$. It is fixed conditional on $\mathcal{F}_M$.

I will compare the implemented statistic with

$$T_M^\circ := \sum_{m \in \mathcal{A}_M} (\xi_{m,M} + \beta_{m,M})^2.$$

In T_M°, the population choice probabilities s_m determine both $\mathcal{A}_M$ and the denominators in $\xi_{m,M}$ and $\beta_{m,M}$. The sampled shares $\widehat{s}_m$ still enter through $\xi_{m,M}$.

Lemma B.4.1 (A positive fraction of markets is retained)**.** *Under E3, ID1(c), ID3(a), and F2, there is $p_* > 0$ such that*

$$\mathbb{P}\{k_M \geq p_* M/2\} \to 1, \qquad d_M \to_p \infty, \qquad M/d_M = O_p(1).$$

Proof. Recall that

$$A_M(w) := \partial_q r(\theta_M^*; w, s(w)), \qquad A_0(w) := \partial_q r(\theta_0; w, s(w)).$$

E3 and Lemma B.1.3 imply $A_M(W) \to A_0(W)$ almost surely whenever $0 < s(W) < 1$, except possibly on

$$\{\lambda_0(X) = 0,\ G_0\{-\alpha_0 - \delta_0 Z\} > 0\}.$$

Lemma B.1.3 gives this event probability zero: ID1(c) makes $\delta_0 \neq 0$, and the continuous conditional distribution of Z in ID3(a) excludes the countably many values at which G_0 puts positive probability on a payoff tie. Thus $A_M(W) \to A_0(W)$ almost surely on $\{0 < s(W) < 1\}$.

Define the event that both trimming inequalities hold strictly:

$$\mathcal{B} := \{\eta < s(W) < 1 - \eta,\ |A_0(W)| > a_*\}.$$

Its probability $p_0 := \mathbb{P}(\mathcal{B})$ is positive by F2. On the event $\mathcal{B}$, the population choice probability already satisfies the population trimming inequalities, and derivative convergence gives $|A_M(W)| \geq a_*$ for all sufficiently large M. Consequently,

$$\liminf_{M \to \infty} \mathbf{1}\{W \in \mathcal{R}_M\} \geq \mathbf{1}\{\mathcal{B}\} \quad \text{almost surely.}$$

Fatou's inequality shows that the probability of satisfying the population trimming conditions stays bounded away from zero:

$$\begin{aligned}\liminf_{M\to\infty}\mathbb{P}\{W\in\mathcal{R}_M\} &= \liminf_{M\to\infty} p_M \\ &= \liminf_{M\to\infty}\mathbb{E}\mathbf{1}\{W\in\mathcal{R}_M\} \\ &\geq \mathbb{E}\liminf_{M\to\infty}\mathbf{1}\{W\in\mathcal{R}_M\} \\ &\geq p_0 > 0,\end{aligned} \tag{B.4.2}$$

where $p_M := \mathbb{P}(W\in\mathcal{R}_M)$. Choose $p_* > 0$ with $p_M \geq p_*$ for all sufficiently large M.

The indicators $\mathbf{1}\{W_m\in\mathcal{R}_M\}$ are i.i.d. within each sample because θ_M^* and $\mathcal{R}_M$ are deterministic and $W_1,\ldots,W_M$ are i.i.d. Therefore

$$\mathbb{E}(k_M/M) = p_M, \qquad \mathrm{Var}(k_M/M) = \frac{p_M(1-p_M)}{M} \leq \frac{1}{4M}.$$

For large M, Chebyshev's inequality implies

$$\mathbb{P}\{k_M < p_*M/2\} \leq \mathbb{P}\{|k_M/M - p_M| > p_*/2\} \leq \frac{1}{p_*^2 M} \longrightarrow 0.$$

Hence the number of retained markets diverges with probability approaching one. In particular,

$$\mathbb{P}(k_M = d_M)\to 1, \qquad d_M \to_p \infty, \qquad M/d_M = O_p(1). \qquad \square$$

For $m\in\mathcal{A}_M$, the population choice probabilities s_m and derivatives $A_M(W_m)$ satisfy $s_m \in [\eta, 1-\eta]$ and $|A_M(W_m)|\geq a_*$. F4 bounds $r(\theta_M^*; W_m, s_m)$ and hence its standardized value $\beta_{m,M}$.

Lemma B.4.2 (The standardized sieve bias is negligible)**.** *Let*

$$\nu_M := \sum_{m\in\mathcal{A}_M}\beta_{m,M}^2,$$

with empty sums and maxima set to zero. Under F4,

$$\max_{m\in\mathcal{A}_M}|\beta_{m,M}|\to 0, \qquad \textit{and} \qquad \nu_M = o(\sqrt{M}).$$

Together with Lemma B.4.1, $\nu_M/\sqrt{d_M}\to_p 0$.

Proof. On $\mathcal{A}_M$, the trimming rules give

$$s_m(1-s_m)\geq\eta(1-\eta), \qquad |A_M(W_m)|\geq a_*.$$

With $\rho_M = \sup_{w\in\mathcal{R}_M} |r(\theta_M^*; w, s(w))|$ from F4, these bounds give

$$|\beta_{m,M}| \le \frac{\sqrt{N_M}\rho_M}{a_*\sqrt{\eta(1-\eta)}}.$$

With constants depending only on the fixed trimming bounds,

$$\max_{m\in\mathcal{A}_M} |\beta_{m,M}| \le C\sqrt{N_M}\rho_M \longrightarrow 0, \qquad \nu_M := \sum_{m\in\mathcal{A}_M} \beta_{m,M}^2 \le CMN_M\rho_M^2 = o(\sqrt{M}). \tag{B.4.3}$$

To verify both rates, F4 states $M^{1/4}\sqrt{N_M}\rho_M \to 0$, so

$$\begin{aligned} &\sqrt{N_M}\rho_M = M^{-1/4}\{M^{1/4}\sqrt{N_M}\rho_M\} \to 0, \\ \text{and} \qquad &\frac{\nu_M}{\sqrt{M}} \le C\sqrt{M}N_M\rho_M^2 = C\{M^{1/4}\sqrt{N_M}\rho_M\}^2 \to 0. \end{aligned}$$

Combining the last bound with Lemma B.4.1 also yields $\nu_M/\sqrt{d_M} \to_p 0$. □

Before studying the statistic's distribution, I show that estimating shares does not change which markets enter it asymptotically.

Lemma B.4.3 (Agreement of population and sample trimming)**.** *Under F1 and F3,*

$$\mathbb{P}\{\widehat{\mathcal{A}}_M \ne \mathcal{A}_M\} \to 0.$$

Proof. Fix a market m and write $e_{m,M} := \widehat{s}_m - s_m$ for its share error. The sample and population trimming rules can disagree for two reasons. First, $\widehat{s}_m$ and s_m can lie on different sides of either η or $1-\eta$. Such a crossing requires

$$\min\{|s_m - \eta|,\ |s_m - (1-\eta)|\} \le |e_{m,M}|.$$

Second, if both $\widehat{s}_m$ and s_m lie in $[\eta, 1-\eta]$, the absolute residual derivative can cross a_*. The uniform Lipschitz bound in Lemma B.1.3 gives

$$\left||A_m(\theta_M^*, \widehat{s}_m)| - |A_M(W_m)|\right| \le K_\eta |e_{m,M}|.$$

A derivative crossing therefore requires

$$\left||A_M(W_m)| - a_*\right| \le K_\eta |e_{m,M}|.$$

For w with $0 < s(w) < 1$, define its distance to the cutoffs by

$$\Delta_M(w) := \min\{|s(w) - \eta|,\ |s(w) - (1-\eta)|, \big||A_M(w)| - a_*\big|\}.$$

When $s(w) \in \{0,1\}$, set $\Delta_M(w) := \eta$. This is the quantity in F3. It is the smallest distance to any trimming cutoff. For a fixed $K \geq \max(1, K_\eta)$, disagreement between the sample and population trimming rules implies

$$\Delta_M(W_m) \leq K|e_{m,M}|.$$

When $s_m \in \{0,1\}$, the observed share equals s_m almost surely and both trimming rules discard the market, so the rules cannot disagree.

Applying Hoeffding (1963, Theorem 2) to the independent Bernoulli choices conditional on $\mathcal{F}_M$, and using a union bound for the two tails, gives

$$\mathbb{P}\{|e_{m,M}| \geq u \mid \mathcal{F}_M\} \leq 2e^{-2N_M u^2}.$$

Substituting $u = \Delta_M(W_m)/K$ and then taking expectations shows

$$\mathbb{P}(\text{the trimming rules disagree for market } m) \leq 2\mathbb{E}e^{-cN_M \Delta_M(W)^2}, \qquad \text{where} \qquad c := 2/K^2.$$

I next bound this expectation using F3. For every nonnegative number u,

$$e^{-cN_M u^2} = \int_u^\infty 2cN_M t e^{-cN_M t^2}\, dt.$$

Taking expectations and interchanging the nonnegative integrals yields

$$\begin{aligned}
\mathbb{E}e^{-cN_M \Delta_M(W)^2} &= \int_0^\infty 2cN_M t e^{-cN_M t^2} \mathbb{P}\{\Delta_M(W) \leq t\}\, dt \\
&\leq 2cCN_M \int_0^{t_0} t^2 e^{-cN_M t^2}\, dt + e^{-cN_M t_0^2} \\
&\leq \frac{2cC}{\sqrt{N_M}} \int_0^\infty u^2 e^{-cu^2}\, du + e^{-cN_M t_0^2} = O(N_M^{-1/2}).
\end{aligned}$$

The second line uses F3 for $t \leq t_0$ and the bound by one for larger t; the third line uses $u = \sqrt{N_M}\, t$. The constants in this bound may depend on the fixed Lipschitz constant K_η.

A union bound over the M markets and F1 now give

$$\mathbb{P}\{\widehat{\mathcal{A}}_M \neq \mathcal{A}_M\} \leq CM/\sqrt{N_M} \longrightarrow 0. \tag{B.4.4}$$

Thus, with probability approaching one, replacing the population choice probabilities with observed shares changes none of the trimming decisions. □

The next result studies T_M°, which uses s_m in the standardization and $\mathcal{A}_M$ to choose the markets. Conditioning on $\mathcal{F}_M$ fixes $\mathcal{A}_M$ and leaves independent share errors as its source of randomness.

Lemma B.4.4 (Normal limit with population standardization)**.** *Under the sampling conditions, E3, ID1(c), ID3(a), and F1–F4,*

$$\frac{T_M^\circ - d_M}{\sqrt{2d_M}} \Rightarrow N(0,1).$$

Proof. Because $\mathbb{E}(\xi_{m,M} \mid \mathcal{F}_M) = 0$ and $\mathbb{E}(\xi_{m,M}^2 \mid \mathcal{F}_M) = 1$,

$$\mathbb{E}\{(\xi_{m,M} + \beta_{m,M})^2 \mid \mathcal{F}_M\} = 1 + \beta_{m,M}^2.$$

Define the centered contribution

$$X_{m,M} := (\xi_{m,M} + \beta_{m,M})^2 - (1 + \beta_{m,M}^2) = (\xi_{m,M}^2 - 1) + 2\beta_{m,M}\xi_{m,M}.$$

Conditional on $\mathcal{F}_M$, these contributions are independent and have mean zero. In particular,

$$T_M^\circ - k_M - \nu_M = \sum_{m \in \mathcal{A}_M} X_{m,M}.$$

To calculate their conditional variances, expand the last expression for $X_{m,M}$:

$$\begin{aligned}\operatorname{Var}(X_{m,M} \mid \mathcal{F}_M) &= \mathbb{E}(\xi_{m,M}^4 \mid \mathcal{F}_M) - 1 + 4\beta_{m,M}^2 \mathbb{E}(\xi_{m,M}^2 \mid \mathcal{F}_M) \\ &\quad + 4\beta_{m,M}\mathbb{E}(\xi_{m,M}^3 - \xi_{m,M} \mid \mathcal{F}_M) \\ &= 2 + 4\beta_{m,M}^2 + 4\beta_{m,M}\mathbb{E}(\xi_{m,M}^3 \mid \mathcal{F}_M) + O(N_M^{-1}).\end{aligned}$$

The final equality uses the binomial formulas in Lemma B.1.5; their bounds are uniform because $s_m \in [\eta, 1-\eta]$ for every $m \in \mathcal{A}_M$. The same lemma gives a uniformly bounded sixth moment of $\xi_{m,M}$. Using $|a+b|^3 \le 4(|a|^3 + |b|^3)$, it follows that

$$\begin{aligned}\mathbb{E}(|X_{m,M}|^3 \mid \mathcal{F}_M) &\le 4\mathbb{E}(|\xi_{m,M}^2 - 1|^3 \mid \mathcal{F}_M) + 32|\beta_{m,M}|^3 \mathbb{E}(|\xi_{m,M}|^3 \mid \mathcal{F}_M) \\ &\le C(1 + |\beta_{m,M}|^3).\end{aligned}$$

Let the total conditional variance be

$$v_M^2 := \sum_{m \in \mathcal{A}_M} \operatorname{Var}(X_{m,M} \mid \mathcal{F}_M).$$

The bound in Lemma B.4.2 supplies the deterministic sequence

$$\bar{\beta}_M := \frac{\sqrt{N_M}\rho_M}{a_*\sqrt{\eta(1-\eta)}} \to 0,$$
$$\text{such that} \qquad |\beta_{m,M}| \le \bar{\beta}_M \quad (m \in \mathcal{A}_M).$$

Lemma B.1.5 also gives $|\mathbb{E}(\xi^3_{m,M} \mid \mathcal{F}_M)| \le C/\sqrt{N_M}$ uniformly over these markets. Substituting both bounds into the variance expansion yields

$$\left|\mathrm{Var}(X_{m,M} \mid \mathcal{F}_M) - 2\right| \le C\left(\bar{\beta}_M^2 + \frac{\bar{\beta}_M}{\sqrt{N_M}} + \frac{1}{N_M}\right).$$

The right-hand side is deterministic, tends to zero, and is the same for every retained market. Summing over $\mathcal{A}_M$ gives

$$|v_M^2 - 2k_M| \le Ck_M\left(\bar{\beta}_M^2 + \frac{\bar{\beta}_M}{\sqrt{N_M}} + \frac{1}{N_M}\right).$$

On the event $k_M > 0$, divide by $2k_M$. Lemma B.4.1 gives $\mathbb{P}(k_M = d_M > 0) \to 1$, so

$$\frac{v_M^2}{2d_M} \to_p 1.$$

For large M, on $\{k_M \ge p_* M/2\}$, $v_M^2 \ge k_M$ and $\sum_{m\in\mathcal{A}_M} \mathbb{E}(|X_{m,M}|^3 \mid \mathcal{F}_M) \le Ck_M$. Therefore

$$\frac{\sum_{m\in\mathcal{A}_M} \mathbb{E}(|X_{m,M}|^3 \mid \mathcal{F}_M)}{v_M^3} \le \frac{Ck_M}{k_M^{3/2}} = \frac{C}{\sqrt{k_M}} \longrightarrow 0$$

on that event. Its probability tends to one, so the ratio converges to zero in probability, and $\mathbb{P}(v_M > 0) \to 1$. As in Lemma B.1.6, the ratio is defined as zero when $v_M = 0$.

To apply that lemma, order the indices in $\mathcal{A}_M$ as

$$\mathcal{A}_M = \{m_{M,1}, \dots, m_{M,k_M}\}, \qquad \text{where } m_{M,1} < \cdots < m_{M,k_M}.$$

There are k_M summands in this sample. Once the market characteristics are fixed, the index list, its length, and the conditional distributions of the contributions $X_{m_{M,1},M}, \dots, X_{m_{M,k_M},M}$ are fixed. The contributions themselves remain random because the choices are sampled. Thus the conditional independence, centering, and third-moment condition just verified are precisely the hypotheses of Lemma B.1.6.

Applying Lemma B.1.6 to $X_{m_{M,1},M},\dots,X_{m_{M,k_M},M}$ gives

$$\frac{T_M^\circ - k_M - \nu_M}{\nu_M} \Rightarrow N(0,1).$$

Here the ratio on the left is defined as zero when $\nu_M = 0$. Since $\nu_M/\sqrt{2d_M} \to_p 1$, Slutsky's theorem (Billingsley, 1995, Section 25, in particular Theorem 25.4 and Example 25.8) yields

$$\frac{T_M^\circ - k_M - \nu_M}{\sqrt{2d_M}} \Rightarrow N(0,1).$$

By Lemma B.4.1, $k_M = d_M$ with probability approaching one. Lemma B.4.2 gives $\nu_M/\sqrt{d_M} \to_p 0$. Hence

$$\frac{T_M^\circ - d_M}{\sqrt{2d_M}} \Rightarrow N(0,1). \tag{B.4.5}$$

□

It remains to check the estimated denominators $A_m(\theta_M^*,\widehat{s}_m)\sqrt{\widehat{s}_m(1-\widehat{s}_m)}$ and residuals $r(\theta_M^*;W_m,\widehat{s}_m)$ in $T_M(\theta_M^*)$. Their replacement error must be small relative to $\sqrt{d_M}$, the scale in the preceding normal limit.

Lemma B.4.5 (Estimated standardization has a negligible effect)**.** *Under the conditions of Lemma B.4.4,*

$$\frac{T_M(\theta_M^*) - T_M^\circ}{\sqrt{d_M}} \to_p 0.$$

Proof. For $m \in \mathcal{A}_M \cap \widehat{\mathcal{A}}_M$, define the share error $e_{m,M} := \widehat{s}_m - s_m$ and the standardized residual

$$Z_{m,M} := \frac{\sqrt{N_M}\, r(\theta_M^*;W_m,\widehat{s}_m)}{A_m(\theta_M^*,\widehat{s}_m)\sqrt{\widehat{s}_m(1-\widehat{s}_m)}}.$$

Its square is the market's contribution to the implemented statistic (4.2). For this calculation I write the population and estimated denominators as

$$D_{m,M} := A_M(W_m)\sqrt{s_m(1-s_m)}, \qquad \widehat{D}_{m,M} := A_m(\theta_M^*,\widehat{s}_m)\sqrt{\widehat{s}_m(1-\widehat{s}_m)}.$$

Both have absolute value at least $a_*\sqrt{\eta(1-\eta)} > 0$. Their signs need not be positive; only this lower bound on their absolute values is used.

Both shares, and the interval between them, lie in $[\eta, 1-\eta]$. Taylor's theorem and the uniform second-derivative bound in Lemma B.1.3 give

$$r(\theta_M^*; W_m, \widehat{s}_m) = r(\theta_M^*; W_m, s_m) + A_M(W_m) e_{m,M} + u_{m,M}/\sqrt{N_M},$$
$$\text{where} \qquad |u_{m,M}| \le C\sqrt{N_M} e_{m,M}^2.$$

By the definitions of $\beta_{m,M}$ and $\xi_{m,M}$, this is

$$\sqrt{N_M}\, r(\theta_M^*; W_m, \widehat{s}_m) = D_{m,M}(\beta_{m,M} + \xi_{m,M}) + u_{m,M}.$$

To bound the change in the denominator, define

$$F_{m,M} : [\eta, 1-\eta] \longrightarrow \mathbb{R}, \qquad q \longmapsto F_{m,M}(q) := A_m(\theta_M^*, q)\sqrt{q(1-q)}.$$

The product rule gives

$$F'_{m,M}(q) = \partial_q^2 r(\theta_M^*; W_m, q)\sqrt{q(1-q)} + A_m(\theta_M^*, q)\frac{1-2q}{2\sqrt{q(1-q)}}.$$

For $q \in [\eta, 1-\eta]$, $q(1-q) \ge \eta(1-\eta) > 0$. Lemma B.1.3 bounds $|\partial_q r(\theta_M^*; W_m, q)|$ and $|\partial_q^2 r(\theta_M^*; W_m, q)|$ by K_η, so $|F'_{m,M}(q)| \le C_\eta$, uniformly over m, M, and q in that interval. The mean-value theorem therefore gives

$$|\widehat{D}_{m,M} - D_{m,M}| = |F_{m,M}(\widehat{s}_m) - F_{m,M}(s_m)| \le C_\eta |e_{m,M}|.$$

Dividing the Taylor expression by $\widehat{D}_{m,M}$ and subtracting $\beta_{m,M} + \xi_{m,M}$ gives the explicit remainder:

$$R_{m,M} := \frac{(D_{m,M} - \widehat{D}_{m,M})(\beta_{m,M} + \xi_{m,M}) + u_{m,M}}{\widehat{D}_{m,M}}.$$

Put $d_* := a_*\sqrt{\eta(1-\eta)} > 0$. Since $|\widehat{D}_{m,M}| \ge d_*$, the preceding bounds imply

$$|R_{m,M}| \le d_*^{-1}\left\{C_\eta |e_{m,M}|(|\beta_{m,M}| + |\xi_{m,M}|) + C\sqrt{N_M} e_{m,M}^2\right\},$$
$$|\xi_{m,M}| = \frac{\sqrt{N_M}|e_{m,M}|}{\sqrt{s_m(1-s_m)}} \le \frac{\sqrt{N_M}|e_{m,M}|}{\sqrt{\eta(1-\eta)}}.$$

Substituting the second inequality into the first shows that the term $|e_{m,M}||\xi_{m,M}|$ is bounded by a constant times $\sqrt{N_M} e_{m,M}^2$. Hence

$$Z_{m,M} = \beta_{m,M} + \xi_{m,M} + R_{m,M}, \qquad \text{with} \qquad |R_{m,M}| \le C\{\sqrt{N_M} e_{m,M}^2 + |\beta_{m,M}|\,|e_{m,M}|\}. \tag{B.4.6}$$

For $m \in \mathcal{A}_M \setminus \widehat{\mathcal{A}}_M$, define $R_{m,M} := 0$. The bound on $|R_{m,M}|$ then holds for every $m \in \mathcal{A}_M$. Using the second and fourth share-error moments from Lemma B.1.5,

$$\begin{aligned}\mathbb{E}(R_{m,M}^2 \mid \mathcal{F}_M) &\le C\{N_M \mathbb{E}(e_{m,M}^4 \mid \mathcal{F}_M) + \beta_{m,M}^2 \mathbb{E}(e_{m,M}^2 \mid \mathcal{F}_M)\} \\ &\le C(1+\beta_{m,M}^2)/N_M.\end{aligned}$$

Let

$$B_M := \sum_{m \in \mathcal{A}_M} R_{m,M}^2, \qquad V_M := 2\sqrt{T_M^\circ B_M} + B_M.$$

On the event $\widehat{\mathcal{A}}_M = \mathcal{A}_M$, the implemented statistic sums $(\beta_{m,M} + \xi_{m,M} + R_{m,M})^2$ over exactly the same indices as T_M°. Expanding those squares and applying Cauchy–Schwarz gives

$$|T_M(\theta_M^*) - T_M^\circ| \le 2\sqrt{T_M^\circ B_M} + B_M = V_M.$$

The conditional expectations satisfy

$$\mathbb{E}(T_M^\circ \mid \mathcal{F}_M) = k_M + \nu_M, \qquad \mathbb{E}(B_M \mid \mathcal{F}_M) \le C(k_M + \nu_M)/N_M.$$

Another application of Cauchy–Schwarz, this time to the conditional expectation, gives

$$\begin{aligned}\mathbb{E}(V_M \mid \mathcal{F}_M) &\le 2\sqrt{\mathbb{E}(T_M^\circ \mid \mathcal{F}_M)\mathbb{E}(B_M \mid \mathcal{F}_M)} + \mathbb{E}(B_M \mid \mathcal{F}_M) \\ &\le C(k_M + \nu_M)/\sqrt{N_M}.\end{aligned}$$

Here $k_M \le M$, and the deterministic bound on ν_M in Lemma B.4.2 is $o(\sqrt{M})$. Taking expectations therefore yields

$$\mathbb{E}V_M \le C\{M + o(\sqrt{M})\}/\sqrt{N_M}.$$

This error must be small compared with the statistic's scale $\sqrt{d_M}$, rather than small in absolute value. For any $\varepsilon > 0$, bound separately the probability that the trimming rules disagree and the probability that too few markets satisfy the population trimming conditions:

$$\begin{aligned}\mathbb{P}\left\{\frac{|T_M(\theta_M^*) - T_M^\circ|}{\sqrt{d_M}} > \varepsilon\right\} \le \mathbb{P}(\widehat{\mathcal{A}}_M \ne \mathcal{A}_M) + \mathbb{P}(k_M < p_* M/2) \\ + \frac{\mathbb{E}V_M}{\varepsilon\sqrt{p_* M/2}}.\end{aligned}$$

The first two terms vanish by Lemmas B.4.1 and B.4.3. The third is $O(\sqrt{M/N_M}) + o(N_M^{-1/2})$ and vanishes because F1 implies $M/N_M \to 0$. Consequently,

$$\frac{T_M(\theta_M^*) - T_M^\circ}{\sqrt{d_M}} \to_p 0. \qquad \square$$

Proof of Theorem 3. Lemmas B.4.4 and B.4.5 give

$$\frac{T_M(\theta_M^*) - d_M}{\sqrt{2d_M}} \Rightarrow N(0,1).$$

I first use this limit to obtain $\mathbb{P}\{\theta_M^* \in \mathcal{C}_{M,\gamma}^s\}$. F5 then shows that, conditional on this acceptance, $[L_M, U_M]$ contains $\phi(\theta_0)$ with probability approaching one. For deterministic integers $k \to \infty$, let $Z_1, \ldots, Z_k$ be i.i.d. $N(0,1)$ and set $C_k := \sum_{j=1}^k Z_j^2 \sim \chi_k^2$. The squared variables Z_j^2 have mean one and variance two, so the ordinary central limit theorem (Billingsley, 1995, Theorem 27.1) gives $(C_k - k)/\sqrt{2k} \Rightarrow N(0,1)$. Since the standard normal CDF is continuous and strictly increasing, the corresponding quantiles satisfy

$$\frac{\chi_{1-\gamma,k}^2 - k}{\sqrt{2k}} \longrightarrow z_{1-\gamma},$$

where $z_{1-\gamma}$ is the standard normal $(1-\gamma)$ quantile. To justify using the random integer d_M, write $c_k := (\chi_{1-\gamma,k}^2 - k)/\sqrt{2k}$. For every $\varepsilon > 0$, choose K such that $|c_k - z_{1-\gamma}| < \varepsilon$ for all $k \geq K$. Then

$$\mathbb{P}\{|c_{d_M} - z_{1-\gamma}| \geq \varepsilon\} \leq \mathbb{P}(d_M < K) \longrightarrow 0,$$

because $d_M \to_p \infty$. This proves

$$\frac{\chi_{1-\gamma,d_M}^2 - d_M}{\sqrt{2d_M}} \to_p z_{1-\gamma}.$$

Define the acceptance event $E_M := \{\theta_M^* \in \mathcal{C}_{M,\gamma}^s\}$. On $\{\widehat{\mathcal{A}}_M = \mathcal{A}_M,\ k_M = d_M\}$, the test's actual count $\widehat{k}_M$ equals d_M, so acceptance is equivalently

$$\frac{T_M(\theta_M^*) - d_M}{\sqrt{2d_M}} \leq \frac{\chi_{1-\gamma,d_M}^2 - d_M}{\sqrt{2d_M}}.$$

The left side converges in distribution to a standard normal and the right side converges in probability to $z_{1-\gamma}$. Slutsky's theorem therefore gives

$$\frac{T_M(\theta_M^*) - \chi^2_{1-\gamma,d_M}}{\sqrt{2d_M}} \Rightarrow N(-z_{1-\gamma}, 1).$$

Since this limiting distribution assigns probability zero to zero, the probability that the displayed difference is nonpositive tends to $1-\gamma$. The event excluded in making the comparison has probability tending to zero. Hence

$$\mathbb{P}(E_M) = \mathbb{P}\{\theta_M^* \in \mathcal{C}^s_{M,\gamma}\} \longrightarrow 1-\gamma. \tag{B.4.7}$$

The test still accepts θ_M^* when $\widehat{\mathcal{A}}_M = \varnothing$. Lemmas B.4.1 and B.4.3 show that the probability of this event tends to zero.

Write $\phi_M^* := \phi(\theta_M^*)$ and define the interval-inclusion event

$$\mathcal{H}_M := \{[\phi_M^* - \varepsilon_M, \phi_M^* + \varepsilon_M] \subseteq [L_M, U_M]\}.$$

F5 gives

$$|\phi_M^* - \phi(\theta_0)| = o(\varepsilon_M), \qquad \mathbb{P}(\mathcal{H}_M \mid E_M) \to 1.$$

The bound $|\phi_M^* - \phi(\theta_0)| = o(\varepsilon_M)$ means that, for all sufficiently large M, $\phi(\theta_0)$ lies in $[\phi_M^* - \varepsilon_M, \phi_M^* + \varepsilon_M]$. On $E_M \cap \mathcal{H}_M$, that interval is contained in the interval $[L_M, U_M]$. Thus, for all sufficiently large M,

$$\begin{aligned}\mathbb{P}\{\phi(\theta_0) \in [L_M, U_M]\} &\geq \mathbb{P}(E_M \cap \mathcal{H}_M) \\ &= \mathbb{P}(E_M)\mathbb{P}(\mathcal{H}_M \mid E_M).\end{aligned}$$

Using (B.4.7) and F5 proves

$$\liminf_{M\to\infty} \mathbb{P}\{\phi(\theta_0) \in [L_M, U_M]\} \geq 1-\gamma.$$

If $\mathcal{C}^s_{M,\gamma} = \varnothing$, the interval $[L_M, U_M]$ is taken to be empty. On E_M, $\theta_M^* \in \mathcal{C}^s_{M,\gamma}$, so this accepted set is nonempty. Compactness of Θ and continuity of ϕ ensure that the values $\phi(\theta)$ for $\theta \in \mathcal{C}^s_{M,\gamma}$ remain bounded. This proves the stated coverage probability for $[L_M, U_M]$ without requiring asymptotic normality of $\widehat{\theta}_M$. □

B.5 Proof of Theorem 4

I use the model and notation in Appendix A. The payoff of alternative 0 is normalized to zero. Write $\alpha := (\alpha_1, \ldots, \alpha_J)'$ and $\delta := (\delta_1, \ldots, \delta_J)'$. The two parameter values $\theta := (\alpha, \delta, \lambda, G)$ and $\widetilde{\theta} := (\widetilde{\alpha}, \widetilde{\delta}, \widetilde{\lambda}, \widetilde{G})$ compared in the proof must each satisfy A1–A6. If $\lambda(x) = 0$ or $\widetilde{\lambda}(x) = 0$, A6(a) requires $\mathcal{Z}_x = \mathbb{R}^J$. Write $p_j(z_j; x)$ for the choice probability of alternative j in the binary RI problem on $\{0, j\}$, using the same marginal distribution of Ω_j under G and unit cost $\lambda(x)$. This probability describes choice when only alternatives 0 and j are available. I aim to recover $p_j(z_j; x)$ for every $z_j \in \mathbb{R}$, including values outside $\operatorname{supp}(Z_j \mid X = x)$. When $\lambda(x) > 0$, $p_j(z_j; x)$ maximizes

$$\phi_j(p; z_j, x) := \lambda(x)\mathbb{E}_G \log\left\{1 - p + pe^{(\alpha_j + \delta_j z_j + \Omega_j)/\lambda(x)}\right\} \quad (p \in [0,1]).$$

When $\lambda(x) = 0$, $p_j(z_j; x) = \mathbb{P}_G\{\alpha_j + \delta_j z_j + \Omega_j > 0\}$. Assumption A6(b) gives probability zero to equality of the two payoffs.

The following lemmas establish properties of the choice probabilities, recover the functions $p_j(\cdot; x)$, and identify the joint taste-shock distribution. The last lemma treats the case $\lambda(x) = 0$ for every $x \in \mathcal{X}$.

Lemma B.5.1 (Choice probabilities and full consideration)**.** *Suppose A2 holds. Fix $x \in \mathcal{X}_+$ and write $\lambda := \lambda(x)$. For $a \in \mathbb{R}^J$, let $\bar{b}(a; x) = (\bar{b}^0(a; x), \ldots, \bar{b}^J(a; x))'$ maximize $q \mapsto \Phi(q; a, x)$ over Δ_J, as in (A.1). This maximizer is unique, and the map $a \mapsto \bar{b}(a; x)$ is continuous. In covariate coordinates, $b(z; x) = \bar{b}(a(z); x)$, where $a(z) := (\alpha_j + \delta_j z_j)_{j=1}^J$. The full-consideration region*

$$\mathcal{W}_x := \{a \in \mathbb{R}^J : \bar{b}^j(a; x) > 0 \text{ for every } j = 0, \ldots, J\}$$

is connected and open. Each function $a \mapsto \bar{b}^j(a; x)$ is real analytic on $\mathcal{W}_x$. The map

$$\begin{aligned} c : \mathcal{W}_x &\longrightarrow \mathbb{R}^J, \\ a &\longmapsto c(a) := (c_1(a), \ldots, c_J(a))', \\ c_j(a) &:= -a_j + \lambda \log \frac{\bar{b}^0(a; x)}{\bar{b}^j(a; x)} \quad (j = 1, \ldots, J) \end{aligned}$$

is a real-analytic diffeomorphism from $\mathcal{W}_x$ onto $\mathbb{R}^J$. For every $j = 1, \ldots, J$ and $a \in \mathcal{W}_x$, $\partial \bar{b}^j(a; x)/\partial a_j > 0$.

Proof. The finite second moments of G stated in Appendix A imply

$$\mathbb{E}_G\|\Omega\|_1 = \sum_{j=1}^{J} \mathbb{E}_G|\Omega_j| \le \sum_{j=1}^{J} (\mathbb{E}_G \Omega_j^2)^{1/2} < \infty.$$

Lemma B.1.8 therefore makes $(q,a) \mapsto \Phi(q;a,x)$ finite and continuous on $\Delta_J \times \mathbb{R}^J$. A2 and Lemma B.1.9 give a unique maximizer $\bar{b}(a;x)$ and continuity of $a \mapsto \bar{b}(a;x)$, including values of a at which the maximizing choice-probability vector lies on the boundary of Δ_J.

Lemmas B.1.10 and B.1.11 show that the map

$$B_x : \mathcal{W}_x \longrightarrow \mathcal{D}_J^\circ, \qquad a \longmapsto (\bar{b}^1(a;x), \ldots, \bar{b}^J(a;x))'$$

is a bijection with a real-analytic inverse $a^*(\cdot;x)$. In particular, $\bar{b}^j(a;x)$ is real analytic on $\mathcal{W}_x$. The set $\mathcal{D}_J^\circ$, consisting of vectors $(q_1,\ldots,q_J)'$ with $q_j > 0$ for every j and $\sum_j q_j < 1$, is convex and hence connected. Its continuous image under $a^*(\cdot;x)$ is $\mathcal{W}_x$, so $\mathcal{W}_x$ is connected (Munkres, 2000, Theorem 23.5). It is open by continuity of $\bar{b}$.

I next verify the asserted inverse for the threshold map c. For any threshold vector $c \in \mathbb{R}^J$, define

$$p_0(\omega;c) := \frac{1}{1+\sum_j e^{(\omega_j - c_j)/\lambda}}, \qquad p_j(\omega;c) := e^{(\omega_j - c_j)/\lambda} p_0(\omega;c), \qquad q_j(c) := \mathbb{E}_G p_j(\Omega;c).$$

Write $q(c) := (q_0(c),\ldots,q_J(c))'$; every component $q_j(c)$ is positive. Define the map $a_x : \mathbb{R}^J \to \mathbb{R}^J$, $c \mapsto a_x(c)$, by

$$a_{x,j}(c) := -c_j + \lambda \log\{q_0(c)/q_j(c)\}. \tag{B.5.1}$$

At $a_x(c)$, the expression inside the logarithm in Φ satisfies

$$q_0(c) + \sum_{j=1}^{J} q_j(c) e^{(a_{x,j}(c)+\Omega_j)/\lambda} = q_0(c)\left\{1 + \sum_{j=1}^{J} e^{(\Omega_j - c_j)/\lambda}\right\}.$$

The conditional probabilities in the share first-order conditions are therefore precisely $p_j(\Omega;c)$. Their expectations equal $q_j(c)$ by definition. As shown in Lemma B.1.10, after substituting $q_0 = 1 - \sum_{k=1}^{J} q_k$ in Φ, these equalities make its derivatives with respect to q_j, $j = 1,\ldots,J$, zero. Strict concavity then gives

$$\bar{b}(a_x(c);x) = q(c), \qquad \text{so} \qquad a_x(c) \in \mathcal{W}_x.$$

Substitution into the definition of $c(a)$ now gives

$$c_j(a_x(c)) = -a_{x,j}(c) + \lambda \log\{q_0(c)/q_j(c)\} = c_j.$$

Thus $c(a_x(c)) = c$ for every $c \in \mathbb{R}^J$.

For the other inverse identity, fix $a \in \mathcal{W}_x$ and write $\bar{q} := \bar{b}(a;x)$. The definition of $c_j(a)$ implies

$$e^{(\Omega_j - c_j(a))/\lambda} = \frac{\bar{q}_j}{\bar{q}_0} e^{(a_j+\Omega_j)/\lambda}.$$

Hence $p_j(\Omega; c(a))$ is the conditional probability associated with the interior optimal vector $\bar{q}$ at a. Lemma B.1.10 gives $\mathbb{E}_G p_j(\Omega; c(a)) = \bar{q}_j$ for every $j = 0, \ldots, J$. Consequently $q(c(a)) = \bar{q}$, and

$$a_{x,j}(c(a)) = -c_j(a) + \lambda \log(\bar{q}_0/\bar{q}_j) = a_j.$$

This proves $a_x(c(a)) = a$ and establishes the bijection between $\mathcal{W}_x$ and $\mathbb{R}^J$.

The functions $q_j(c)$ are real analytic by Iaria and Wang (2025, Theorem 1(a)). Thus $c \mapsto a_x(c)$ is real analytic by (B.5.1). Its inverse $a \mapsto c(a)$ is real analytic because the shares are analytic and strictly positive on $\mathcal{W}_x$. These explicit inverse maps prove the asserted analytic diffeomorphism. For completeness, with $\mathbf{q}(c) := (q_1(c), \ldots, q_J(c))'$, differentiating $B_x(a_x(c)) = \mathbf{q}(c)$ gives

$$D_c a_x(c) = \{D_a B_x(a_x(c))\}^{-1} D_c \mathbf{q}(c). \tag{B.5.2}$$

The first factor is nonsingular by Lemma B.1.11. To verify the second, put

$$\mathbf{p}(\Omega; c) := (p_1(\Omega; c), \ldots, p_J(\Omega; c))'.$$

Differentiating the probabilities under the expectation gives

$$D_c \mathbf{q}(c) = -\frac{1}{\lambda} \mathbb{E}_G\{\operatorname{diag}(\mathbf{p}(\Omega; c)) - \mathbf{p}(\Omega; c)\mathbf{p}(\Omega; c)'\}.$$

For any nonzero $u = (u_1, \ldots, u_J)' \in \mathbb{R}^J$, $-u' D_c \mathbf{q}(c) u$ is $1/\lambda$ times the expected variance of $0, u_1, \ldots, u_J$ under the positive probabilities $p_0(\Omega; c), \ldots, p_J(\Omega; c)$. That variance is positive by the argument in Lemma B.1.10. Thus $D_c \mathbf{q}(c)$ is negative definite and nonsingular.

Finally, Lemma B.1.11 gives a positive-definite share Jacobian $D_a B_x(a)$. For the coordinate vector $e_j \in \mathbb{R}^J$,

$$\frac{\partial \bar{b}^j(a;x)}{\partial a_j} = e_j' D_a B_x(a) e_j > 0.$$

Thus increasing a_j while holding the other payoff indices fixed locally increases $\bar{b}^j(a;x)$ at every $a \in \mathcal{W}_x$. □

The next lemma recovers the probability $p_j(z_j; x)$ for choice from $\{0, j\}$ using probabilities from the model with all alternatives available. At positive information cost, fix $u \in \mathbb{R}$ with $p := p_j(u; x) \in (0, 1)$. I construct covariate vectors z_n with $z_{n,j} = u$ for every n and $b^i(z_n; x) > 0$ for every $i = 0, \ldots, J$, such that $b^j(z_n; x) \to p$, $b^0(z_n; x) \to 1 - p$, and the probabilities of all other alternatives tend to zero. The comparison uses equality of $b(z_n; x)$ and $\widetilde{b}(z_n; x)$ only at points where every alternative has positive probability.

Lemma B.5.2 (Choice probabilities when only alternatives 0 and j are available)**.** *Two parameter values satisfying A1–A6 and generating the same $\mathcal{B}_{\text{obs}}$ have the same functions $p_j(\cdot; x)$:*

$$p_j(z_j; x) = \widetilde{p}_j(z_j; x) \quad \textit{for every } z_j \in \mathbb{R},\ j = 1, \ldots, J,\ x \in \mathcal{X}.$$

Proof. I first show that δ_j and $\widetilde{\delta}_j$ have the same sign for each $j = 1, \ldots, J$. Suppose first that at least one parameter value has positive cost at some $x \in \mathcal{X}$. Interchanging the two parameter values if necessary, choose x° with $\lambda(x^\circ) > 0$. Assumption A3 supplies a covariate vector

$$z^\circ \in \operatorname{int}(\mathcal{Z}_{x^\circ}), \qquad \text{such that} \qquad b^i(z^\circ; x^\circ) > 0 \quad (i = 0, \ldots, J).$$

Since z° lies inside the conditional support, the choice probabilities are identified in a neighborhood of z°. Continuity keeps every probability positive in a sufficiently small such neighborhood. Holding z_k fixed for $k \neq j$, Lemma B.5.1 and the chain rule give

$$\frac{\partial\, b^j(z; x^\circ)}{\partial z_j} = \delta_j \frac{\partial\, \bar{b}^j(a(z); x^\circ)}{\partial a_j}, \qquad \frac{\partial\, \bar{b}^j(a(z); x^\circ)}{\partial a_j} > 0.$$

Thus $b^j(z; x^\circ)$ strictly increases with z_j when $\delta_j > 0$ and strictly decreases when $\delta_j < 0$ in this neighborhood.

The other parameter value gives the same observed probabilities there. If $\widetilde{\lambda}(x^\circ) > 0$, the same derivative calculation applies to $\widetilde{b}^j$ and $\widetilde{\delta}_j$. If $\widetilde{\lambda}(x^\circ) = 0$, increasing z_j changes only alternative j's payoff, by $\widetilde{\delta}_j$ times the increase in z_j. Hence $\widetilde{b}^j$ is nondecreasing in z_j when $\widetilde{\delta}_j > 0$ and nonincreasing when $\widetilde{\delta}_j < 0$. It cannot agree with b^j on this neighborhood if the two coefficients have opposite signs. Therefore the two coefficients have the same sign whenever at least one of the two cost functions takes a positive value.

If instead $\lambda(x) = \widetilde{\lambda}(x) = 0$ for every $x \in \mathcal{X}$, fix any $x^* \in \mathcal{X}$ and hold $z_k = 0$ for $k \neq j$. A6(a) gives $\mathcal{Z}_{x^*} = \mathbb{R}^J$. Write $e_j \in \mathbb{R}^J$ for the jth coordinate vector. When $\delta_j > 0$, for every realization of the taste-shock vector, alternative j eventually has the largest payoff as $t \to +\infty$ at $z = te_j$, and eventually loses to alternative 0 as $t \to -\infty$. Bounded convergence gives

$$\lim_{t \to +\infty} b^j(te_j; x^*) = 1, \qquad \lim_{t \to -\infty} b^j(te_j; x^*) = 0.$$

When $\delta_j < 0$, the two limits are reversed. The same argument applies to $\widetilde{\delta}_j$. Equality of the observed probabilities at every te_j therefore rules out opposite signs in this case as well. Thus, in either case,

$$\operatorname{sign}(\widetilde{\delta}_j) = \operatorname{sign}(\delta_j) \quad (j = 1, \ldots, J).$$

Replacing Z_j by $-Z_j$ whenever this common sign is negative makes both δ_j and $\widetilde{\delta}_j$ positive. I use these shifters throughout the rest of the proof.

Fix x with $\lambda(x) > 0$ and $\widetilde{\lambda}(x) > 0$. Define

$$W := \{z \in \mathbb{R}^J : b^j(z; x) > 0 \text{ for every } j = 0, \ldots, J\},$$
$$\widetilde{W} := \{z \in \mathbb{R}^J : \widetilde{b}^j(z; x) > 0 \text{ for every } j = 0, \ldots, J\}.$$

By A3, there is $z_x^\circ \in \operatorname{int}(\mathcal{Z}_x)$ at which every $b^j(z_x^\circ; x)$ is positive. The observed equality $b(z_x^\circ; x) = \widetilde{b}(z_x^\circ; x)$ makes every candidate probability positive as well. Continuity therefore supplies an open ball $D \subseteq \operatorname{int}(\mathcal{Z}_x) \cap W \cap \widetilde{W}$ around z_x°. Since D lies in the observed support, $b(z; x) = \widetilde{b}(z; x)$ for every $z \in D$. The regions W and $\widetilde{W}$ are open and connected, and $z \mapsto b(z; x)$ and $z \mapsto \widetilde{b}(z; x)$ are real analytic there, by Lemma B.5.1. Let V be the connected component of $W \cap \widetilde{W}$ containing D. Since an open Euclidean set is locally connected, its components are open (Munkres, 2000, Theorem 25.3); thus V is open.

I now show that $b(z; x) = \widetilde{b}(z; x)$ for every $z \in V$. For a fixed $j \in \{0, \ldots, J\}$, put $f(z) := b^j(z; x) - \widetilde{b}^j(z; x)$ for $z \in V$, and define

$$E_f := \{z \in V : D^\nu f(z) = 0 \text{ for every } \nu \in \mathbb{N}_0^J\},$$

where D^ν is the partial derivative with multi-index ν. For each multi-index ν, the set $\{z \in V : D^\nu f(z) = 0\}$ is closed relative to V because $D^\nu f$ is continuous. Their intersection E_f is therefore closed relative to V. If $z \in E_f$, every coefficient in the Taylor series of f at z is zero. Real analyticity makes f identically zero on a neighborhood of z; every derivative is then zero on that neighborhood, so it lies in E_f. Thus E_f is open relative to V as well. Since $f = 0$ on the open ball D, all its derivatives vanish there, and $D \subseteq E_f$. Connectedness of V now gives $E_f = V$. In particular, the derivative with multi-index zero is f itself, so $b^j(z; x) = \widetilde{b}^j(z; x)$ for every $z \in V$. Applying this to all j gives $b(z; x) = \widetilde{b}(z; x)$ on V. This is the same open-and-closed argument used for the one-dimensional identity result in Krantz and Parks (2002, Corollary 1.2.6); the connectedness criterion is given in Munkres (2000, Section 23).

This component is also closed relative to W. Indeed, if $z_n \in V$ and $z_n \to z \in W$, continuity gives $\widetilde{b}(z; x) = b(z; x)$ with every coordinate positive. Thus $z \in W \cap \widetilde{W}$. Because this intersection is open, choose $r > 0$ with $B(z, r) \subseteq W \cap \widetilde{W}$. For sufficiently large n, $z_n \in B(z, r) \cap V$. The

ball and V are connected and intersect, so their union is connected. Maximality of the connected component V implies $B(z,r) \subseteq V$, and in particular $z \in V$. This proves that V is closed relative to W. Because V is nonempty and both open and closed relative to the connected set W, $V = W$ (Munkres, 2000, Section 23); interchanging the two parameter values gives $V = \widetilde{W}$. Consequently

$$W = \widetilde{W}, \qquad b(z;x) = \widetilde{b}(z;x) \quad (z \in W). \tag{B.5.3}$$

The equality in (B.5.3) has been proved only for $z \in W$, where every $b^j(z;x)$ is positive.

Now fix $j \in \{1,\dots,J\}$ and $u \in \mathbb{R}$ with $0 < p_j(u;x) < 1$. Write

$$p \coloneqq p_j(u;x), \qquad q_0 \coloneqq 1-p, \qquad a_j^* \coloneqq \alpha_j + \delta_j u, \qquad \lambda \coloneqq \lambda(x), \qquad c^* \coloneqq \lambda \log(q_0/p) - a_j^*.$$

I will keep $z_{n,j} = u$ by choosing a threshold near c^* at each n. For $s \in \mathbb{R}$ and $0 \le \varepsilon \le 1$, define

$$R_k \coloneqq e^{\Omega_k/\lambda}, \qquad L_\varepsilon(\Omega;s) \coloneqq 1 + e^{(\Omega_j - s)/\lambda} + \varepsilon \sum_{k \notin \{0,j\}} R_k,$$

and

$$\begin{aligned} Q_0(s,\varepsilon) &\coloneqq \mathbb{E}_G\{1/L_\varepsilon(\Omega;s)\}, \\ Q_j(s,\varepsilon) &\coloneqq \mathbb{E}_G\{e^{(\Omega_j - s)/\lambda}/L_\varepsilon(\Omega;s)\}, \\ Q_k(s,\varepsilon) &\coloneqq \mathbb{E}_G\{\varepsilon R_k/L_\varepsilon(\Omega;s)\} \quad (k \notin \{0,j\}). \end{aligned}$$

Each integrand lies in $[0,1]$. The probabilities $Q_0(s,\varepsilon)$ and $Q_j(s,\varepsilon)$ are strictly positive, and when $\varepsilon > 0$ so is every $Q_k(s,\varepsilon)$. These probabilities sum to one. The payoff index for alternative j reconstructed from these probabilities and the threshold s is

$$f_\varepsilon(s) \coloneqq -s + \lambda \log \frac{Q_0(s,\varepsilon)}{Q_j(s,\varepsilon)}.$$

Bounded convergence makes f_ε continuous in s and gives $f_\varepsilon(s) \to f_0(s)$ as $\varepsilon \downarrow 0$ for each $s \in \mathbb{R}$.

The binary first-order condition at u gives

$$p = \mathbb{E}_G \frac{p e^{(a_j^* + \Omega_j)/\lambda}}{q_0 + p e^{(a_j^* + \Omega_j)/\lambda}}.$$

Since $e^{(\Omega_j - c^*)/\lambda} = (p/q_0) e^{(a_j^* + \Omega_j)/\lambda}$, this equation and its complement imply

$$Q_j(c^*,0) = p \qquad \text{and} \qquad Q_0(c^*,0) = q_0, \qquad \text{so} \qquad f_0(c^*) = a_j^*.$$

To solve $f_\varepsilon(s) = a_j^*$ for small positive ε, I first show that f_0 is strictly decreasing. Put

$$P_s(\omega) := \frac{e^{(\omega - s)/\lambda}}{1 + e^{(\omega - s)/\lambda}}.$$

The derivative of $P_s(\omega)$ with respect to s is bounded in absolute value by $1/(4\lambda)$. Differentiating under the expectation and using $Q_0(s,0) = 1 - Q_j(s,0)$ therefore gives

$$\begin{aligned} f_0'(s) &= -1 + \frac{\mathbb{E}_G[P_s(\Omega_j)\{1 - P_s(\Omega_j)\}]}{Q_0(s,0)Q_j(s,0)} \\ &= -\frac{\mathrm{Var}_G(P_s(\Omega_j))}{Q_0(s,0)Q_j(s,0)} < 0. \end{aligned}$$

The strict inequality follows because A2 rules out a constant Ω_j and P_s is strictly increasing in its argument.

For each $n \geq 1$, choose $0 < \varepsilon_n < 1/n$ small enough that

$$f_{\varepsilon_n}(c^* - 1/n) > a_j^* > f_{\varepsilon_n}(c^* + 1/n).$$

Such a choice exists because f_0 is strictly decreasing, $f_0(c^*) = a_j^*$, and f_ε converges to f_0 at the two displayed arguments. Continuity and the intermediate value theorem then give $c_n \in (c^* - 1/n, c^* + 1/n)$ satisfying

$$f_{\varepsilon_n}(c_n) = a_j^*, \qquad \text{and} \qquad c_n \to c^*.$$

Define

$$\begin{gathered} Q_{i,n} := Q_i(c_n, \varepsilon_n) \quad (i = 0, \ldots, J), \qquad Q_n := (Q_{0,n}, \ldots, Q_{J,n})', \\ c_{j,n} := c_n, \qquad c_{k,n} := -\lambda \log \varepsilon_n \quad (k \notin \{0, j\}), \\ a_{i,n} := -c_{i,n} + \lambda \log(Q_{0,n}/Q_{i,n}), \qquad z_{n,i} := (a_{i,n} - \alpha_i)/\delta_i \quad (i = 1, \ldots, J). \end{gathered}$$

Lemma B.5.1 gives $z_n \in W$ and $b(z_n; x) = Q_n$. The defining equation for c_n gives, for every n,

$$a_{j,n} = f_{\varepsilon_n}(c_n) = a_j^*, \qquad \text{so} \qquad z_{n,j} = u.$$

Since $c_n \to c^*$ and $\varepsilon_n \to 0$, bounded convergence in the formulas for Q_i yields

$$Q_{0,n} \to q_0, \qquad Q_{j,n} \to p, \qquad Q_{k,n} \to 0 \quad (k \notin \{0, j\}). \tag{B.5.4}$$

I also need an upper bound on the payoff indices $a_{k,n}$ for $k \notin \{0, j\}$. Because $c_n > c^* - 1$ and $\varepsilon_n \le 1$,

$$L_{\varepsilon_n}(\Omega; c_n) \le \overline{L}(\Omega),$$
$$\text{where} \qquad \overline{L}(\Omega) := 1 + e^{(\Omega_j - c^* + 1)/\lambda} + \sum_{k \notin \{0,j\}} R_k.$$

For each such k, let $m_k := \mathbb{E}_G\{R_k / \overline{L}(\Omega)\} \in (0, 1]$. The definitions of $Q_{k,n}$ and $a_{k,n}$ then give

$$\frac{Q_{k,n}}{\varepsilon_n} = \mathbb{E}_G \frac{R_k}{L_{\varepsilon_n}(\Omega; c_n)} \ge m_k, \qquad a_{k,n} = \lambda \log \frac{Q_{0,n}}{Q_{k,n}/\varepsilon_n} \le -\lambda \log m_k. \tag{B.5.5}$$

By (B.5.3), the same Q_n maximizes the objective under $\widetilde{\theta}$ at z_n. Write $\widetilde{\lambda} := \widetilde{\lambda}(x)$ and $\widetilde{\Omega} \sim \widetilde{G}$. The payoff index of alternative j under $\widetilde{\theta}$ is $\widetilde{a}_j := \widetilde{\alpha}_j + \widetilde{\delta}_j u$ for every n. The other indices are bounded above because

$$\widetilde{a}_{k,n} = \frac{\widetilde{\delta}_k}{\delta_k} a_{k,n} + \widetilde{\alpha}_k - \frac{\widetilde{\delta}_k}{\delta_k} \alpha_k$$

and the slope ratios are positive. Set $\widetilde{a}_{j,n} := \widetilde{a}_j$ and write

$$\widetilde{D}_n := Q_{0,n} + \sum_{i=1}^{J} Q_{i,n} e^{(\widetilde{a}_{i,n} + \widetilde{\Omega}_i)/\widetilde{\lambda}}, \qquad \widetilde{\Phi}_n(Q_n) := \widetilde{\lambda} \mathbb{E}_{\widetilde{G}} \log \widetilde{D}_n.$$

For every realization of $\widetilde{\Omega}$, (B.5.4) and the upper bounds on the indices give

$$\widetilde{D}_n \longrightarrow q_0 + p e^{(\widetilde{a}_j + \widetilde{\Omega}_j)/\widetilde{\lambda}}.$$

For all sufficiently large n, $\widetilde{\lambda} \log \widetilde{D}_n \ge \widetilde{\lambda} \log(q_0/2)$. Choose constants $\bar{a}_i$ bounding $\widetilde{a}_{i,n}$ above. Since Q_n is a probability vector,

$$\widetilde{\lambda} \log \widetilde{D}_n \le \max\{0, \max_{i \ge 1}(\bar{a}_i + \widetilde{\Omega}_i)\}.$$

Consequently

$$\left|\widetilde{\lambda} \log \widetilde{D}_n\right| \le \left|\widetilde{\lambda} \log(q_0/2)\right| + \max_i |\bar{a}_i| + \sum_{i=1}^{J} |\widetilde{\Omega}_i|.$$

The right-hand side is integrable under $\widetilde{G}$ because each $\widetilde{\Omega}_i$ has a finite first moment. Dominated convergence therefore gives

$$\widetilde{\Phi}_n(Q_n) \longrightarrow \widetilde{\phi}_j(p; u, x).$$

For any $p' \in [0,1]$, the probability vector assigning $1-p'$ to alternative 0, p' to alternative j, and zero to every other alternative is feasible in every problem under $\widetilde{\theta}$. Because $z_{n,j} = u$ for every n, its objective is exactly $\widetilde{\phi}_j(p'; u, x)$. Optimality of Q_n and the preceding limit give

$$\widetilde{\Phi}_n(Q_n) \geq \widetilde{\phi}_j(p'; u, x) \quad \Longrightarrow \quad \widetilde{\phi}_j(p; u, x) \geq \widetilde{\phi}_j(p'; u, x) \quad (p' \in [0,1]).$$

Assumption A2 implies that $\widetilde{\Omega}_j$ is nondegenerate. Lemma B.1.9, applied with $J = 1$ to this marginal distribution, makes $\widetilde{p}_j(u; x)$ the unique maximizer of $\widetilde{\phi}_j(\cdot; u, x)$. Thus $p_j(u; x) = \widetilde{p}_j(u; x)$ whenever $0 < p_j(u; x) < 1$. By Lemma B.1.1, both functions are continuous, nondecreasing CDFs with limits zero and one. Their equality on $\{u : 0 < p_j(u; x) < 1\}$ forces equality also where $p_j(u; x)$ equals zero or one, by monotonicity. Therefore $p_j(u; x) = \widetilde{p}_j(u; x)$ for every $u \in \mathbb{R}$.

If exactly one of $\lambda(x)$ and $\widetilde{\lambda}(x)$ is zero, A6(a) gives $\mathcal{Z}_x = \mathbb{R}^J$. Thus the observed equality $b(z; x) = \widetilde{b}(z; x)$ holds for every $z \in \mathbb{R}^J$. Write $p_j^+(z_j; x)$ and $p_j^{\text{zero}}(z_j; x)$ for the probabilities of choosing j when only j and 0 are available, under the parameter values with positive and zero information costs, respectively. Fix $u \in \mathbb{R}$ with $p := p_j^+(u; x) \in (0,1)$ and apply the preceding construction to the parameter value with positive cost. It gives z_n and Q_n satisfying (B.5.4), with $z_{n,j} = u$ for every n. Under the parameter value with zero cost, choosing j when all alternatives are available implies that its payoff exceeds alternative 0's. If j would be chosen from $\{0, j\}$ but is not chosen when all alternatives are available, another alternative must be chosen. Assumption A6(b) excludes ties. Since the observed choice probabilities agree at z_n,

$$0 \leq p_j^{\text{zero}}(u; x) - Q_{j,n} \leq \sum_{k \notin \{0,j\}} Q_{k,n} \longrightarrow 0.$$

Together with $Q_{j,n} \to p$, this gives $p_j^{\text{zero}}(u; x) = p_j^+(u; x)$. Both functions are continuous, nondecreasing CDFs with limits zero and one, by A6(b) and Lemma B.1.1. Their equality therefore extends to points where $p_j^+(u; x)$ equals zero or one. Thus $p_j^{\text{zero}}(u; x) = p_j^+(u; x)$ for every $u \in \mathbb{R}$.

If $\lambda(x) = \widetilde{\lambda}(x) = 0$, their choice probabilities with all alternatives available are identified on $\mathbb{R}^J$ by A6(a). For $J = 1$, these are already the probabilities for choice from $\{0, 1\}$, so the conclusion follows directly. For $J \geq 2$, fix z_j and write $a_j := \alpha_j + \delta_j z_j$. Define the covariate vector z_n by $z_{j,n} := z_j$ and $z_{k,n} := -n$ for $k \in \{1, \ldots, J\} \setminus \{j\}$. Under either parameter value, each rival's payoff tends to $-\infty$ for every realization of the taste-shock vector because its slope is positive. Assumption A6(b) gives $\mathbb{P}_G(a_j + \Omega_j = 0) = 0$.

Outside this event,

$$\mathbf{1}\{a_j + \Omega_j > \max(0, \max_{k \notin \{0,j\}} (\alpha_k - \delta_k n + \Omega_k))\} \longrightarrow \mathbf{1}\{a_j + \Omega_j > 0\}.$$

Bounded convergence gives $b^j(z_n; x) \to p_j(z_j; x)$. Applying the same argument under $\widetilde{\theta}$ and using equality of the identified choice probabilities gives $p_j(z_j; x) = \widetilde{p}_j(z_j; x)$ for every $z_j \in \mathbb{R}$. □

Lemma B.5.3 (The probability of choosing alternative 0 identifies the joint taste-shock distribution)**.** *Suppose the payoff coefficients (α, δ) and the cost function $\lambda : \mathcal{X} \to [0, \infty)$ are known and A2–A3 hold. Fix $x \in \mathcal{X}_+$. On an observed nonempty open set $D_x \subseteq \operatorname{int}(\mathcal{Z}_x)$ with $b^j(z; x) > 0$ for every $j = 0, \ldots, J$, the choice-probability vector $z \mapsto b(z; x)$ identifies the joint taste-shock distribution G through the probability of choosing alternative 0 $z \mapsto b^0(z; x)$.*

Proof. Fix such an x, write $\lambda := \lambda(x)$, and take independent standard Gumbel variables $\eta_0, \ldots, \eta_J$, independent of Ω. At z with all $b^j(z; x) > 0$, the conditional logit formula gives the probability that alternative zero has the largest value among

$$a_j(z) + \Omega_j + \lambda \log b^j(z; x) + \lambda \eta_j \quad (j = 0, \ldots, J), \qquad \text{where } a_0(z) = \Omega_0 = 0.$$

Let $\zeta_j := \eta_j - \eta_0$ and $H := \Omega + \lambda\zeta$. Let D_x be the observed open set in the lemma, on which $b^j(z; x) > 0$ for every $j = 0, \ldots, J$. Assumption A3 and continuity ensure that such a nonempty set exists. Define the known map

$$\begin{aligned} c_x : D_x &\longrightarrow \mathbb{R}^J, \\ z &\longmapsto c_x(z) := (c_{x,1}(z), \ldots, c_{x,J}(z))', \\ c_{x,j}(z) &:= -\alpha_j - \delta_j z_j + \lambda \log \frac{b^0(z; x)}{b^j(z; x)}. \end{aligned}$$

The probability of choosing alternative 0 satisfies

$$b^0(z; x) = \mathbb{P}\{H_j \le c_{x,j}(z) \text{ for every } j\} = F_H(c_x(z)).$$

The two threshold maps satisfy $c_x(z) = c(a(z))$. The affine map $z \mapsto a(z)$ has nonsingular Jacobian $\operatorname{diag}(\delta_1, \ldots, \delta_J)$, and Lemma B.5.1 makes $a \mapsto c(a)$ a diffeomorphism. Thus $c_x(D_x)$ is a nonempty open subset of $\mathbb{R}^J$. The values $b^0(z; x)$ therefore determine F_H on $c_x(D_x)$.

By Lemma B.1.7, H has a real-analytic density f_H. Taking one derivative of its CDF in each coordinate determines that density f_H on $c_x(D_x)$. If two taste-shock distributions G and $\widetilde{G}$ give the same observed choice probabilities, their implied densities of H agree on this open set.

The identity argument just used in Lemma B.5.2 extends equality to the connected domain $\mathbb{R}^J$. Thus the complete distribution of H is determined. For $t \in \mathbb{R}^J$, write $\varphi_G(t) := \int e^{it'\omega}\, dG(\omega)$ and $\varphi_H(t) := \mathbb{E}e^{it'H}$ for the characteristic functions of Ω and H. Independence and (B.1.11) then give

$$\varphi_H(t) = \varphi_G(t)\varphi_\zeta(\lambda t), \qquad \varphi_G(t) = \frac{\varphi_H(t)}{\varphi_\zeta(\lambda t)} \quad (t \in \mathbb{R}^J).$$

The denominator never vanishes. Uniqueness of multivariate characteristic functions (Billingsley, 1995, Section 29, p. 382) identifies the joint distribution G, including dependence among $\Omega_1, \ldots, \Omega_J$. □

Lemma B.5.4 (Payoff coefficients and the taste-shock distribution when information is free)**.** *Suppose A1 and A6 hold, $\lambda(x) = 0$ for every $x \in \mathcal{X}$, $\mathbb{E}_G\Omega = 0$, $\mathbb{E}_G\|\Omega\|^2 < \infty$, and $\delta_j > 0$ for $j = 1, \ldots, J$. The observed family $\mathcal{B}_{\text{obs}}$ identifies α_j/δ_j for $j = 1, \ldots, J$ and δ_j/δ_1 for $j = 2, \ldots, J$. Once δ_1 is identified, it therefore identifies (α, δ, G).*

Proof. Fix $x^* \in \mathcal{X}$. A6(a) gives $\mathcal{Z}_{x^*} = \mathbb{R}^J$, so every vector $z \in \mathbb{R}^J$ lies in the observed conditional support. A6(b) excludes ties between any two payoffs. For a fixed $z_j = v$, sending every rival shifter z_k, $k \neq j$, to $-\infty$ gives the probability of choosing j when only j and 0 are available, by bounded convergence, as in the zero-cost case of Lemma B.5.2. When $J = 1$, no alternatives need to be excluded. The function $p_j(\cdot; x^*)$ is the CDF of the switch point under full information:

$$S^{(j)} := \frac{-\alpha_j - \Omega_j}{\delta_j}, \qquad \text{so} \qquad p_j(v; x^*) = \mathbb{P}_G\{S^{(j)} \leq v\}, \qquad v \in \mathbb{R}.$$

Its mean exists by the finite second-moment restriction, and

$$\mu_j := \int_{\mathbb{R}} v\, dp_j(v; x^*) = \mathbb{E}_G S^{(j)} = -\frac{\alpha_j}{\delta_j}.$$

Thus the functions $p_j(\cdot; x^*)$ determine α_j/δ_j for each j.

To recover the slope ratios when $J \geq 2$, fix $j \in \{2, \ldots, J\}$. For $r > 0$, $u \in \mathbb{R}$, and $t > 0$, define the covariate vector $z^{(j)}(t; r, u) \in \mathbb{R}^J$ by

$$z_1^{(j)}(t; r, u) := t, \qquad z_j^{(j)}(t; r, u) := rt + u, \qquad z_k^{(j)}(t; r, u) := 0 \quad (k \notin \{1, j\}).$$

The payoffs of alternatives 1 and j both tend to $+\infty$ for every realization of the taste-shock vector as $t \to \infty$. Every other payoff, including alternative 0's, stays fixed along this sequence.

Consequently, eventually the chosen alternative is either 1 or j, except when their payoffs tie. Their payoff difference is

$$v_j(\Omega; z^{(j)}(t;r,u)) - v_1(\Omega; z^{(j)}(t;r,u)) = (\delta_j r - \delta_1)t + \alpha_j - \alpha_1 + \delta_j u + \Omega_j - \Omega_1.$$

If $\delta_j r > \delta_1$, this difference tends to $+\infty$; if $\delta_j r < \delta_1$, it tends to $-\infty$. If $\delta_j r = \delta_1$, the difference does not depend on t, and A6(b) gives probability zero to its being zero. Bounded convergence therefore establishes the observable limit

$$\begin{aligned} L_j(r,u) &:= \lim_{t\to\infty} b^j(z^{(j)}(t;r,u); x^*) \\ &= \begin{cases} 0, & r < \delta_1/\delta_j, \\ \mathbb{P}_G\{\Omega_1 - \Omega_j < \alpha_j - \alpha_1 + \delta_j u\}, & r = \delta_1/\delta_j, \\ 1, & r > \delta_1/\delta_j. \end{cases} \end{aligned}$$

The distribution of $\Omega_1 - \Omega_j$ is continuous by A6(b). Its CDF has limits zero and one at the two ends of $\mathbb{R}$. Because $\delta_j > 0$, there is a value $u \in \mathbb{R}$ for which the middle probability is strictly between zero and one. At every other value of r, the limiting probability is zero or one for every u. Thus the observed limits determine the unique slope ratio through

$$\{r > 0 : \exists u \in \mathbb{R} \text{ with } 0 < L_j(r,u) < 1\} = \{\delta_1/\delta_j\}.$$

Once δ_1 is known, these ratios give every δ_j. The already identified means give $\alpha_j = -\delta_j \mu_j$. For $J = 1$, no additional slope ratio is needed, and the same mean formula identifies α_1.

It remains to recover G. Alternative 0 is chosen when all other payoffs are nonpositive; using weak rather than strict inequalities does not affect this probability by A6(b). Therefore

$$b^0(z; x^*) = \mathbb{P}_G\{\Omega_j \le -\alpha_j - \delta_j z_j \text{ for every } j = 1, \ldots, J\}.$$

For any $h = (h_1, \ldots, h_J)' \in \mathbb{R}^J$, set $z_j(h) := (-\alpha_j - h_j)/\delta_j$ and $z(h) := (z_1(h), \ldots, z_J(h))'$. This vector lies in the observed support, and

$$G(h) = \mathbb{P}_G\{\Omega_j \le h_j \text{ for every } j\} = b^0(z(h); x^*).$$

Hence the probabilities of choosing alternative 0 identify the joint CDF on all of $\mathbb{R}^J$, including dependence among the taste shocks. □

Proof of Theorem 4. Compare $\theta := (\alpha, \delta, \lambda, G)$ and $\widetilde{\theta} := (\widetilde{\alpha}, \widetilde{\delta}, \widetilde{\lambda}, \widetilde{G})$ satisfying the assumptions and generating the same $\mathcal{B}_{\text{obs}}$. Lemma B.5.2 gives $p_j(z_j; x) = \widetilde{p}_j(z_j; x)$ for every $z_j \in \mathbb{R}$, j, and x. For each j and x, let $S_x^{(j)}$ be the switch point from Lemma B.1.2, with CDF $p_j(\cdot; x)$ given

by the probability of choosing j when only j and 0 are available. At positive cost, this CDF is continuous by Lemma B.1.1; at zero cost, A6(b), applied to $(j,0)$, gives continuity. The identified functions $p_j(\cdot;x)$ therefore determine

$$\mu_j(x) := \mathbb{E}S_x^{(j)}, \qquad V_j(x) := \mathrm{Var}(S_x^{(j)}), \qquad C_j(x) := \mathrm{Cov}\big(S_x^{(j)}, \ell(p_j(S_x^{(j)};x))\big).$$

If $\mathcal{X}_+ \neq \varnothing$, A2 rules out a constant Ω_j: otherwise 1 and $e^{\Omega_j/\lambda(x)}$ would be linearly dependent at any $x \in \mathcal{X}_+$. If $\mathcal{X}_+ = \varnothing$, A6(b), applied to the pair $(j,0)$, gives a continuous distribution for Ω_j, which also rules out a constant. Thus $\sigma_j^2 := \mathrm{Var}_G(\Omega_j) \in (0,\infty)$ in either case. Lemma B.1.2 gives

$$\mu_j(x) = -\frac{\alpha_j}{\delta_j}, \qquad \delta_j^2 V_j(x) + 2\delta_j \lambda(x) C_j(x) = \sigma_j^2, \qquad C_j(x) > 0. \tag{B.5.6}$$

I first recover δ_1, α_1, and λ using the normalization $\sigma_1^2 = \mathrm{Var}_G(\Omega_1) = 1$. A4 gives a sequence of cost-shifter values x_M with $\lambda(x_M) < 1/M$. The cost may equal zero at some of these values. The variance bound in Lemma B.1.2 gives $V_1(x_M) \to 1/\delta_1^2$, while every variance is at most $1/\delta_1^2$. Hence the functions $p_1(\cdot;x)$, $x \in \mathcal{X}$, identify

$$\delta_1 = (\sup_x V_1(x))^{-1/2}, \qquad \alpha_1 = -\delta_1 \mu_1(x), \qquad \lambda(x) = \frac{1-\delta_1^2 V_1(x)}{2\delta_1 C_1(x)}.$$

Applying the same formulas to the other parameter value gives $\widetilde{\delta}_1 = \delta_1$, $\widetilde{\alpha}_1 = \alpha_1$, and $\widetilde{\lambda}(x) = \lambda(x)$ for every $x \in \mathcal{X}$, including values at which $\lambda(x) = 0$.

It remains to identify the remaining payoff coefficients and G.

First suppose $\mathcal{X}_+ \neq \varnothing$ and choose $\bar{x} \in \mathcal{X}_+$. For $j > 1$, the variance σ_j^2 is not normalized, but λ is now known. At $\bar{x}$, $V_j(\bar{x}) < \sigma_j^2/\delta_j^2$. The same low-cost sequence satisfies $V_j(x_M) \to \sigma_j^2/\delta_j^2$, so for some x', $V_j(x') > V_j(\bar{x})$. Subtracting (B.5.6) at these two cost-shifter values cancels the unknown taste variance:

$$\delta_j^2\{V_j(\bar{x}) - V_j(x')\} + 2\delta_j\{\lambda(\bar{x})C_j(\bar{x}) - \lambda(x')C_j(x')\} = 0.$$

Since the variance difference is nonzero and $\delta_j > 0$,

$$\delta_j = \frac{2\{\lambda(x')C_j(x') - \lambda(\bar{x})C_j(\bar{x})\}}{V_j(\bar{x}) - V_j(x')}.$$

The mean identity then gives

$$\alpha_j = -\delta_j \mu_j(x).$$

These common moments give $\widetilde{\alpha}_j = \alpha_j$ and $\widetilde{\delta}_j = \delta_j$ for every $j > 1$. Substitution also identifies each σ_j^2.

Only the joint taste-shock distribution G remains to be identified. Lemma B.5.3, applied at $\bar{x}$, recovers it from $z \mapsto b(z;\bar{x})$ on the observed open set $D_{\bar{x}} \subseteq \text{int}(\mathcal{Z}_{\bar{x}})$, where $b^j(z;\bar{x}) > 0$ for every $j = 0,\dots,J$. The argument does not require the choice probabilities at unobserved covariate values. Thus $G = \widetilde{G}$ and $\theta = \widetilde{\theta}$ when $\mathcal{X}_+ \neq \varnothing$.

If $\mathcal{X}_+ = \varnothing$, the already identified common cost function satisfies $\lambda(x) = \widetilde{\lambda}(x) = 0$ for every $x \in \mathcal{X}$. The value of δ_1 is also already known. Lemma B.5.4 then identifies every remaining slope δ_j, every intercept α_j, and the joint distribution G from the same observed family $\mathcal{B}_{\text{obs}}$. Consequently $\theta = \widetilde{\theta}$ in this case as well. The two cases exhaust the nonnegative cost functions allowed in the model and prove the theorem.

Since $\theta = \widetilde{\theta}$, the choice probabilities satisfy

$$b(z;x) = \widetilde{b}(z;x) \quad \text{for every } z \in \mathbb{R}^J,\ x \in \mathcal{X}.$$

When $\lambda(x) > 0$, both choice-probability vectors maximize the same strictly concave objective $q \mapsto \Phi(q; a(z), x)$. When $\lambda(x) = 0$, both are generated by the same payoff comparisons under the same distribution G; A6(b) excludes payoff ties. This equality also holds at z where $b^0(z;x) = 0$. □